

Jour 411 The Politics of Cancer

Numbered Transcript

JOUR 411 • Gerald Posner

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Lecture 1: Cancer: From Mystery to Industry

At a glance — Posner traces cancer from mystery to industry.

- 1.1 Thank you for being here. This will be a [course] I call The Politics of Cancer. It's not about cancer, but of course, it is about cancer. What I mean by that is, I'm not doing a course in which we're going to get into the nuances necessarily—sometimes we do—of the disease itself, but instead, what I'm doing is a study as a classic investigative journalist into the industry and what shapes it over a period of time, from the ancient beliefs about what had caused cancer, to how something becomes eventually what I call an industrial empire.
- 1.2 Now, the fact you hear industrial empire, you immediately think of public health and you think bad—not necessarily so, but you understand how an industry over time develops and how it gets its own focus, which in terms of cancer, as you will see as we go through these lectures, becomes increasingly—and that, emphasize that point—it becomes increasingly about treatment rather than prevention. So I think that that's what we find—we find an industry that often looks at, how do we treat it, not how do we prevent it? And we'll explore these practices coming up.
- 1.3 So, unmasking the cancer industry. There is no secret here. It's not as though suddenly I'm going to come in and tell you, by the way, there's a cure for cancer that you can get down at Walgreens, and just run down there before they run out of it. There is no such thing. I don't mean unmasking the cancer industry that way. We're sort of exposing how it's developed into this machinery.
- 1.4 So, there's the financial and political influences. We're going to talk about how some of those financial and political interests impacted treatment and research priorities. It will not surprise those of you who listened to me before about the pharmaceutical industry, that there is a through line here sometimes about follow the money, because that's inevitable. It is a business that's doing treatment. At the same time, it's also making sure that it's funding itself. And those priorities matter.
- 1.5 The evolution of treatments, how they've evolved from very, very dubious and terrible remedies that were fake, to biotech solutions that have really shifted the entire industry and will in the future coming forward, as well. And then what I call profit versus public health, the tension that exists between public health—and this is the same thing you get in the pharmaceutical industry, it's the same thing you get in other medical areas, you certainly get it in the cancer industry, this idea of influence of patient care and policy decisions.
- 1.6 So, how is it that I ended up doing a deep dive on cancer? It was because my wife, Trisha—we've been together for 45 years, and my partner on all of my projects, and she's an author in her own right—5 years ago this coming August, got diagnosed with

breast cancer. Now, a surprise, yes, it's always a surprise. Until you have a cancer diagnosis yourself or somebody very close to you, cancer seems a little bit distant.

- 1.7 She came from a family—all the women on her side, from her mother to her two maternal aunts to her grandmother, going back, had had breast cancer. Her first cousins, they were all survivors. She thought maybe it would skip her. She did not have the BRCA gene, which is an indicator that you have a much higher genetic risk for cancer. So she had been tested for that early on. But still, she got diagnosed with that.
- 1.8 And I think that one of the interesting things was, once that diagnosis—you cannot fake what I call a positive attitude about going through a health crisis. You either have it or you don't. If I could bottle her attitude, I would bottle it and sell it because her attitude was fantastic, which is, "I didn't want this. I'm not happy to get it, but now I have to do whatever I have to do to get it in my rearview mirror and get it behind me." And that was her approach, and she was fantastic.
- 1.9 We would run into, a year or two later, friends we hadn't seen for a while, and we would start to talk to them about life in general and catching up. Then when we'd be leaving, she'd say, "Oh, yeah, I forgot to tell you, by the way, I had cancer a couple of years ago," because it wasn't the first thing on her agenda, which was fantastic. It did become the first thing on my to-do list. So I did a deep dive, as though I was doing a book—the same thing— and I became a pain for the doctors we were sometimes dealing with, because I wanted to know everything.
- 1.10 I didn't want her to do all the research necessary. She was doing enough, but she had to avoid the research that led me down this rabbit hole. And this rabbit hole took me back to the origins of the disease, took me back to the early days of how it was done, and I started to get a better understanding for the industry. And that's what really started this process.
- 1.11 Now, cancer's heroes and villains, boy, we're going to find them. There's no doubt about it. And you'll see them here, what I call these relentless cancer researchers. They're there, they discover it, they are looking for cures, and they're dealing with corporate interests. What I call corporate and government conflicts, no surprise about that—you won't be surprised to find out that the interests in the cancer industry are sometimes fighting between what I call private industry and the influence of the key stakeholders, regulators, nonprofits, medical professionals, and others who are challenging the cancer care system.
- 1.12 Why politics of cancer? I told you that I was with her on the day that she went to get a mammogram, and the doctor said, "By the way, I see a spot, and I think that we should do a biopsy on it, and I'm pretty sure that it's going to be cancerous." It was very interesting, and that turned out to be right. She became on that day, I said, one of more than 1.8 million Americans who get an annual diagnosis of cancer during their lifetime, which is pretty remarkable, about 5,600 a day diagnosed, 600,000 a year

dying. So we're talking about a disease that still has a tremendous impact. And it does have a way of changing everything instantly.

- 1.13** But we did what we do, which is research. Now, there's a sidelight to this. I do love this. That's the day you want in cancer treatment, the day you get to ring the bell indicating the end of treatment. But that was the best part of it. So it jumps to the top of your list.
- 1.14** Now, Trisha ended up writing—by the way, this is a sidelight—an article in The Wall Street Journal, "When Did 'Woman' Become a Dirty Word?" And she was in her cancer treatment, and a nurse said to her, "The prognosis for your chest cancer is very good." And she said, "What? Chest cancer?" He said, "We like to be inclusive. We don't want to make anybody feel as though they're being ostracized by saying breast cancer." It turns out, 1% of breast cancer cases are men, so it is a small number. But Trisha said, "Well, I think I have just breast cancer." And again, the nurse said, "We don't want to make anyone uncomfortable." Trisha said, "You're actually making me feel uncomfortable by not calling it breast cancer." That happens to be a sidelight.
- 1.15** She wrote a piece in The Wall Street Journal, and I do like this. I'm going to just read you this one paragraph. "There may be no universal female experience, but there are certain biological and physical differences that affect only women. I'm sorry if this offends anyone, but men don't menstruate, give birth, experience menopause, or develop endometriosis, ovarian cancer, or cervical cancer. Men do get breast cancer, but it's extremely rare.
- 1.16** "For women like me, it seems as if we are incrementally being erased in a rush of political correctness to ensure that no one's offended. What about my rights? Isn't it possible that women will feel marginalized if institutions rely on gender-neutral language? It seems to be the death of common sense. Many of my female friends and colleagues have privately told me they would like to speak up, but they're worried about public consequences. Self-censorship is one of the reasons that our language is changing to meet the demands sometimes of only a small minority. Silence is interpreted as acquiescence.
- 1.17** "Over 100 years ago, women were fighting for a right to vote. When I was a girl, my aunt Rose regaled me with stories of public protests of demanding suffrage, which England granted in only 1928. Rose was only 16 and took to the streets against her immigrant parents' wishes. 'There are some things too important to sit back,' Rose instilled in me as a child. We make our own future. It's Orwellian today, to me, to feel compelled to remain silent about our female bodies, motherhood, and our health as women."
- 1.18** I love the fact that she was willing to write that as a result of having had that encounter about chest cancer. It had an impact that it should have had. And it was the woman that I fell in love with—the woman who was not just going to be quiet. So, she was not quiet at that moment.

- 1.19** Now, modern cancer care, big business. The \$200 billion spent last year, that's a 50% surge from just a decade ago and double the spending on COVID-19 vaccines. So, for those of you who were here for the talk about Big Pharma and the money that was made from COVID-19 with the contracts the government bought with over \$26 billion, double this on cancer care. Expected to reach—you think \$200 billion is big—\$320 billion by 2026. Those are the estimates within the industry and by the American Cancer Society, focused on patient insurance, company billings, and growth fueled by drug development, some of which is very expensive.
- 1.20** The ancient fears. So cancer isn't new. It actually goes back—they're being described as having no treatment, essentially a death sentence, on a piece of papyrus found from ancient Egypt, described on a 5,000-year-old piece of papyrus about surgeries for these lumps that we now interpret and know as cancer. 2500 years later, the father of medicine, the Greek physician Hippocrates, ends up dubbing them—the tentacles of the tumors—as karkinos, Greek for "crab." Another Greek doctor referred to those growing tumors as masses. A Roman philosopher early comes in and dubs those Greek terms in a medical encyclopedia, and lastly, carcinoma as cancer, onkos eventually becomes oncologist. And that's how we get the shorthands for those who study tumors.
- 1.21** I'm going to talk to you a little bit more about some of the early beliefs, but the causes were, they thought, black bile, divine punishment, bad climate, and an imbalanced sex life. So it's the same thing about early diseases. They thought tuberculosis was coming from the environment, and they weren't sure what immigrants were bringing in as diseases. They had no idea about what caused early diseases. They didn't even know about germ theory early on.
- 1.22** So, for thousands of years, cancer is basically looked at as this mysterious and fatal disease with all of these mythic explanations coming up and moral judgments around it. So, we go from the ancient Greeks and Romans, this idea of what I call black bile inside the system somehow, a concept rooted in the four humors of medicine, to the Renaissance, where it's viewed as divine punishment, a more religious idea. So, you're getting somehow retribution. This is bad karma for all your sins. This is sort of this period in which people think that they're suffering as a result of that.
- 1.23** In the late Middle Ages, it becomes popular as sort of what I call lifestyle. There's even a theory for a while that catches hold that says it's extreme climates. If you live in extreme climates, bad climates or very cold climates, it could be the problem. And for those who thought it was too much sex, the cure for cancer was easy. It was celibacy. And the idea that nuns and priests were not developing cancer at a higher rate was proof of that, thought the theoreticians.
- 1.24** Environmental causes. Boy, I'll tell you, a German professor—by the way, at this time—blamed it on mammary duct blockages. The first who got it right was a British surgeon in 1775. He linked the high rates of cancer of the scrotum among English chimney

sweepers to constantly being exposed to soot, the ash that they were working with all the time.

- 1.25** German chimney sweepers had a lower rate of tumors based upon this comparison that he had done, and he wondered why. He found out that German chimney sweepers wore a leather—what looked like an apron—and they bathed more regularly. And that, he thought, might be the difference. So he thought maybe the exposure to the soot, the constant accumulation of that, was some type of causative factor. That's the first time you start to get somebody thinking of it in a way that makes sense.
- 1.26** Early milestones, you have in the early 1800s—now, this is really early. Before you really get the idea, in America, at this time, there's no licensing required for doctors. There's no AMA, there's nothing else. You can end up putting up a shingle for any reason at all. In England, a little bit stricter, they do have a surgeon, and he pioneers what I call very early-days, rudimentary surgery. Out of 60 of his patients, only two survived. It was, as I say, sort of a torturous procedure because there was no anesthesia at this time. This is not a time you want to be what I call, essentially, an experiment before one of the surgeons, but he's the first one in the industry doing this.
- 1.27** In 1902, this German zoologist discovered a chromosome link to cancer. This is a major thing. Mutations. He's working with animals, and he thinks that among the laboratory animals, both rabbits and mice that he's working with, he thinks that there's this chromosomal change that causes a mutation, a change in their cells. They can't quite see yet with the microscope, but it's unchecked cell growth. He's blaming, essentially, microbes and chemicals. It turns out that Theodor Boveri, that German zoologist, was right. It was a breakthrough. But doctors and the medical profession, as it existed then in 1902, dismissed him widely and said, "He's a zoologist. What can he really know about the body?" So, you get the first breakthrough from somebody who's not working with humans, but the human doctors say, "I don't think that can be right."
- 1.28** You do get the formation of the American Cancer Society, 15 doctors and businessmen united in New York. The aim? To pool knowledge on cancer and seek better understanding of the disease and its causes. So, what I call the start of that isn't about business; it's about the disease. And that's actually a good start.
- 1.29** Janet Elizabeth [Lane-Claypon]. If you don't know her name, you should. She's a British physician who really was, I say, a trailblazer in cancer treatment. She's the one who decides to focus on breast cancer. And she does this study, and it really is a meticulous study at the time, of 500 women who had significant links. She establishes between 500 women who had cancer, and she's looking at 500 women who had not had cancer. She believes that there is a link at this point between childbirth— not having children puts you at a higher risk of breast cancer, she concludes—and women who have not breastfed, slightly higher risk of breast cancer.

- 1.30** And this was, A, overlooked because although she was one of the very few female doctors at the time, men were not paying as much attention to her. She produced the first risk and life expectancy tables, and at that time, they dismissed the idea. If you look at some of the contemporaneous statements about her work when she first puts it out, they say it's just about impossible that the question of whether you have a child or not as a woman is going to affect your breast cancer risk. It can't have a link.
- 1.31** We now know, in fact, that women who don't have children have a slightly higher risk of breast cancer. The things that she had pointed out in her study, back at this time, turned out to be true. It took nearly 100 years for her to get the acknowledgment that she deserved as being ahead of the rest of science. She had established the first, what I call the equivalent of a study that made sense between women who had cancer and women who did not, and looking for the difference in the causes in terms of lifestyle, and that often gets overlooked. Few doctors believed her when she said it. It's now an accepted factor for breast cancer, literally a century ahead of her time.
- 1.32** Now, William Stewart Halsted, a radical solution. So this surgeon, William Stewart Halsted, he's American, he sought to improve surgeries. We talked before about the very crude surgeries that had been done without anesthesia. Now, you're at a point where he's at Johns Hopkins, and he is taking the idea that we could do something with this. He's this chief of surgery. You've had a tumor. Can the tumor be removed so that it will not return, and then you're cured and free of cancer? So this is Halsted's idea.
- 1.33** The interesting thing about him is, he takes over Johns Hopkins. He's known for being very quiet and understated. And when he had gained his fame beforehand, the story that used to make the rounds about him, and it turns out to be true, is, he had performed surgery on his mother and also saved his own sister with a blood transfusion that he made.
- 1.34** It turns out even to be more dramatic than that. He had removed his mother's gall bladder on her kitchen table, so somebody who doesn't shy away from doing something. His sister was bleeding profusely at one point after childbirth, and he withdrew his own blood and set it up and infused it into her. So he's a bit of a risk-taker, if you might be able to say that. He would work these long hours, and he suffered a breakdown in medical school before he goes on to Johns Hopkins.
- 1.35** Now, 10 years before he's at Hopkins, he became addicted to cocaine. Interesting thing, cocaine was still around. Pharmaceutical cocaine, by the way. He was getting it prescribed. Cocaine was available, so I don't want you to think he was going on the street and buying it. He was addicted to pharmaceutical cocaine. And he used to inject it, which at times would make him seem a little bit manic, but after battling cocaine addiction for 5 years, he entered a sanatorium and successfully treated that addiction.
- 1.36** The problem is, by the time he left, he had a morphine addiction. So he joins Johns Hopkins a year later on the surgical staff, before he becomes head of surgery. He is a

morphine addict. The great thing is, he decided to go back to cocaine as well. So he would mix cocaine and morphine together. What addicts described decades later as a speedball that killed people like John Belushi, the actor, comedian, and others, that type of mixture, because you're mixing a stimulant together with a very, very heavy respiratory downer, that pushed your heart rate at a low rate, at the same time another stimulant is bringing it up, it's a perfect thing for a heart attack.

- 1.37** But he managed to hide that addiction so well that people didn't know he had it. And he was working, and he performed, and he did his surgeries. He gets married in Baltimore to a nurse who didn't even know he was an addict until she started to live with him. So he was hiding that very, very well.
- 1.38** And what he ends up doing, what Halsted ends up doing, is something interesting. He's the one who decides and finally makes what I call a surgery. This surgery that he does is, he names it radical mastectomy, after the [Latin] word meaning "root" in this sense. He uses a scalpel to remove the root of the cancer, as he views it. He's convinced that he could cure breast cancer if he goes in and is able to arrange to take those tumors out. He cuts through the pectoralis muscle. It's a procedure through the collarbone, removes a strip of the lymph nodes. It's a very invasive procedure.
- 1.39** But he's keeping a list of what happens as you go along during this time. He has at Johns Hopkins a story of every woman who goes through this very, very radical procedure in which the breasts are removed and tissue is removed underneath, and he says it'll never come back. That's the end of cancer. Half of his patients had cancer return, and they were dead within 3 years. So, clearly, to him, he could not understand that.
- 1.40** That's because, at this point, when he's inventing the radical mastectomy, there's no understanding of how cancer metastasizes. People like Halsted don't understand that it can get into the lymph nodes, it could travel to other parts in the body, that you might have it present as a breast tumor, but it might already be in other organs. So he's thinking, if he can remove that one tumor, that that's the end of the process, but it's not.
- 1.41** He presents his findings at a meeting of surgeons in 1898 in New Orleans, and dozens of surgeons start to experiment with his idea of a radical surgery, and they develop their own techniques for it. They look at it. But this, from Halsted, becomes the first surgical technique for cancer that takes hold in the industry. In 1898, we're talking about a fairly old time in terms of how this works.
- 1.42** From death sentence to surgical intervention, so 18th and 19th centuries, doctors started to do autopsies, and that revealed some new patterns, but cancer was still seen basically as a death sentence. In the 1800s, first attempts to remove—surgery [without] anesthesia and hygiene— those were the really torturous days. Then you get William Halsted sort of pioneering radical mastectomy, cutting through breast and muscle. In

the early 1900s, his technique becomes the gold standard despite the high mortality rates, despite half of his patients dying within 3 years.

- 1.43** And by the mid-1900s, something has taken place. Aggressive treatment is seen as both status and revenue, even if it's ineffective, because people like Halsted and others were able to turn it around. Glass half full, glass half empty. So he thinks he's going to be able to do this surgery, 100% survival rate. Oh, half your patients are dead within 3 years? Oh, that means half my patients are alive. I'm getting pretty good results. So it's reversed. Would they have been alive anyway, if the surgery hadn't been done? We don't know.
- 1.44** One of the problems with cancer, by the way, just so you know this, and one of the most frustrating parts with cancer, is that some cancers are very slow-growing. Some cancers are fast-growing. And oncologists, even today, cannot tell the difference when they see a tumor between a slow-growing cancer and a fast-growing one. So the reaction is, and the medical policy, is to act, to try to remove the cancer, because you're not quite sure.
- 1.45** But they cannot tell you at any point whether, especially if you're older and develop cancer, it is a slow-growing cancer. It's possible that you might outlive that cancer. You might live with that cancer. Autopsies done on men over the age of 85 who have passed away, it turns out that over two-thirds of them have some remnant of prostate cancer, but many of them didn't know they had that. They had a cancer and they outlived it, essentially. But they can't tell you still today. It's not as though they can do a quick test and say, "Oh, fast-growing and slow-growing."
- 1.46** Now, into this mix come what I call the pioneers of invisible rays, Röntgen, Curie, and Ewing. This is groundbreaking material. So we have surgeons on the one hand, and now we have this period with a German physicist, Wilhelm Röntgen. He discovers a new type of electromagnetic radiation that passes through human tissue. He's actually working at this point—it was interesting—on whether cathode rays could pass through glass. And he covered a cathode tube with a black canvas. And then he noticed that he could see a light through that tube, and he noticed that, on an adjacent fluorescent screen, there were parts of that light that were escaping.
- 1.47** He eventually realized that it could go through solid objects. And if it went through solid objects, he would try things like his hand and others. There's this picture which he puts up and tries to see if he could do that. He can see the bones underneath the tissue. He has discovered something unusual. What's great about it, he didn't know what type of light rays they were. So what did he call them? He called them X, meaning X—I don't understand. You put an X down for something, X-rays, and that stuck. And that's how we get the beginning of X-rays.
- 1.48** So Röntgen realized the importance. Sometimes doctors—great things in medical history—somebody stumbles across an accidental find and they don't know the

importance of it; he knew the importance of it. And within a year, doctors in Europe and the US were using rudimentary devices to deploy X-rays for locating bone fractures. And he went on, in 1901, just 6 years later, to win the first Nobel Prize in Physics for that discovery.

- 1.49** Marie and Pierre Curie, Polish [and French] physicists and chemists who conducted literally pioneering research on radioactivity. She and her husband led the way on these elements, and they discovered, while they were doing all of this work, two new elements, radium and polonium. She's the first woman to win a Nobel Prize for her work. She actually ends up winning it twice. And she wins it in 1903, just a couple of years after Röntgen does, and comes back and gets the Nobel Prize in Chemistry in 1911. She's the first person to win it twice [in two disciplines], the only one, and the first woman ever to win the Nobel Prize.
- 1.50** They are taking radium and polonium, by the way, the Curies are, which is more radioactive than uranium, because they're trying to get the radioactive material here from pitchblende, this material and dirt that they were getting it from, at this time, a Czechoslovakian mine. But you had to mine that out. And she named polonium herself. She gave it that name after her native Poland.
- 1.51** So the radium boom, World War I, created a frenzy. Now we're talking, think of this, 1914, '15, '16, '17, into 1918, a frenzy for radium. The military on both sides wanted watches and instruments that had luminized dials, so they want to be able to see things at night. It makes perfect sense. So they're using radium, and they're literally having it applied by hand to dials and other instruments. Overnight, radium becomes one of the most expensive and sought-after minerals on the Earth.
- 1.52** So, one of my favorite little trivia facts—because of its scarcity and how hard it was to find radium and the demand for it in World War I, the price booms. Early in the war, the price has peaked at \$150,000 a gram. Now, think of this. You're talking 1/27th of an ounce. That's about \$68 million a pound back then in World War I money. That's \$1.95 billion today. You'd pay almost \$2 billion if the same demand was there for a pound of radium. In comparison, gold was \$20.67, so radium was really the value.
- 1.53** During the Great Depression—we would find out, in just a few minutes—Memorial Hospital, which becomes Memorial Sloan Kettering, their radium department became the most profitable and helped them survive the Depression. And by 1937, radiation therapy, not recognized yet as a proven cancer treatment, finally gets the acknowledgment from the medical association: yes, you can use X-rays, you can use radiation to shrink tumors.
- 1.54** Surgeons, by the way, Halsted and the rest of the surgeons who had developed radical mastectomy, talk very badly about radiation. Why, because they think it's dangerous? No, because it's competing for surgery. Maybe somebody might just go to get a tumor reduced through radiation, which at that time, nobody realizes the long-term

consequences, that it could be cancer-causing. But it seems a less invasive and easier solution.

- 1.55** Radium King. James Douglas, the CEO of the Phelps Dodge copper mining company—the DuPont Company would later compete with him—he created the National Radium Institute. And with his good friend, a Cornell professor and pathologist, James Ewing, who you'll meet in a minute, he gave \$100,000, \$3 million in today's money, to Memorial Hospital. And he wanted them to be able to become a cancer hospital. When he gave the \$100,000, he said, "You become a cancer hospital." This was back in, [1912] and you have to be able to do research on radium and X-rays and see what you can do.
- 1.56** They set up a 20-bed clinical study. His motivation was personal. His [daughter], Naomi, had had breast cancer. They traveled to Europe. They went to see the Curies. They had gone everywhere for five different treatments, and she still died. So he was motivated by that personal loss, the inability to save her. He thought that radium could cure not just cancer, but other illnesses as well. So he donated 3.75 grams of radium, which was then really sought after. And on his death, he gave the last amount he owned, 5 grams, everything he owned, to Memorial Hospital.
- 1.57** In the rise of radiation therapy, his partner in this, James Ewing, the pathologist at Memorial, had tested, and he did test, radium. He's sort of this pathologist who plays a crucial role in the promotion of radium therapy, and he establishes Memorial Hospital in New York as the leading cancer research center, just at a time when the American Cancer Society is coming around to say this is something we should use. He's testing it at Memorial on patients for prostate, skin, gynecological cancers. They're looking at radium for possibly anything.
- 1.58** The way they would do it then, early days, they would take a little bit of radium, just literally specks of radium. They would put it into a box, often a wood box, not realizing that that didn't stop the radiation coming out of the radium. They would have a small hole in the bottom of the box, and then they would hold the box over the patient in the area where the tumor was, in the hope that that little hole would direct the radium rays and the radiation coming from the radium there.
- 1.59** They also had a second method. They would collect radium and what they call radon gas in glass tubes, and then they would actually implant the small glass tubes directly into the patient in surgery— not the most successful way to be able to pass it along. There were some doctors, when I look at the contemporaneous medical literature, who criticized what Memorial was doing, not out of jealousy, but because they thought it was really a form of human experimentation. But it was the only way that they were going to be able to advance the limits of what the hospital could do.
- 1.60** He had more control as a hospital medical director than today's hospital medical directors, who have to go through a committee, what James Ewing has decided to do. And he ends up founding the American Society for the Control of Cancer. He publishes

a book about radium and diseases and cancer, which becomes the textbook, so he's setting the groundwork on this. And his textbook on tumors, an instant classic.

- 1.61 There's actually a—I had never met anybody who has it, but you will find patients today who have something called Ewing's sarcoma. It's a bone cancer; it's named after him. This is this period in which they're going through. You hear the name of a cancer, you have no idea what it's from. This is from James Ewing.
- 1.62 When I say that radium saved Memorial Hospital during the Depression, that's because they put a hefty surcharge on radium emanations, as they called it. That was the practice. Desperate cancer patients, you will not be surprised, you're looking at a disease that you think is going to be death. People would sell their life savings sometimes to go over and try to get the radiation, the radon emanations.
- 1.63 The tragic cost of radiation discovery. So these pioneers who were involved in the early days of discovering radiation and radon, they paid a price because they didn't realize how deadly it was. Radium-226, that's the isotope that Marie Curie worked with. There were a few early warning signs. Nikola Tesla, Thomas Edison, Pierre Curie, they all reported injuries. They'd gotten burns at some point on their skin from exposure to radiation, so they knew that they suffered these skin irritations.
- 1.64 There was a report in 1902 that listed 171 instances of accidental burns from the radiation. A German physician was the first case in that year of radiation causing a cancer. But the problem was this. The isotope that she was working with, that other doctors and other researchers are working with, has a very long half-life. We're talking like 1,600 years. You don't put it out, and then all of a sudden it's gone, and you can throw it away in the dustbin. So significant amounts even remain today from some of that.
- 1.65 Persistent exposure to it, the isotope that she worked with, continues to produce radiation today. She handled that radium extensively, unknowingly contaminating herself. Here's what those early pioneers did not know, and that was that it wasn't—they believed that if you didn't have a radiation burn, you were all right. When I told you a moment ago that 171 people in a report, doctors and other researchers working with radiation reported skin irritations or burns, they thought that was the risk.
- 1.66 If you weren't careful with it, if you handled it too long, and you didn't take enough of a precaution, then you could get a skin burn. They didn't realize anything about cumulative exposure, that it was that cumulative exposure over a period of time that would add up and cause cellular changes inside. I think that they refused—many of the doctors and researchers of the time also refused to take the warnings seriously. They didn't understand the latency period. But more than that, I think that they really believed that they were safe because they didn't feel ill.
- 1.67 And that's one of the other things. If you have a poison, you ingest a poison, you get an infection, you feel bad. You're getting exposed to radiation at low levels, not at a high

level, at low levels, you don't feel bad necessarily. And therefore, they would say, "These warnings, these reports, other doctors, they must have done something wrong because I don't feel bad." And then they would continue to work with it, without taking the precautions.

- 1.68** The Curie family, Irène and her husband— Marie Curie's daughter, Irène, marries another [scientist], and they continue the work of her mother and her father. Her father's been killed in a [wagon] accident, by the way. They go on to win a Nobel. Talk about a talented family who's really making a difference: father, mother, and the daughter. But the entire Curie family, unfortunately, suffered the consequences of this period. So they weren't the only ones. I'm coming back to them for one moment.
- 1.69** But the young women who were painting luminous watch dials, tens of thousands of women worked in factories during the war, in World War I, and they're putting the paint on, and it continued in the 1920s. There was a time, I remember my father had a watch, which was fantastic. The dials would glow. I could see it at night. It was great. I didn't realize that they were painting, essentially, these dials, these radium-glow watch dials, glow-in-the-dark, a novelty.
- 1.70** The number of people working in factories, putting them on, over 95% were women at the time. They would often lick their brushes to fine-point them, so they could get a small amount of radium on the dial itself, nobody telling them at the time it might cause a problem. This is what I call the effect. The ingestion of radium led to an enormous amount of bone cancers and severe radiation injury for those women who worked in those factories, one of the highest percentages we've ever seen for just coming out of an industry, all by radium poisoning.
- 1.71** And what's being sold to the public at this time? There it is. You can open up a newspaper and see an ad, "Radium is restoring health to thousands. No medicine or drugs, just a light, small, comfortable, inexpensive, radioactive pad, worn on the back and over the stomach at night, sold on trial. You can be sure it's helping you before you buy it. 150,000 sold on the plan."
- 1.72** Now, it's not just for cancer, because we know this is a period in which there's yet no FDA and nobody's really trying to control it or anything else. This is for rheumatism, high blood pressure, constipation. Put a little bit of radium on yourself for constipation, nervous prostrations, asthma, other respiratory disorders, heart, liver, kidney, bladder trouble. "No matter what you have tried, you may be able to find it here in Degnen's Radio-Active Solar Pad at our risk, trial offer."
- 1.73** And this isn't just a one-off. There are ads after ads after ads in which this is being marketed as the equivalent of what I call snake-oil salesmen, except now they're selling something that isn't making you sick, as many of the patent medicines were at the turn of the century, because they were loaded with morphine and they had cocaine and

other items that were legal. This is something that you put onto yourself and you don't necessarily feel ill. You feel good, and that's part of the problem.

- 1.74** The grim toll of the early part of radiation exposure. Marie Curie, she ends up with aplastic anemia. That's extensive exposure to radium and polonium. That's an incurable blood [disorder] she has at that time. Henri Becquerel, severe burns. He carried radioactive materials in his pocket. Thomas Edison, skin cancer, frequent use of X-rays for experimentation. William Röntgen, the developer of X-rays, the person who found it, severe hand burns, used it all the time. He was taking pictures, X-rays of his own hand constantly to show others how you could see through skin, and eventually it keeps burning them, time and time again. And Walter Ken and one of the early leukemia, worked extensively with radioactive materials. All those early warning signs ignored there and put not just the researchers and the doctors at risk, but also the patients at risk.
- 1.75** So you get Nikola Tesla, Thomas Edison, Pierre Curie, they reported injuries, but they believe their injuries are from their X-ray experiments. They knew it, but they didn't realize the long-term effects. Patients suffered serious skin irritations. There's a list at Memorial Hospital at that time of how many skin reactions patients who got this treatment had, and it was just thought of, this was one of the side effects of it.
- 1.76** In 1911, the report lists 94 cases of radiation-induced cancers. Now, half of those were doctors or technicians. Hard to imagine because we think of radiation possibly leading to cancer as a development much later, but so early here that there's this report in 1911 that says, just under 100 cases, we believe that these tumors are a direct result of the radiation exposure from these researchers. Then the refusal of doctors to take the warnings seriously. They worked with X-rays. They were unaware of the cumulative risk.
- 1.77** Marie Curie's lead coffin and tomb. You didn't think when you were coming in for the politics of cancer, I was going to tell you about Marie Curie's tomb, which is in Paris, as a matter of fact, at the Pantheon. It's a very famous thing because she's very famous. It shows you the power of the radiation that she was dealing with. They buried her in a wooden coffin. And then her notebooks and other items are stored in a special lead-lined box, and they require protective gear if anybody's handling them to look at them for research.
- 1.78** This lead-lined box, the wood coffin, is then placed inside an inch-thick lead-lined box to provide additional shielding from radioactive contamination, because by the time she dies, they're worried that she might be radioactive herself, the corpse might, and they're putting her into this great center in Paris, this historical place where people will come as tourists to look at it. Lead-lined tomb at the Pantheon, Marie Curie's final resting place has extra layers of lead shielding to protect the public from the layer of radiation emitted from her remains.

- 1.79** Now, you might say, isn't that a little bit over the top? I mean, haven't they gone from one extreme of handling this material all the time without any gloves or protective material, to then burying one of the pioneers of the industry inside of that? Well, guess what? In 1995, they exhumed her remains. It's this period in which they finally decide to do it. And her remains were exhumed in '95, 60 years after her death, and what did they find? Radioactive contamination.
- 1.80** Detectable levels of alpha and beta radiation, both of them, both dangerous, still saying that her body, 60 years after it had been buried, was still dangerous for anybody who was with it without protective material. That's rather remarkable. So, when you think about it, not only were the early researchers poisoning themselves, they were also, in many ways, affecting those who were near them. Talk about second-hand smoke. This is second-hand radiation exposure, a little bit different.
- 1.81** Now the push for federal cancer funding. Doctors were lobbying for a national cancer research lab. They felt as though, if they could get federal money, they might be able to get—what they were really looking for is money. And who had the money? The federal government. So you could get philanthropists to go ahead and do it, and James Ewing might get a copper miner who was going to give him some money, but in the end, you needed the government to get behind the idea of cancer.
- 1.82** And they also wanted another thing. They wanted—these doctors, who were asking for cancer to be regulated at the federal level—let's get rid of the quacks. Let's get rid of all the nostrums and everything else that say cure for cancer. And there were plenty of these fake peddlers still at a time before the government has any ability to regulate for safety. The government really can't regulate for safety until 1938, when they passed the Food, Drug, and Cosmetic Act. So until this time, they have no ability to step in. The FDA exists, but it's toothless, totally toothless.
- 1.83** So, there are people offering cancer cures, and you can't do anything about it. They're clearly fake. And people who are desperate will try anything, and they did. And so as a result, they want those cures exposed or at least controlled. And those doctors' efforts to secure the funding mark the first time there becomes a debate in the federal government about whether they should get involved or not on doing something about cancer care.
- 1.84** Now enters Senator Matthew Neely. It often takes one—West Virginia—one senator to decide to change the mode, and this senator decides to take on the cause. So here's what happens. It transitions to Congress first. The shift moves from hospitals to surgeons to the US Congress. He comes in and becomes the advocate in Congress to say, "Let's do something about this." In 1916, there's a report that says—for the first time, it gets a lot of coverage—that cancer has become the second largest killer in the United States. It's moved up 50% after heart disease. That's stunning to people, because they always think of cancer as something—but they weren't measuring the cancer rate at that point.

- 1.85** Now, Time magazine and Newsweek and all those early magazines and newspapers are all reporting that. So there's some incentive for politicians to react to it. Neely has a proposal. In 1927, he offers a \$5-million reward to anyone who comes up with a cancer cure. \$93 million today in 2026 money.
- 1.86** What do you think that results in? Oh, my God. Police who are looking for a suspect and they say, "By the way, there's a \$100,000 reward," know that they're going to get some good tips, but they're also going to be flooded with fake tips and bad tips and everything else, and they have to go through it. Congress, Neely's office, all of the senators get flooded from all of the quacks who are making cancer cures, to say, "I deserve the \$5-million reward because I already have a product out that cures cancer." They were all sending it in, so that went nowhere. No one collected on that particular reward.
- 1.87** So he tries to go ahead. He gives this impassioned speech inside of Congress, and he's trying to get a bill passed at one point. And his persistence, he puts in a bill year after year and can't get any money for it. So at one point, he gives a speech after the \$5 million was up. And he says here, he calls in this Senate speech, cancer—I'm going to quote here—"humanity's greatest scourge. If the rapid increase in cancer fatality should persist in the future, the cancer curse would, in a few centuries, depopulate the Earth."
- 1.88** Now, that might be somewhat dramatic, but it still got the idea across, that he wanted some funding for cancer. And it's very interesting that, at this point, he's looking for what would be the equivalent of \$500,000. Much bigger money was being used on research for other diseases and other health matters, but it gets turned down every year. He can't get it out of committee. There just isn't an ear in Congress.
- 1.89** And then the media comes to his aid, unintentionally, but its coverage of cancer—we all know how something becomes top of the line. It becomes a subject just on the radar for the media, and then the media gets on top of it. And this is what happens in 1936 and 1937. Time magazine had a cover story, and they reported that he had a chance, finally, maybe, of getting it over the cover line, because its cover story was, "US Science Wars Against an Unknown Enemy, Cancer."
- 1.90** Life magazine did a cover story in which they talked about it being the second most lethal cause of death in the United States and predicted it would not be long before cancer killed one in five Americans. One in five Americans, meaning voters. So now Congress is paying a little bit more attention to that. The editors at Fortune ran an extensive cover story and published an 80-page companion book called Cancer: The Great Darkness, and it criticized the fact that the government had stayed out of the funding.
- 1.91** So finally, Neely thinks he has a 50/50 chance of getting it through, and he does. In 1937, the National Cancer Institute Act is passed. So this is the first time the federal

government decides to take a role, and it provides a modest amount of funding, not very much, but better than nothing, for federal cancer research in Bethesda. That's where the National Cancer Institute will be set up, and it marks the first time the government's involved in this research.

- 1.92 The Bethesda lab became a hospital during World War II. So 1937, it gets passed. You have this funding. It's going to do cancer research. It's fantastic. Doctors want to go over there, and then suddenly World War II takes place, and they shelved that plan because they make it into a World War II hospital, and soldiers returning who are wounded in the war end up coming back and often being treated there.
- 1.93 It also becomes the future [site], and we'll talk about this in another lecture of the chemical weapons research that was started by the military. And cancer is a federal mandate. So the implications of moving cancer care from just private practice to federal funding is a major step. And now you're involved, as I say, for the first time with private enterprise going ahead and trying to join hands with the federal government and seeing how they can move the topic forward.
- 1.94 In many ways, these cancer statistics that Neely used and were used by magazines to say, "Second biggest cause of death, one out of five Americans could possibly die from that," statistics become a major driver in what becomes an industry eventually. Those statistics about, A, on the one hand, fear, the spread of cancer and how many people are dying; and on the second hand, it becomes eventually the tool.
- 1.95 Remember when I was saying that Halsted, the surgeon, half his patients die in 3 years after he's taken this tumor out through a radical mastectomy? And then he says, "Well, but half of them have survived," so now you look for standards to be able to say, "We have success in this effort against cancer." So what do we say? People have survived 2 years, they've survived 3 years. They will see how the standard became 5 years. And then once that becomes the standard, you're able to cite that to show progress. And then you say, "We have progress on cancer. The only thing stopping us from curing it is more money. So we need more money to be able to cure it." And the federal government comes in time and time again on that.
- 1.96 So, to recap, we moved from what I call ancient fear, misguided surgery, the radiation and radium economy, and the first federal role, the early quacks as gatekeepers. One of the things that surprised me, may surprise nobody else, is that when Trisha got diagnosed, coming on 5 years ago, we were looking at options. The options there are still surgery, radical mastectomy or lumpectomy, which develops later, and radiation.
- 1.97 There are some pills that, for the long term, on a hormonal-based cancer, will help to reduce the chances of recurrence. The pills are new, but the surgery is still back to the days of 1898. It's been refined a little bit. The anesthesia is better, but it's the same. The radiation has been refined. The machines are better. The technician has gone on the other side of big, thick glass walls and it's delivered. And you're taught as a patient

beforehand how to breathe in a way that if your cancer is on the left side—of breast cancer—the radiation will not harm your heart if you're breathing and the heart is at a different location as they're giving the rays.

- 1.98** But we were sort of back to what one doctor said to me. "We're still back to slash-and-burn." I thought that was very interesting when they said that to me, because that's stayed the same. It's not like I can say to you—by the way, we'll see later—there are some theories of what's happening in terms of biologics that are pushing the envelope and may change the future of cancer care in later lectures. But for the bulk of people who end up with cancer, the options are very similar to what they were nearly 100 years ago.
- 1.99** Core patterns to watch for. Aggressive treatment as the default, even sometimes if the evidence is weak. That was certainly the case with Halsted, but what he got kudos for was aggressive treatment. The thought was—and you see this in some of the contemporaneous doctors who wrote about his technique—they said, "You know what? He is very aggressive with that surgery, but cancer is aggressive. So you've got to meet it head-on." And so, being aggressive with your technique gets a reward here.
- 1.100** Money and prestige clustered around specific technologies. That clearly happens with Memorial Hospital. We'll see later how it becomes Memorial Sloan Kettering. And institutions say, "We're protecting patients," but at the same time, they're also protecting their turfs. So the surgeons are arguing against radiation because they think surgery is the better option. Not necessarily because it's a better option, but because the radiation is a competitor.
- 1.101** Cancer is an industrial product, as I call it. By the mid-20th century, it's no longer just a disease. It has increasingly become what I call an industrial product. It's the synthesis of pesticides, petrochemicals, possible causes for cancer, carcinogenics in the environment, in the workplace, that might be causing a spiraling cancer rate after World War II, but nobody's made the connection yet.
- 1.102** And preventable. Well, I'll tell you, from the very start, the federal budgets—it makes sense—and all the resources were put in, were really to deliver treatments. So it took a long time for the federal government to think that it might be able to provide enough money to help treat cancer, but it was a [bridge] too far. They could not get their head around the idea that maybe they could also look at possible causes of cancer that might reduce those rates as well before any money gets there.
- 1.103** And this is where you really come into, I think, one of the core issues that takes place in what I call the politics of cancer. A case of cancer prevented, because you have somehow been able to avoid somebody's exposure to environmental or workplace carcinogenics, is a statistic that nobody knows about. Just that that person didn't develop a cancer they might otherwise have had. You can't boast about it. You can't

say, "By the way, we kept 432,000 people free of cancer because we eliminated exposure to asbestos." You don't know what that number is.

- 1.104 So prevention isn't something that you can list in an annual report that the American Cancer Society can put forward. But if you're talking about treatment, you can talk about the statistics we had before—longer life span, you're living longer with treatment, the number of people who are able to look back and say, "Now I'm a cancer survivor." And so, those are statistics; you can say those are improving. We need more money to be able to fund that. But your focus is on treatment, not on prevention.
- 1.105 The next lecture, if you think that we're still back in the dark ages, well, we're out of the dark ages a little bit. We're going from mustard gas to chemotherapy. You think, "Oh, no, he's just saying that. Chemotherapy had nothing to do with mustard gas." You might be surprised, and I look forward to telling you about that. Thank you very much.

Lecture 2: Origins of Chemotherapy

At a glance — The origins of chemotherapy in chemical-warfare research.

- 2.1 It's like one of my favorite stories, how chemical warfare research led into this breakthrough with some very powerful personalities into a cancer treatment that becomes a mainstay for cancer going forward.
- 2.2 I always say—we talked about this in another course about the pharmaceutical industry—the pharmaceutical industry will brag about its current achievements. And if you look at a 10-K or annual report for one of the public companies like Merck or Pfizer or whatever else, they say, "Our Story." If you go on their website, it says, "Our Story." And they start with their story: World War II and the penicillin discovery. And it's a great story. They don't go back to when the companies were founded in the middle of the Civil War, and they were basically just peddling morphine. So the story is in the past, and then they get to be able to create it in a different way.
- 2.3 And the same thing here, very, very, very few patients, if any, have any reason to know when they're getting chemotherapy, the origins of chemotherapy, and the hospitals don't necessarily tell you about it. But this was the start, what I call the pre-official press-relations story of chemo.
- 2.4 And that's not to say it's not effective when it's used for its purpose, but this is where it gets its start from. So you know how we got here, right? The radical surgery of William Halsted, the first solution, the X-rays, the federal stirrings, the National Cancer Act. This pivot from cutting out tumors or blasting it with radiation, the idea that you might be able to poison the cancer is the next solution. We were talking before about slash, meaning surgery; burn, meaning radiation, and now you're at the point of maybe we could poison it.
- 2.5 Something, by the way, treatment-focused approach. Again, this is all about treating it. This is how you'll be able to stop it. It's something that you can go back to lawmakers and those who are funding it.
- 2.6 Something Trisha said to me on the break before—those of you who have been with me before in a lecture know that I sometimes go back in a new lecture and then I say, "Oh, by the way, I meant to tell you on the previous lecture also this." She reminded me of one of the most fundamental aspects of the entire lecture series, which is that not all tumors are cancerous.
- 2.7 Of course, which makes it so much more difficult in the early days for doctors to figure out because if they saw a tumor, if they were doing a surgery for somebody who had a gallbladder being removed and they saw what they thought was a growth inside, they would automatically say, "That tumor has to be excised." No equivalent then of a

biopsy, no looking for what would be cancer cells. So tumors were automatically viewed as something that had to be surgically excised.

- 2.8** I guarantee you, can I prove it to you? No, because you can't go back and look at the examples of those tumors. But in many of the most radical surgeries that were done back in the late 1800s and early 1900s, they were removing tumors without any cancer, but they wouldn't know that. They thought the tumor itself was a sign that it was a deadly disease.
- 2.9** Now, how did we shift from cutting tumors and blasting them with radiation to get over here to being able to poison them? It's a story that's interesting. So it's the wartime context, first of all. It had only been 20 years when this gets going from the start of World War I, when mustard gas had literally killed and maimed thousands of soldiers in World War I.
- 2.10** So mustard gas, invented by the very same person that had come up with some of the medical cures, a German doctor who went on to do mustard gas. He had also come up with some pharmaceutical cures that had been good and discovered them over time. He was the one responsible for discovering mustard gas, and it's used widely in World War I.
- 2.11** There's a wartime urgency. World War II gets going. We know the Penicillin Project was big for the federal government. We want to get penicillin done. They're funding all the research that the Oxford researchers have brought over to get penicillin on the battlefield to save soldiers from dying from infections. But at the same time, they're looking into whether they might be able to accelerate what's happening inside their own chemical weapons programs.
- 2.12** When I say a lack of preventative focus, boy, this really is a time in which there's little effort toward preventing exposure to the carcinogens. Again, you're looking at developing drugs. So the same properties that make a chemical agent—believe it or not—so good for killing soldiers, also can attack rapidly dividing cells and make it, therefore, a candidate to administer to a patient that has cancer cells proliferating.
- 2.13** If that nerve agent, that gas, that chemical agent, can stop cells from replicating, maybe it could also be used as a cancer treatment is the idea behind this. In 1943, there's an American [cargo ship] that's mustard gas laden. It's filled with mustard gas. It has an explosion on board. Some of the sailors die, and some survive and are treated. In the treatment of the survivors and in those who had died in autopsies that were taken, the scientists discover that their white blood cell counts, especially on the survivors, were greatly suppressed across the board.
- 2.14** The exposure to the mustard gas when they lived meant their white blood cell counts were very, very low, opening them up, possibly for infection. But some researchers said, "You know what? That means we might be able to use it on treating leukemia, a blood cancer. Maybe if it's going after those white blood cell counts, it could go after

the cancer cells and leukemia." It unintentionally pushes forward the idea that you could use some derivative of this.

- 2.15** The nitrogen mustard gas experiment. It takes Alfred Gilman Sr. He is a [pharmacologist] at Yale, and he's part of a secret deal the US government has. So the US government is doing these programs to develop the atomic weapon, to try to do a penicillin, which was at that time still a secret project because we didn't want the Germans or the Japanese doing ahead of us. And they're also looking at this, what I call chemical weapons program. So he has a deal, Yale does with the United States.
- 2.16** And he had unintentionally discovered this liquid while he's doing his experiments that he called nitrogen mustard, a variation of mustard gas. He substituted one nitrogen atom for a sulfur one. Now, we know from the drug industry, sometimes it would move out one molecule and be able to do a me-too drug of somebody else's successful antibiotic. So in this case, he's not changing it to be able to patent a drug and make money off of it. He's trying to come up with the derivative of what mustard gas is. Then he takes that product—this, as he calls it, nitrogen mustard—and he starts experimenting with it by injecting it into mice in his lab. Only 20 years since we know the dangers of mustard gas, but he's working with a new variation of it.
- 2.17** And what happens? Inside the lab, he sees tumors shrink, and one mouse lives much longer than expected. That is what becomes the news out of that particular group of experiments. We had one mouse live much longer than the control group after being injected with this, and they had tumors. So now doctors in other areas are wondering, could it be applied to people? And that discovery is the first hint that a chemical poison could temporarily suppress cancer. First time you get it in a lab, it's the beginning of what I call that initial baby step forward.
- 2.18** In comes Cornelius Rhoads. Boy, he's a power in chemotherapy and cancer development. He's a Harvard-trained physician, worked for the Rockefeller Foundation. He is chief of the US Chemical Warfare Service Medical Division during World War II. So he's correctly placed in an area of the US government that's doing these types of studies. He coordinates these toxicity studies using mustard-type agents like had been done and developed at Yale on soldiers, volunteers, and others. When I say it blurs the line between medical research and human experimentation, you're right at the edge on this.
- 2.19** He's doing it all, by the way, [before] this period in Puerto Rico, not in the United States. Most of the volunteers, except for some of the soldiers, are Puerto Ricans. As we talked the other day in a course about the pharmaceutical industry, they often do their testing and clinical studies in foreign countries because they don't want to end up in America with a spotlight on it all the time.
- 2.20** And after the war, by the way, during those, what I call treatments that he was doing inside of Puerto Rico ends up that 13 people die at one point over a period of time. So

they are pushing the limit to find out what will work and what won't work. And he's very aggressive in that.

- 2.21** After the war, what does he become? He leaves the Chemical Warfare Service division inside the US government to become the director of Memorial Hospital, which is a perfect fit for him. And he helps transform. I don't mean just helping by a little bit. He's a moving force in making it into Memorial Sloan Kettering, a chemo research center. Interesting that Memorial had been the place to survive the Depression based upon radium treatments, and now it's going to be the first to move into chemotherapy.
- 2.22** He really is what I call this tireless salesman, telling all the time that he's ready to move on to the new frontier. He's convinced—I will say this. It's one thing if you're a charlatan and you don't believe in what you're pushing and you just go back at home at night and say, "By the way, I wonder if I can convince people chemotherapy or this idea that we're developing will be good. I don't know. I don't care. We'll just make a lot of money."
- 2.23** That wasn't Rhoads. Rhoads was a true believer. He really thought he had come to believe from his own work that chemotherapy would be the cure for cancer without realizing yet how difficult a medication it would be. And I think that he sees it not just as an anti-cancer tool, but that might be used in other diseases as well. He becomes an evangelist for it. And part of his success is the creation of this industrial enterprise that takes place at Memorial and then spreads out. So what I call the rise of chemotherapy: promises and pitfalls. And there's certainly we're both along the way. So we have December '42.
- 2.24** Now, middle of the war, everyone wants to find something. We go beyond the mice experiment at Yale, which had had the first remission in mice, and we start doing human trials. And first patient who is really, really sick, near death with lymphoma, gets nitrogen mustard, the same chemical product that had been given to the mouse in the Yale experiments. Now we're going to try it on a human. What happens? Good for science and good for those who believe in it, a remarkable remission. So goes from death's door to really feeling strong. And then, as happened on the first test trial for penicillin, years earlier in the [1940s], it returns. This time, it's only a few weeks; it returns and kills him in less than a month. And they're not quite sure why.
- 2.25** So they don't know at this stage, and think about this, if you're one of these cutting-edge researchers, you've given this nitrogen mustard to a deathly ill lymphoma patient, that patient suddenly starts to come back to life and seems as though they're well, and then they fall ill again, and then they die. Did we not give enough medication? Did we give too little? Did we do the process wrong? Should it have been given over a different type of schedule? Was his particular form of cancer different than the one that could be treated, that maybe somebody goes into remission and comes back? So all this does is whet the appetite for the researchers to say, "We have something promising here."

- 2.26** Subsequent trials. Yale and other schools, prominent schools with contracts with the government, treat more than 160 very ill cancer patients with versions of nitrogen mustard. And the trial results, one after the other after the other: temporary improvements. I go through a list. A few don't respond at all, but that's literally only a few. Most of them show signs of what we will then later call remission, but feeling better, looking better, and no cures but toxic side effects.
- 2.27** You're starting to see some of those side effects. And this is early days, so think about this. Today, chemo is still very powerful, but it's been through a series of changes that have made it less toxic than the original nitrogen mustard. So they're seeing now and recording the side effects: all the patients die, and not five years later or three years later or two years later, within months. So they treat them, they see the remission. They have 160 cases now, but all different types of cancer, and that remarkable remission doesn't stay. That's the problem.
- 2.28** And the reason, by the way, they're choosing nitrogen mustard as the chemical agent to give is that's the agent with which they are most familiar. So they don't have others to be able to pull off the shelf and say, "Let's try that." They feel comfortable with humans receiving only this variation of mustard gas. That's how they might be able to do something.
- 2.29** Lesson learned. It really does, for me, this period, introduce the basic trade-off with chemo that exists even today, which is, it kills cancer cells. The trade-off is damage to other tissue, including—they didn't know this—bone marrow at the time. By the way, as a sidelight, it's patients with Hodgkin's disease had responded the best for some reason to the nitrogen mustard, and they just couldn't figure out why. So they put that down as a footnote to look at later. But this is the beginning of what I call this trade-off.
- 2.30** Rhoads the evangelist. So his experience in Puerto Rico, resulting in what were suspicious deaths, at the very least, it showcases to me this problem in oncology going forward, and that is the prototype he is of the confident, well-connected doctor, never brokers any doubt about what he's doing because he's convinced it's the right thing.
- 2.31** Like Halsted was convinced that the radical mastectomy would cure cancer, Rhoads becomes the evangelist for what he thinks will be the change in the cancer industry forever. So Rhoads is an early example of what I call the establishment, and I don't mean just the cancer establishment. This is the same in the drug industry, it's the same if you're dealing with the defense industry or other industries, the tech industry today. It's the aggressive institution-building personalities who make their mark. They're effective at attracting prestige. They're effective at attracting money because donors who have been affected by cancer are looking for somebody they think might be at the forefront of being able to cure it. He really becomes a salesman for it for the next cancer treatment.

- 2.32** He builds a permanent chemo research center at Memorial and recruits these industry partners who might be part of the chemo machine eventually, but also are people who will be able to use it in other clinics and hospitals around the country. So he has a combination of government money and elite bankers at the same time. And this is this marriage that I call the first chemo lab is the template going forward for hospitals, industry, federal funds, and philanthropy.
- 2.33** And you see it repeated time and time again down the road when it comes to cancer care. This is early days. This is the National Cancer Institute around 1950. Dr. Jonathan Hartwell and his assistant, Sylvvy Kornberg, these are some of the earliest studies being done on chemotherapy at the NCI. The National Cancer Institute was in this pretty early. We're only talking here five years after the end of the war. The government is putting chemotherapy on its top-to-look-at list. It's made that, it's got funding, and it's inside the NCI.
- 2.34** Multicenter experiment. So now you have Memorial doing it, but you also have University of Chicago. You have other hospitals in New York. They're using mustard gas derivatives to see if they can push along some of the results from Yale. In some cases, because these are experiments, they're not quite sure what works, they would actually do an external application to breast tumors. They would think that maybe if they, not giving it intravenously, if they made this nitrogen mustard into a salve and they applied it to the tumor, that would do it. They were doing that, and the results were not good at all. Small temporary gains overall from the application of the nitrogen mustards, but the effects are small and short-lived.
- 2.35** Severe toxicity. I will tell you that if you go through those early reports of the 160 people who were treated who had the temporary remission. Do you know how a pharmaceutical company will run an advertisement? At the end of the advertisement on TV, it has to tell you about side effects, and it tells you about 36 different side effects, and the last one is sometimes death?
- 2.36** Well, that would be the same here in the 160 cases of side effects from the application of chemo. So much so that you would ask yourself, is it really going to be what Rhoads thinks it is, this miracle drug that will change the nature of cancer? Short survival benefit for the patients from a few weeks to several months, and that's it.
- 2.37** And stacking drugs to boost the response rates. It's very interesting. The National Cancer Institute, the federally funded institute, says, "These experiments aren't very good. Maybe we could actually combine different versions of chemo drugs." We stacked them the way that the drug industry, years earlier, had taken antibiotics and had done fixed-dose combinations, saying, if one antibiotic is good, if streptomycin is good, or aramycin or penicillin, why don't we put them together?
- 2.38** More must be better, which it wasn't. All it caused was more side effects and didn't cause any more cures. Here they're looking for what they call stacking these chemo

agents in the hopes of boosting the response rate. But often what's added, and you will not be surprised, is no change in how long the patients who receive it live, but certainly additional side effects, and that becomes part of the problem.

- 2.39** This is what I call the birth of an industrial approach. This shift from knives, rays to chemicals, it guarantees you that the industry, and I don't mean industry in a disparaging way. I say cancer industry, people think, "Oh, you're disparaging cancer treatment." No. Trisha is alive because of the treatment. I'll just give you an example of that later. I understand that it can save lives, and it does save lives, but it's an industry at the same time. It's a machine. I mean it like that. This idea of prevention gets pushed aside. Drugs are patentable.
- 2.40** I forgot about that. You can't process and patent drugs. You can't patent the surgery. Anybody can do the technique. There were some patents on the radiation machines, the particular ones, the X-ray machines that were delivering radiation. But for the most part, anybody could come up with a new machine. But if you had a particular chemo, a form of chemo as a company, that you said, "We've combined it this way and created this way in the lab, you could put a patent on it." That was something that was billable and politically easier to sell than confronting the carcinogen.
- 2.41** So Rhoads's vision is a permanent chemo research machine. He thinks that the problems in those 160 cases when they first apply it and people are really getting sick and they're living a little bit longer, if you were able to travel back in time and ask Rhoads about that, he would just say, "That's because we haven't developed the right product yet. This is the right base. We just have to have the research to develop the right product."
- 2.42** Integrating hospital, philanthropy, industry, and federal funds. That's the magic mix. It's the Heinz 57 ketchup or Coca-Cola's secret formula. That's the not-so-secret formula. It's a success in this. Lays the foundation for the modern cancer treatment industry. And into all of this comes a fellow, Sidney Farber. Now, many of you have heard of the Farber Cancer Institute. It's famous. People don't know it's necessarily named after this pioneering pathologist.
- 2.43** So he is a pathologist at Boston Children's Hospital, and he's known as "the doctor of the dead." He's not doing research. He's not a cancer researcher. He's not somebody in a lab doing it. He's doing autopsies.
- 2.44** And he is particularly angry, frustrated, upset, talks about it, talks to others about it, over the fact that especially with children, when he did the autopsies on children who died from leukemia, that they were dead and there was nothing that could be done to solve it. So he decides on his own that "maybe I can go ahead and come up with"—he knows about the work being done on chemo—"I can come up with a chemical treatment for leukemia that could save these children's lives."

- 2.45 Now, why does leukemia matter in science? Before I said to you that there had been an explosion on that destroyer that was laden with mustard gas, and in treating the survivors, doctors noticed that the white blood cell counts had dropped, and this is why it's important. Blood cancers, like leukemia, are visible under a microscope. The cancers can't. You have a tumor, a breast tumor or something else.
- 2.46 You look at it under a microscope, you can't tell. That's why a biopsy is more invasive. They need to find out. Whereas blood cancer cells look different under a microscope. So solid tumors, hard to measure without imaging, and there's very little imaging at this time or surgery, the blood cancer is directly observed and monitored. You can see them growing over a period of time. So leukemia cells can be traced. You think this is important in terms of medicine in real time?
- 2.47 He found that if he applied certain chemical treatments, as these blood cells were proliferating, it would allow him to adjust dosages. He could see if they were receding, if they appeared to be under attack, if it was changing as these chemical agents were being put in. These blood cancers provide the experiment, the testing ground for Farber to be able to move forward. Others are not focusing just on leukemia.
- 2.48 Now, the odd thing is it's a pathologist, somebody who's dealing mostly with dead patients. Those are his patients who's decided to do this and look into this. Others aren't yet pursuing it. It becomes in some way the ideal testing ground for Farber to experiment with chemicals, leukemia, and it has an advantage the others who are dealing with solid tumors don't have.
- 2.49 Now, I call this the pivot to antifolates. Farber's idea, if he thinks there's a folic acid that he applies, it helps the blood. Maybe he thinks if it boosts the blood, gives it more red blood cell counts than that, maybe it will help with leukemia, make the blood stronger so it could fight the cancer cells inside the blood. So he thinks if it can make somebody who's anemic, not anemic, maybe can normalize the blood itself.
- 2.50 So he decides that's going to be one of the first things he ends up giving. Folic acid, this cancer trial. It turns out that when you give folic acid to somebody who has leukemia, including the children, it accelerates the rate of the cancer cells. You don't know that. It's one of those things that logically doesn't follow. You don't know why it's an antagonist as opposed to being something that turns out to be effective, but it is.
- 2.51 It turns out that when he gives it to those children, a synthetic form of folic, the leukemia progresses faster and they die quicker. Now, Farber takes that pretty hard. He's not just a callous person who thinks on the side. He's doing this in part because he's tired of doing autopsies on children who died of leukemia. So he's trying to come up actually with a fix. And the first thing that he does makes it worse, not the best thing.
- 2.52 So now he says, "Okay, if folic acid is no good, maybe I should move ahead to antifolates." Something that fights the folic normally would make you weaker if you

had anemia. Maybe that will block the folic acid that's causing the cancer growth. So it makes sense Farber decides to try a new antifolate. He has these created. They send them into his lab, and that's the pivot that takes place. It really is that moment where he has the insight into blocking the fast-dividing cancer cells that could suppress leukemia. It's the first of what I call a successful experiment coming up on remission for leukemia, and that's from Sidney Farber.

- 2.53** Now, the first leukemia remission. In the story of antibiotics and drugs, there's a moment in which a British policeman who cut himself on a thorn while pruning roses and gets it infected and loses an eye and he's infected, he's near death's door, saved by penicillin, which is given to him as an experiment.
- 2.54** Those moments in which a patient who's really sick turns around based upon a new treatment, they're legendary in medicine. And this is one of those moments. He saw this child that he's treating, the leukemic cells, the cancer cells plateau, and then they start to drop. They did return months later. So his view of it at that point is "chemo is gaining me time, but it's not necessarily a cure."
- 2.55** That remission that he develops really is a term, becomes the key term in cancer care. Is your cancer in remission? People think if you've never had cancer, you've never dealt with it in the family or anything else, you hear that somebody's in remission. That's good news. You like the word "remission" with a cancer patient. Bad news is remission may or may not be forever. You just don't know. So cancers can return, and that's always the difficulty of this particular disease.
- 2.56** So he publishes the results in the New England Journal of Medicine. It's called Temporary Remissions in Acute Leukemia in Children Produced by Folic Acid Antagonists. And it's the first time that that is put out now.
- 2.57** Colleagues, what about the rest of the medical professionals? What about the Rhoadses of the world and everybody else? They see this. This is in the New England Journal of Medicine. And they say, "He's not an oncologist? We've never heard of this guy, Farber. Got to be kidding. He's an autopsist? He's a pathologist? He could be making some breakthrough that we haven't already seen here in our very expensive research labs." So there's an initial skepticism about it because he's not part of the club. He doesn't have the right credentials when he's coming forward. And they think maybe he's come up with some fluke.
- 2.58** Quietly, however, behind the scenes, although they won't congratulate him in person and nobody is going over to give him a medal, they do start to work with some of the same material, and they start to think about the same things.
- 2.59** This is what I call, after Sidney Farber has done this, I call it the golden age of chemotherapy. Now, it may be hard to think of chemotherapy having a golden age, but the golden age in these terms, this is when doctors really believe that it can make a

difference. This is also the time when the five-year survival rate becomes key as a metric.

- 2.60** An earlier diagnosis, you have a few years of remission combined with a curve that mortality may be unchanged. You may still end up dying at the same time you would have died before, but you've had cancer remission for a while. You've been well, and it may come back. But if you survive five years, that's considered the magic; you're a cancer survivor. It's not industry-wide yet, but it's becoming one of the key metrics later for the so-called war on cancer.
- 2.61** Does desperation shape ethical judgments? Well, I'll tell you, if you're a patient in 1948, and this is when Farber is doing it, and you had a child who had leukemia, would you allow somebody like Sidney Farber to use an experimental drug that you didn't know anything about that might buy a few months of time with unknown long-term effects?
- 2.62** I ask parents that. Ninety-nine out of 100 say, "Yes, of course." If I think that my child is going to die of leukemia, it's a death sentence, and now somebody's trying an experimental drug. Even if you say to me, it might only buy them a few months, maybe that drug's better than expected and buys them years. You don't know. It'll have toxic side effects, more toxic than death. So there's a tendency for parents to line up to be able to help Sidney Farber work on their children.
- 2.63** This is also a time, by the way, very interesting, that after the war, at Nuremberg, at the big trials of Nazi war criminals, concentration camp victims, Bayer and other companies, IG Farben, had used people to test drugs on, and without their consent, of course, because they had sent them to these concentration camps enormous work camps. So the whole idea of informed consent becomes very big in American medicine. This was before informed consent was yet set so that Farber was able to do these experiments right at the cusp of a time when medicine was changing, and it might have been more difficult to have done those experiments on children.
- 2.64** The accidental DNA insight. I love accidental discoveries in medicine. It turns out, Farber knows what's happening. He can see that the antifolate, the chemical that he is giving, is causing the cancer cells in a child to plateau and then drop. He doesn't know why. He doesn't know that it's actually blocking a chemical step that the cells need to copy their DNA.
- 2.65** So cancer cells. One cell inside the body goes off-kilter and becomes essentially a cell that keeps replicating uncontrollably. Then as it's replicating uncontrollably, it eventually contributes to the formation of a mass or blood cancers. It's a cell that has stopped replicating itself in the normal way. How do you stop that? In this case, the uncontrolled replication was stopped because the cells were missing a DNA element that they needed to continue to do that, and it's a result of the antifolate.
- 2.66** Now, fast-dividing cells, that means bone marrow, gut, essentially, and those become the fundamental logic of chemo because they're the ones that are fast-growing, they're

easiest to see. You can see them underneath this microscope at the same time. And by targeting the fast-dividing cells, it becomes the fundamental logic of chemotherapy, which is to target the cancer cells and try to avoid all the healthy organs and tissues around it.

- 2.67** Now, into all of this mix comes Mary Lasker, who many people may never have heard from, but she turns out to be a dynamic force inside the emergency cancer industry. She's smart. Her husband is a very wealthy advertising executive who helps to sell cigarettes to America, by the way, but that's all right. One of the things that made so many cancer patients. Her family's cook, so to say that she's wealthy, ended up having ovarian cancer and died.
- 2.68** Mary Lasker couldn't believe that there wasn't anybody that was able to treat her effectively. Ovarian cancer, a silent cancer, we say, because you sometimes don't know you have it until very late in the process. So it wasn't as though she was dealing with something that's even unusual today, decades later. But at that time, she was frustrated by that.
- 2.69** So she went over to see the American Cancer Society. She wanted to visit it and see what they were doing about the types of cancer that had killed her employee. Then it was called the American Society for [the Control of Cancer]. It becomes the American Cancer Society in 1943. She goes over there—I mean, '45, excuse me. She goes over in 1943 in the middle of the war.
- 2.70** She visits her and she said, "What are you doing for research?" They say, "Not much. We don't have that much money." "What do you mean? So what are you doing?" "We're just raising awareness. We're just trying to make people aware of cancer being very, very dangerous and make sure politicians are aware of it so they continue to fund the NCI."
- 2.71** And she was horrified at that. She said, "You're kidding." Because her husband had been in the ad business. So this is a quote from her at the time. She later writes about it. She said, "I became infuriated when I read there was no single place which had as much as \$500,000 for cancer research." You can tell she comes from a wealthy background. She's talking about \$500,000 in 1943. She just can't believe they don't have \$500,000 for cancer research. "This wouldn't even be a suitable sum for an advertising campaign for a toothpaste."
- 2.72** She goes back to her husband and she said, "You've got to be kidding. They don't have any fundraising at all. They're not doing any research. They can't launch a campaign." He said, "Why don't we do something about that?" She then wants to go on the radio and he gets her a local radio spot so she can talk about this and try to raise some money. It turns out, she's told by the radio show host and producers before she goes on, "By the way, you can't mention the word 'cancer' while we're talking about this."

You just have to talk the big C. It's something you can't say—the word 'cancer.' We don't talk about it on our radio station." That made her even more furious.

- 2.73** Now she gets her husband together. They enlist the help of other philanthropists and wealthy friends. They get influential officials. Her husband ends up dying of cancer in 1952. You think that would take the wind out of her sails? Not at all. It redoubles for her the idea of what she's going to do. So Mary Lasker becomes this force. And you'll see later on, time and time again, she's convinced that only the federal government—she can provide money. Wealthy philanthropists can send in money.
- 2.74** But you know what? You can't fight a disease on a national level and make real breakthroughs with private money. They had to help. No question about it. Can have a new hospital waiting, can have a cancer center, whatever else. But you need federal funding. That idea that Neely, the West Virginia senator, had. So she's really hammering away at the feds. You can do it state by state. So maybe California ends up doing more work than Florida does or Florida does more work than Tennessee. That's not going to matter. Only for the people living in those states. So do it at a federal level.
- 2.75** She becomes what I call a lobbyist in pearls and gloves, and that's her approach. So she's a little—she has success in part because she's talking about a serious disease, but people often underestimate her. She's a dynamo, but they're expecting somebody to come in in this suit and tie, and she comes in looking like she's ready for a society luncheon, and then she's hammering away on the facts.
- 2.76** She connects scientists and legislators, and she champions Sidney Farber as the face of cancer research. He actually is a great public speaker. It turns out who would know that a pathologist would be so lively in front of a group? Sidney Farber was. He was missing one of his callings. She discovered that and she said, "You're fantastic. You're going around with me on this national show, city by city, to make sure that people are aware of what's happening."
- 2.77** Now, the broken ACS. She visits the ACS, the American Cancer Society, finds it. Not only, by the way, weren't they spending any money on research. I forgot to tell you that she also found it totally disorganized in a complete mess. She couldn't believe the way that they operated. Nobody in there was worried about where the files were and what they were done. She really transformed. She doesn't—
- 2.78** One of the interesting things here, there are some people who have to have, if they're doing this type of work and they have the money that they're contributing and they're putting in their time. They want their name on it. You see it all the time. Cancer centers around the country named after somebody who's given \$50 million, \$60 million, \$70 million to a hospital. Fantastic. I don't mean to denigrate that. Somebody who wants their name on it, and they've given that money, that's great.
- 2.79** But she didn't want that. She didn't want the Lasker Foundation or that. She worked through the American Cancer Society, which is interesting. She decided to use them as

the tool. So unless you know the story of Mary Lasker, you don't know that there, which is really rather astonishing given a day and age in which people like to see as the Sackler family did with OxyContin and their name in granite on a lot of buildings.

- 2.80** Critics at this time were saying, "The American Cancer Society is based too much on treatment and not enough on prevention." True. But let me tell you, it's not a criticism I take to heart because in the politics of cancer, as I call it, you had to get them first going on one track or another, and they were doing nothing. She at least gets them focused on research for treatment. Better than nothing; you can't take it all on at once. Maybe decades later, we can criticize them for not focusing more on prevention.
- 2.81** She has a fundraising revolution. She recruits top ad executives, people, even after her husband died, who knew her, and she taps into them in business. Especially, she's smart. Some of you may remember from the lecture series on Big Pharma that there had been a patent drug that was basically a third opium, and it was being sold to new mothers to let their children sleep at night: Dr. Kopp's [Baby's Friend] Soother. The company would look through the local newspapers to see the birth announcements. Then when they found that somebody had a new baby, they would send them a letter saying, "By the way, we have this miracle drug that will allow your child to sleep at night," and it turned out to be this.
- 2.82** She went not just to regular philanthropists, people with money, but she went to people who she knew had been affected by cancer. They lost somebody to cancer. A husband or a wife died or a child died of cancer, and they were very successful in their careers. So they were her target audience initially, and she transformed the way the American Cancer Society did fundraising. My God, she's teaching them. She's giving them a course, 101 in fundraising on a disease. You go back to the wealthy families, if you want to get some private donations that have been affected by the disease. And that turned out to be a major source of revenue.
- 2.83** She pays for and these pitches that I call in mass media, from newspapers to national magazines, really making—and I can go through a list of these at some point—these very effective appeals for cancer awareness and reaching a much wider audience—the audience, the general reading public, at that time, hard to imagine people actually read newspapers and magazines. It's true. And so they did, and they would open them up, and they would see these appeals about cancer, and that worked very effectively. Today, she would be flooding TikTok and Instagram, I'm sure.
- 2.84** And most importantly, because she believes this, she doesn't think she's just sounding the alarm about cancer and that nothing's going to be done on it. She views it as others who got involved, like Rhoads, as a curable, solvable problem. We're facing it. We just haven't put enough money into it, and we can change it from a death sentence to a challenge if we get enough money to research and development. So that's one of the things that drives her. And I think that that becomes absolutely critical as you go along here. It's her strategic fundraising efforts. I will tell you that if you look at the increase

in the funding that comes into the American Cancer Society starting in the early '50s, it skyrockets. It's truly a surge, all as a result of what she's working behind the scenes.

- 2.85** Now, Lasker goes around and they go to different medical conferences. They have public hearings. They go up to Washington to ask for more money. She's the one who calls Farber the father of chemotherapy. She's telling everybody it's an odd name, but he wore it with some pride. I'm sure that Cornelius Rhoads at Memorial was upset that he wasn't the father of chemotherapy. But Farber, the pathologist, not the oncologist, is now the father of chemotherapy in terms of the public face of cancer.
- 2.86** The great part is marketing, marketing, very important. Mary Lasker, trying to raise funding and make cancer something that people feel they could have a stake in that's solvable, they talk about it in terms of penicillin. So chemotherapy could be like penicillin. Fantastic. Penicillin, one of the most important discoveries in human history. Stop simple infections that used to kill people beforehand. We know all the side effects afterwards about overexposure and the development of super [bugs], but on its own, penicillin—a major discovery.
- 2.87** So if you think of chemotherapy, you haven't yet had cancer, you haven't been through it, you don't know somebody. Not a lot of Americans knew yet somebody who had been through cancer and chemotherapy. You thought to yourself, "Oh, my goodness, they're working on a penicillin for cancer. That's fantastic. Where did I send my \$5?" That was very, very effective. The public narrative, I think, shaped in many ways this enthusiasm for the funding of chemotherapy. She was focused on it. As a result, if Mary Lasker had been focused on radiation or on surgery as the solution, maybe that would have been the driver, but she wasn't. She was focused very much on Sidney Farber and chemotherapy.
- 2.88** What happens? A couple of charismatic doctors in the strategy of the future, how likely is it that you're going to get dissenting views from politicians, especially? There's not a politician who's standing up who's saying—and there shouldn't be—"I'm against this. I don't want to give any more money to it because I just don't think it's worthwhile. You're talking against patients, you're talking against voters, you're talking against the advance of medicine as something white."
- 2.89** So when I say she matched science and politics, she becomes a lobbyist, the connector, the champion behind Farber, promotes him as the face and then leverages her high-powered impact. There is a medal that you can get named after her. That's the only thing. She did not ask for it. But Mary Lasker really turns the American Cancer Society into a fundraising and PR powerhouse.
- 2.90** Now, James Shannon, somebody you may not have heard of before, he was researching malaria in World War II, and he becomes director of the National Institutes of Health, which start to feel some power after World War II. He shared Mary Lasker's belief that government-funded biomedical research was the way to go for the future.

He helped expand the NIH's budget multiple times inside the bureaucracy. He really sets up the basis for the war on cancer when it starts to come up in the '70s after Richard Nixon, and we'll talk about that. But James Shannon becomes this key figure at this point.

- 2.91** So the proto-empire emerges. It's not a term of art. It's just how I view this part. Tripling of National Cancer Institute funding. So you have Lasker, Shannon, Farber. In the 1950s, the National Cancer Institute, which had been started, triples its budget. Pretty good at a time when not all budgets were being expanded nearly as fast as that. That's a sign of success inside the bureaucracy.
- 2.92** Growing competition for chemo dollars. A lot of places are doing research now on chemotherapy, and there's competition in the labs and in the hospitals to get the grants from the government to do the chemo development. So now you have a whole bunch of units chasing in the 1950s, at a time when the drug companies, the pharmaceutical companies, were in their golden age of selling antibiotics, the cancer industry is looking and vying for federal knowledge to develop new chemo agents. That's where the focus is.
- 2.93** A culture of celebrating toxic drugs. Now, the extensions of life, the remissions that took place in the earliest stages of chemo through the '50s, were almost always short-lived. For the children who had leukemia would often come back. They still weren't able to have long-term remissions. They were looking at five years as the outside goal.
- 2.94** But in terms of medical conferences, the New England Journal of Medicine, in terms of medical journals, every small extension of a remission was viewed as a triumph. The same way that the drug industry came in and would talk about miracle drugs, the development of something that would change your life and would have no side effects. They didn't say no side effects here, but they'd say, "We came up with a chemo agent that is going to change and revolutionize the cancer industry."
- 2.95** Can't tell you how many times in the 1950s, there was talk about that celebrating the revolutionary change in chemo, and they were incremental steps. There was no major change here. There's nothing that changed in the '50s in terms of all this work.
- 2.96** But it did do one thing, and that's it, institutionalized chemotherapy. It became a centerpiece of the federal funding, and questioning that model made you an outlier. If you said, "Gee, are you sure poisoning the cancer cells is the way to go?" They would say, "What do you want? You want people to die? You don't want to be able to defeat cancer." Chemo gets a foothold here that's very solid.
- 2.97** We trace the origins of chemotherapy from these pioneers like Rhoads, Farber, and Lasker into a centerpiece of cancer treatment. Think about this. These are mustard gas being used from World War II. It's nitrogen mustard being used. It's just a molecule different in terms of these experiments. And now you have the federal government paying for studies of this type of poisonous chemical. The pattern is the same. It's

money and prestige. You get enough money, you get enough funding, you have the prestige to do it. It's hard to question that model.

- 2.98** It was Memorial Sloan Kettering at this point. It's the leading hospital in the world on this chemotherapy front, and people aren't criticizing that for the fact that they're focused so heavily on chemotherapy. People view Memorial Sloan Kettering as at the frontier. They're at the cutting edge. Cutting may be the wrong thing to use in terms of tumors, but they are at the leading edge of cancer research.
- 2.99** Modest incremental survivor benefits. I'll tell you, for solid tumors, chemotherapy benefits were really, really tiny, and in many cases, not at all. But the system continued to think it doesn't just have to be blood cancers. We're going to use these for a wider variety. I think that what we're going to see is this pattern of entrenched interest in protecting the treatment. It becomes key when we look at lecture three about how this industry moves forward on these early tests from the 1950s.
- 2.100** Now, remember I said to you before that I was surprised some ways that when Trisha ended up going in and had to treat her cancer, that the options were still the surgery? A variation of that surgery that was less intrusive than had been developed, but still people are still doing radical mastectomies, no question, and radiation, and then the option of chemotherapy, which now, by the way, this is down the road, far down the road, they can do some initial testing on the tumor once they've done a biopsy, and they can determine whether you would be a candidate for what they call chemotherapy that may prevent recurrence down the road. Trisha was not a candidate for that. So chemo never became part of her treatment regimen as she went forward. But that was determined based upon a test that came back.
- 2.101** It's still a process in which you look at it, it's surgery, radiation, and chemo. You would tend to think that after all this time, we may be further along than we were in the 1950s. Not necessarily so. We'll talk about that more in lecture three. Thank you very much.

Lecture 3: Who Controls Treatment?

At a glance — Who controls cancer treatment.

- 3.1 Before, we were looking at—I'm going to recap it a little bit, but I always go back as an investigator to the early parts of history, because in part, I think that where we end up today when we're talking about an industry, a powerful industry like the cancer care industry, the DNA, the origins, the templates are set back in the day. So if you start to understand how the origins of the industry grew, you also understand after a while how they become what I call gatekeepers to the industry as it gets along, because they don't necessarily want it challenged. And I understand that.
- 3.2 It's a business, like in the drug business, pharmaceutical business, of healing and providing treatment, but at the same time, it is also an industry. And so, keeping those two things in mind doesn't mean that there's any contradiction between them. It's very much the case. Remember before we were talking about the size of the industry: over \$200 billion a year in cancer care, going to \$320 billion. It's a big industry.
- 3.3 So, a quick note, a reminder. It's historical and political analysis. I don't want you to think it's medical advice. I'm old enough to remember Dr. Marcus Welby, who was a very popular TV show at one point and wildly popular, an MD. Then that person, that actor who played Dr. Marcus Welby, was later used by a pharmaceutical company in an ad, and actually said in the ad, "I'm not a real doctor. I just play one on TV." I don't even play one on TV. So rest assured, I'm giving you the background of this industry that treats cancer, but I don't want you to take it as any medical advice because I'm clearly not a doctor.
- 3.4 Recap—the shift from cancer as a mysterious illness to something as measurable as an epidemic, the birth of chemotherapy and mustard gas. And today's lecture really is, what is a fraud and what are the interests protected in identifying something as a fraud? Now, who gets to decide what's a cure? Who gets to decide what's quackery? And when we're talking about cancer, we're talking about fake cures that really have more than just an effect of taking somebody's money.
- 3.5 So, you might just take somebody's money if you're fleecing them on some idea that they're going to have a vacation home in an exotic location. If they put down a deposit, they'll be able to buy something that's worth \$200,000 for only \$30,000 today, and you never see your money again. That's bad enough. That's just the money fraud. When you're selling somebody a fake cure for cancer, you're selling based upon desperation and fear.
- 3.6 And the people who take advantage of that are people who understand that fear is a great motivator. We all saw the early days of COVID, when we were all out of fear of what this new virus would be, and society starts to lock down. And the concern—people reacted often on that basis of that fear. And so, you're talking about cancer,

certainly in days in which people didn't understand what was behind it. That comes out time and time again.

- 3.7** And again, this idea of treatment over prevention, so there's some public confusion about cancer. People don't really understand in the '50s, especially, and even into the '60s, even today sometimes, what's causing it. They're still confused by that. I do think that the institutions that are forming here very much are focused on the treatment end. They're keeping the eye off of what might be the cause.
- 3.8** Now, there are two things that happen. It's not until 1964—and we'll talk about this later—in which the Surgeon General issues a report that says smoking might be dangerous. It's two years later before they put a warning label on the side of cigarettes. It's all the way into the [1990s] when cigarette CEOs, tobacco CEOs, Big Tobacco, are going before Congress and saying, "We don't have an addictive product, and we're still contesting the idea that it could be causing cancers."
- 3.9** So, it's interesting that if you think that's a difficult one, imagine what happens in terms of other industries in which somebody raises the idea that an environmental product or an industrial product might be a carcinogenic. We're going to talk about some of those in a later lecture. But one of the things that we now know from having investigated, there was a time in which some of the companies who were afraid of their products that might, in fact, be causing—like vinyl chloride and others—cancers, were looking at cancerous tumors on autopsies and reporting them as benign. So they weren't really cancerous.
- 3.10** And therefore, the figures didn't look as though their chemicals were causing cancers. Later, those findings would be overturned by pathologists who reviewed it, and then their chemicals and carcinogenics would be under a microscope. One of the things that happens as well, you'll see this in the industry, when tobacco eventually falls prey to being a—everybody says, "We agree tobacco can cause cancer." Then it's as though the American Cancer Society and other major cancer organizations say, "Okay, we have somebody to blame. It's tobacco."
- 3.11** And that sort of takes the pressure off of them of any other prevention. They're saying, "Just don't smoke." And they almost stop in some ways—I'm not saying this universally—they stop looking for other carcinogenics. They say, "Tobacco is it, and change it." And the fall in tobacco, the fall in smoking, there's always a new report by the American Cancer Society saying, by the way—and you've seen this, you've lived long enough—every three years, every five years, there's a new report that says, "Cancer rates are falling dramatically. Cancer survival rates are better than ever." It's great. It tracks the fall in smoking.
- 3.12** So I can take two different charts with you and show how, from the 1970s and on, as smoking rates fall in the United States, cancer rates start to fall as well. Much of the

success in the efforts to fight cancer has been as a drop in smoking, not necessarily as an improvement always in treatments. So, it's very hard to think of those.

- 3.13** So I'm going back for half a second to these early 20th-century cancer cures, and I'll tell you why. These treatments, what I call the clearly fraudulent treatments, the ones that are being put out as snake oil salesmen, these ineffective ingredients that people are saying, by the way, "It will cure tumors, it'll get rid of it all," and they know it won't, these are really people who deserve to be in jail.
- 3.14** The American Medical Association launches campaigns against the early quack medicines, the material that's being sold, the cancer clinics, motivated in part not only because they think they have the right answers in terms of medicine for how to treat cancer, but because they are making sure that the market that they have, that's a large market, isn't affected by somebody coming out with a claim who's outside of the system.
- 3.15** So look, back in the day, there's no question about it. You had all types of Radol, cure for cancer. There were people putting this out. The American Medical Association would go after them and say, time and time again, "These are quacks." And that was what was used more than anything else. I think that one of the things that happens is, the American Medical Association put out a 500-page volume, much of it directed toward fake cancer cures, and they call them Nostrums and Quackery.
- 3.16** One of my personal favorites is a fellow called—and I don't mean favorite, as in good—Dennis Dupuis, who called himself a doctor. He said Dr. Rupert Wells. We don't know where he came up with that name. So he marketed a cure for cancer, and this is in the early 1900s, and his product was called Radol treatment. You could drink it or you could go ahead and put it on your skin— an area, if you thought you had a tumor, you could apply it there.
- 3.17** Now, here's how he sold it. In advertisements in newspapers, on the sides of billboards, in placards in cities, he would say, "I can cure cancer at home without pain or operation. I have uncovered a new and seemingly unfailing remedy for this deadly disease. I have made an astonishing cure. Marvelous radiotized fluid did it. No matter what your condition may be, do not hesitate to write." So, he encouraged people to write in, like it says, "Call the 800 number to get more information," that was the equivalent of it then, with snail mail. "Write in, and I'll tell you more."
- 3.18** So, people would write to him. And then he would say, "Oh, I have to tell you that there's a cost to this. It's not just free. I'm not just saving lives." So, at this point, it was \$10 a month. Now, the part of it that's so fascinating, you can see the sales technique working here. He had a big warning on the bottom of the box in which Radol treatment was sent. Not warning you about using it or anything else, don't worry about that. It would say, "This fluid will not retain its radioactivity beyond 40 days from the date of this label."

- 3.19** Now, we don't even know how much radiation it had in it, but he's telling you, you have to be on a subscription plan. So it's \$10 a month for the rest of your life because, of course, you don't know if that tumor that I'm going to get rid of is going to come back. Now, the best part of this is that people would sometimes get his notice and pass and say, "Thank you very much. I don't want to buy it."
- 3.20** And then there's a whole list of things inside the AMA that eventually uncovers him, in which he would write back to them, and he would say, "Well, okay, I feel sorry for you. I'll give it to you for \$5 a month." And you'd say, "No." "I'll give it to you for \$2.50 a month." So, I love the fact that he's cured cancer, but he's willing to negotiate all the way down to try to induce you to come in as a special this month only. This was a bestseller. This made millions of dollars, millions of dollars, at a time in the early 1900s, because you're preying off of people's worst fears.
- 3.21** So, defining quackery and gatekeeping. These treatments, which are clearly outrageous—and we look at them when we think those people belong in prison—this set a low bar for the American Medical Association, because when they said "quackery," we all agree. No doubt about it. There's no debate. The word "quack," as a quack treatment, carries a lot of weight. Back then, it was true.
- 3.22** But anytime it's used in the future to describe a treatment that comes out of the ordinary, conventional medical system, if you're able to attach the word "quackery" or "quack" to a doctor or a treatment, it means they are unlikely to get insurance coverage for that treatment. They're unlikely to get federal funding for research on that treatment. They're unlikely to be used at a clinic or hospital that is using a standard form of chemotherapy or anything else. So it's possible that, just by delegating it as quackery, you start to automatically put it on its back heels.
- 3.23** It becomes what I call one of the mechanisms for gatekeeping, to make sure that those people from the outside who aren't part of this new system developing of oncology are able to come in and offer it without anything. And legal and financial consequences is what I was saying: licensing, insurance, reimbursement, and even possibly at times, criminal exposure. So, the selective use of evidence: the same evidence can be used to justify both suppression and supportive treatments, depending on the interpretation. You'll see this in a moment on a couple of examples.
- 3.24** And quackery is a political weapon, no doubt about it. Many of these alternative cures are, in fact, completely exposed as fakes, and they are indeed quackery. But the very use of it raises immediately the idea that nobody should even be looking at it, and it's used aggressively by the AMA and others.
- 3.25** So, early 20th-century cancer cures, the rise of patent medicines and secret cancer tonics, like Dupuis's Radol treatment. Then from the 1910s to the 1930s, we got the promotion of radium water and other ineffective cures. Remember, no law yet in the

United States to regulate the safety of medications; hasn't been passed yet in Congress, so in this period, still a bit of the Wild West.

3.26 In the 1930s to 1940s, into the war period, the AMA is conducting campaigns against unlicensed cancer clinics. Now, hard to imagine. In the 1960s, the country was riddled with thousands of diet clinics, which were essentially selling just amphetamines, but they were legal prescriptions. And that caused—they were being diverted to the public. It caused an addiction crisis. And then Congress got involved. And finally, the diet clinics were closed up, just like in the early 2000s, there were pain clinics.

3.27 I live in Florida. Florida and Broward County, just north of where I live in Miami, was ground zero for pain clinics. You'd see signs, "1-800 call for pain." And there'd be some clinic in which doctors were dispensing OxyContin with an armed guard outside and taking cash only, instead of taking insurance.

3.28 So we've seen this before. In the '30s and '40s, cancer clinics are popping up. Not Sloan Kettering, not Boston Children's Hospital, but independent, in a strip mall, cancer clinics that would say, "Come in, we can cure you." And in each instance, those were, in fact, fakes. The AMA was very good about going out, one by one, and finally closing up. In the '40s and '50s, there becomes a series of what I call more pronounced wars between the American Medical Association and the proponents of these alternative therapies, and they set the groundwork for what comes in the future.

3.29 The story of Royal Raymond Rife. How's that for a name? Pretty good. It's not even something that you could say later I made that up for the purpose of my resume and going into Hollywood. That was his real name, Royal Raymond Rife. He was working—he's an optical engineer who is creating a microscope that is—you'll see in a second—60,000 times the amplification of any microscope that existed at the time. It was revolutionary in terms of what it was able to do.

3.30 The same way that if you take a picture, take a photograph, and you digitize it, and then you expand it, and you make it very, very large and zoom in, all of the photograph starts to look like pixels. At a certain point, it loses its definition. Well, here, it was almost the equivalent of that because you were seeing things at 60,000 times magnification that was hard to imagine in any other microscopic setting.

3.31 There's no doubt that his inventions in terms of optical sensors, in terms of what he was doing in microscopes, are ahead of the entire field. So I call him brilliant, and I think it fits with him, this ultra-powerful microscope. Now, the part that set everybody's hair on fire was that he said that, when he was looking at tumors—mice in his lab that had cancers—he was able to see this bacteria. He calls it the BX. He says, "I can see that in every cancer slide, and I don't see it in slides of mice without tumors. So, I think that I can see what—"

3.32 Now, there are two things. Is that bacteria that he's seeing part of a side effect of what is developing in the mice with cancer? So they developed that. Is it the cart before the

horse? Is it pulling the cart, or does it come afterwards, or is it causing the cancer? Rife says, "I think that it's causing the cancer. So I've found the microorganism that causes cancer. And I can—" He didn't say cure it. He said, "I can devitalize it." That's a great word. It's like one of those words you would think, "How did you come up with that?"

- 3.33** The marketing people could not have come up with a better word because it doesn't hit the cancer industry on the head with a cure, "devitalize" this unusual little microorganism that only Rife has really seen. When he devitalizes it in animal experiments, the tumors disappear. Now, these are small clinical studies with a very high remission rate, and so that's what's coming out of it. Now, of course, no human studies at this point with Rife. The microscope, again, 60,000 times magnification, 20 times more powerful than today's microscopes. So this is really a one-off.
- 3.34** After a while, they realize they don't need a magnification this big, but this is how he's been studying what had previously been invisible microbes. What's important about that is, when he initially gets some pushback, Rife's answer is, "Well, I don't blame you for not having realized that these microorganisms cause cancer, because you people, Rhoads, Farber, etc., you couldn't even see these. You didn't have the instrument to be able to see it. So, instead of saying, 'I don't think so,' maybe you should be more open to the fact that I finally made a discovery and stumbled on it."
- 3.35** Part of the problem for that is, remember before, when I was saying the cancer establishment initially didn't like Sidney Farber because he was a pathologist, not an oncologist? They said, "Okay, he can't be making a discovery." Well, take that and multiply it by 100 because Rife isn't even an MD. He's not a doctor. He's an optical engineer. You have an optical engineer who's got a discovery in a microscope, telling you that he may have found the cause of cancer and he can devitalize it.
- 3.36** You don't think there was some resistance from people who have been spending a decade or more on chemotherapy and radiation and on surgical alternatives? He ran into a wall from the very beginning. I think that his device, if you were looking at it from the outside, people who were a little bit removed from the cancer industry would say it falls somewhere in between promising and implausible. Nobody was saying he's figured it out, but at the best, they might say that it was promising.
- 3.37** So, here he is. He identifies this cancer-causing microbe. He observes it. He dubs it BX in every cancer tissue that he studies, so this is across the board inside his lab. He creates tumors from that microbe. He says this is interesting. Four hundred tumors in lab mice with a single culture of that BX microbe. Now, he's gone beyond the point of saying, "Okay, I spot this BX microbe," as he calls it, in every mouse that has a tumor. He's saying, "Then I took an extract of that BX microbe and I inject it into healthy mice, and in the 400 I inject into, they end up getting tumors."
- 3.38** So now you think, oh, cause and effect? Could be. Everybody understands how the logic works on this. The idea that a single microorganism might be behind all cancers

certainly ran against the idea of what conventional cancer doctors thought about it. And again, he's very careful that he's curing it only in [mice]. He's not saying he's curing it; he's devitalizing it. Now, what happens with this information?

- 3.39** So, he develops a beam ray, literally a light beam that he comes up with, that he's going to be able to use this oscillating beam, and that's what destroys this BX, cancer-causing microbe. The oscillating beam belongs to Rife, so he has the patent on it. He has the ability, therefore, to profit from it, if that's going forward, if he's right. He destroys them in 10 consecutive exposures. So you have to take his beam ray machine and use it in these directions that he has over 10 times, and eventually it will eliminate the microbes that are causing cancer. And he now starts to get some reports from terminally ill cancer patients that they're having remission and they're feeling better.
- 3.40** The fallout from his beam ray and his claim. "Dread Disease Germs Destroyed by Rays, Claims Scientist." Boy, you don't think that sets the hair on fire on people who are part of what I call the cancer industry for a long time? So, a team of physicians at the University of Southern California, they end up taking 16 very, very sick, terminally ill cancer patients, and they move them out of the hospital. They give them the ray beam according to directions that Rife has given them. What they report is that 14 of the patients had been clinically cured after 70 days, and the remaining two, number 15 and 16, cured after another 20 days, after they got the recommended frequencies from the ray beam.
- 3.41** Now, of course, what are you interested in, in terms of cancer care? What am I interested in if I see a remission like that? I was talking before about Sidney Farber having given the folates to children. It made it worse. Then he gave anti-folates, and then it cured them, but then they died afterwards. This process where you get remission, but you're not quite sure. But still, the fact that you even have remission from a group of doctors, the University of Southern California, raises everybody's eyebrows.
- 3.42** Two prominent bacteriologists from the Mayo Clinic and from Northwestern University try Rife's machine with directions, and they don't get as spectacular results as the University of Southern California, but they still get results that say, "We saw some improvement." Pretty amazing. Now, if you want to get under the skin of the establishment, have some other part of the establishment seem to not endorse it, but give you an opportunity to talk about it. And that's what happens when the Royal Society of Medicine in London invites Rife over to talk about, in a keynote address, his new beam ray and what it might mean for the world of cancer.
- 3.43** The internal memos at places like Memorial Sloan Kettering and others about what was going on, and who the hell is Rife, and why is he going before the Royal Society of Medicine? There were many of them. And then the Smithsonian wrote this really wonderfully favorable and fawning piece about him and his lab. Popular Science

featured him in an extensive profile. So at this point, the AMA is going to get involved. They've had enough of it.

- 3.44** This is not a quack like the nostrums we were talking about before, the Radol treatments, in which somebody's selling a fake cure. They know they are, and they know you're going to die anyway. This is somebody who thinks he's actually made a real innovation, believes it, has a device that he thinks treats it, and is offering that, and is allowing others to use it in treatment as well. But the AMA is convinced internally that it has to be wrong.
- 3.45** And so, what they do is, they get involved. They take notice of his beam technology and its impact on the medical establishment. What's interesting about that is, when the AMA is first discussing this, nobody inside the AMA says at any moment that I'm ever able to find—a memo—"By the way, maybe this works. And then it should become part of our conventional treatments, together with chemo and surgery and radiation. Maybe this will be the fourth element."
- 3.46** They're convinced from the beginning that it's wrong, in part because he is not a doctor, and it just seems too simple. Think of all the things they're using to fight cancer, and they're difficult, and they have toxic side effects. And then you mean some guy has a beam of light, and you don't feel anything at all, and it's gone? It can't be. There's a natural skepticism of it inside of there.
- 3.47** So they put Morris Fishbein on it. He is [the editor] of the AMA, but he had been around for 25 years by then, and he had a specialty. When there was a fake cancer clinic—these clinics would pop up—or there was some quack putting out a new type of drink that was supposed to solve cancer, Fishbein was the one who would lead the effort against it. So he knew how to go after the fakes. He knew how to go after the quacks. They put him onto Raymond Rife. So first of all, they start to attack Rife's credibility. And I know how this works. As somebody who has seen journalism from the inside and has written 13 books, there's what I call a whisper campaign.
- 3.48** So reporters and health reporters and others start to get tips about Rife, and he may not be that reliable, and you have to be careful about him. Science writers get it. In one case, they even paid a technician to file a lawsuit against Rife, saying, "By the way, you're skewing your documents." They're taking away the luster of the new golden boy, so that's the first thing that happens, is tarnishing the reputation.
- 3.49** And then—Fishbein knows this very well; the AMA is very good at this—they block any further scientific research. By the end of World War II, despite prevailing in the legal battles that have been brought by the technician and others, they're cutting off the funding. They're going to the federal government and to others and say, "By the way, don't waste your money on this. We know, unofficially, it's a waste of time. So if you're going to send money, send it to traditional cancer cures. Don't send it over here."

- 3.50** Now, before we get here, sorry about this. In the end, Rife's machine is shown not to be a cancer cure. And the BX microbe that he had determined was there is an artifact that you can find in many cancer patients, but isn't unique just to cancer patients. It isn't indistinguishable and new. When you go through his reports of how he did his studies and how the tumors grew in mice, there's all types of operational problems. And so, in the end, there's no doubt that Rife's beam ray was not a miracle solution to cancer, nor had he found the reason that cancer exists.
- 3.51** But what was interesting to me, when I talk about something establishing a template or how an industry gets going, the AMA and Fishbein's response to Rife is the template that they established when there was something else that they thought might be a threat to established cancer treatment. And that was: you went after the person behind it, you sort of denigrated them a little bit, you cut off the funding, and then you hoped that it was wrong.
- 3.52** Here's the part I don't like. I'd have no problem at all—this is just my own personal conclusion—with the AMA doing all the things it did against Rife, if it had conducted its own studies and determined that that machine was not promising. It turns out that the remissions—the University of Southern California and what they had seen at Northwestern—it turned out to be temporary remissions. It could be, there are even people afterwards who said, "It's the power of positive thinking." Everybody thought they were being treated by a new beam therapy, and for a while, they felt much better. Possible, we don't know why, but it clearly wasn't a cure for them. In other cases in which it was applied, it didn't even give that remission.
- 3.53** Now, if the AMA had had a group of doctors do that type of research, and they were able to independently show that it was wrong, even though he wasn't doing a fraud, they could have discredited him for the underlying, but they did it without knowing, because their instinctive response was, "It can't come in from out of the blue and be right. It's just too simple." And that sort of establishes that framework, that template, as I call it, going forward.
- 3.54** Krebiozen. Krebiozen is a cancer treatment that becomes, really, a hot subject inside the cancer industry in the 1950s. We were talking before in the earlier lecture about that chemotherapy: the golden age of chemotherapy was the 1950s, as they're starting to evolve on new patents and variations of chemotherapy. Well, into the heart of this comes this drug called Krebiozen.
- 3.55** To me, it's an example of the competing data and how you interpret it, but also how institutional power can exclude somebody on the outside. This becomes a fight eventually over the size of tumors. They get into the minutiae: the size of tumors, the growth rate. Can I see the underlying slides? In this case, the AMA and those who are fighting it are doing much more of trying to get into the details of it, to try to prove it to be a fraud.

- 3.56** How it begins. Stevan Durovic, he is a Yugoslavian doctor. He immigrates to the United States in 1949. Just at the point, 1950, I'd said in a previous course about Big Pharma, that in 1940, German pharmaceutical companies wrote one of every two prescriptions around the world. In 1950, five years after the end of [World War II], American companies had 80% of all the prescriptions worldwide and 95% of the profits.
- 3.57** The change in 10 years had been absolutely dramatic because of the fact that World War II had decimated the German pharmaceutical industry. Much of it had been bombed out, and the American industry had this great head start in terms of the penicillin project. He, Durovic, arrives in 1949. And he has a substance that he has dubbed as [Kositerin]. High blood pressure—that's what it's treated for. But no doctors are interested. He tries to interest doctors at Northwestern University, and they're not interested. He says it helps to lower blood pressure. That was the original claim that Durovic had.
- 3.58** He approaches this Dr. Andrew Ivy at the University of Illinois and announces he had an anti-cancer drug that he calls substance X, later calls it Krebiozen. He says that, in [Argentina], he had seen some success with this compound that he had come up with, with patients who had cancer. That's what he tells this Dr. Ivy. Now, he refuses to disclose the manufacturing process, and the medical community, of course, is skeptical about that from the get-go.
- 3.59** So, if you want to say that something might have anti-cancer effects, you tell us how you make it, you can go ahead, and we can test it. But of course, if he says how he makes it, then he loses control over the drug, and he understands that very well in a day and age in the 1950s, in which patented drugs are becoming very, very important. Durovic did not need to be in the United States very long to understand you have to own your drug to make money from it. So the controversy kicks into high gear, as I say.
- 3.60** In 1951, he's only been here for two years, and he has a press conference at Chicago's Drake Hotel— excuse me—in which he says, five women, they've been brought back from near death. They were all dying of cancer. And he doesn't just say that. He's got five patients there with him, who say, "We took Krebiozen. We were told by our doctors we were near end of days, and it's miraculous. We feel great, we feel healthy, and we feel like our cancer is vanquished."
- 3.61** He doesn't do it in terms of an academic study. He doesn't publish in the Journal of the American Medical Association. He's not giving it to a group of doctors at Memorial Sloan Kettering. He's announcing it at a press conference. You know how the industry hates press conferences that they're unprepared for. So they're caught really blindsided by that press conference, because what do you think happens as a result of that press conference? Do you think the press picked up on that story? Big time. Headlines everywhere. Chicago papers, New York papers, everything else. "Doctor announced a miraculous cure, patented material, Krebiozen, and five patients say how it's changed their life."

- 3.62** Now, he says, of course, Durovic says he created the drug from the blood of horses in Argentina. I love that. He's pressed more. And what's behind this? He still won't give it out, but he says, "I'll tell you what I did. I created it from the blood of these horses. I was able to do an extract. It's completely unique. But you're all skeptical. So therefore, I will provide it to any cancer center that wants to try it. You want to try it? Go ahead and try it. See what the results are. Conduct your own studies."
- 3.63** Now, the problem with that is, it really does cause what I call an inflection point for the industry itself, because you would think that they might rush to go ahead and start testing it. But the word that goes out from the AMA and from the big cancer centers is, "Don't waste your time. We know Krebiozen can't work. There's no such thing as a blood extract from horses in Argentina that might possibly work. This doctor from Yugoslavia, he seems a little shady to us. We're just not sure. Those patients might be real, but they might just be going through a period in which they feel better, and he's been able to optimize that." So, the general consensus is, don't go ahead and even test it.
- 3.64** Again, this is the same thing that happens with Rife and his microscope. My problem that I have is, not that there aren't alternatives that don't work, but why aren't we testing them more rigorously if it falls outside the confines? If you get to define it as quackery—we talked about that before—you've already tarred and feathered it. So he offers it and says that.
- 3.65** Now, the AMA and the cancer institution, they get a writer who puts together a fast book, what I call a speedy book, that's ready to come out from Beacon Press that takes apart Krebiozen, that tells you why Durovic is not to be trusted and why it's just a bad drug. And so, what does he do? Instead of just allowing the book to come out, ruin his reputation, and stop the sales of Krebiozen, he sues Beacon Press, trying to get an injunction against its publication. Very hard to do in the United States.
- 3.66** He said, "This is false. It's defamatory. It's going to ruin me. It's meant as a hit piece in order to stop the development of this drug." So this is the Krebiozen Research Foundation, which he has formed by this point, and Beacon Press, regarding the publication of this book that essentially takes on his cancer drug. And the therapeutic merits are still under investigation at the time of the dispute, but the book takes it to be that there's no additional research needed to call this a fraud. So, to stop the publication, he sues the publisher, and the court denies the injunction and says, "No, free speech reigns."
- 3.67** It's almost impossible to get a preliminary injunction to stop the publication of a book in America because— and I know this from having seen and worked with lawyers at publishers, and I've put out my own books—courts assume that if you are damaged by the publication of a book, so let's say that Durovic is right, and that this book was just a hit piece, and that the information in it was false, and it's going to upend his drug and stop its sales, then the courts say, "You have a right to come back and sue for damages."

You can prove your case later in a libel action or defamation action, but we won't stop the book from being published."

- 3.68** So they let it come out, and that book starts to change the mind of what I call some of the science writers and health writers. So it highlights the balance between what I call protecting individual reputations, which is his, and then upholding the constitutional right to free speech. Personally, if you ask me, the book has a right to be published, even though I think it was a hit piece. So it's both. And then he has to worry about how he's going to defend himself.
- 3.69** So this is the establishment's response. The American Cancer Society and Sloan Kettering, in 1951, opposed conducting clinical trials for Krebiozen. It's pretty remarkable. So you have a doctor and you have five people at a press conference saying this. I understand you're not going to go for everything that comes in, and they have their money already allocated. But Sloan Kettering says, "We're not going to take him up on his free offer." Sloan Kettering is one of the big names now in the business. So Sloan Kettering says, "We don't think it's worthy even of spending any time in a lab on." That means other major cancer institutions and research centers and academic centers will start to follow.
- 3.70** In addition, you have Mary Lasker, American Cancer Society—not that she owns it—but the American Cancer Society has some funding now. It's getting the word out. She's been helping them. The American Cancer Society, focused largely on treatment, says, "We don't think it's worth testing." Very interesting. They're not all doctors there, but some of them are. So that's the first thing.
- 3.71** In '51, they do a little bit more. They urge the FDA to block interstate sales of Krebiozen. Now, how about that? So it's one thing to say, "We're not going to test it." Now you're Sloan Kettering and the American Cancer Society. You go to the FDA and you say, "Hey, why don't you stop this drug from being sold? It's being sold across state lines." Meanwhile, there hasn't been a reported death, haven't been any side effects. Nobody's left a trial of Krebiozen paralyzed or with any of the toxic side effects that chemotherapy will sometimes produce.
- 3.72** But without that, how do you stop a drug that had a press conference and could be popular, with a doctor who's very outspoken? Cut the sales off. You've got the Beacon Press book. That's doing the hit job. Now, if that doesn't put him under and take Krebiozen off the market, then what you could do is cut off his ability to have sales. If the FDA blocks it from being sold interstate, bingo, you've just knocked the drug off the market without any other research.
- 3.73** From 1951 to '63. Now, boy, 12 years. Did I make a typo on that? No, it really is '63. For 12 years, organizations and top thought leaders, and those who had influence inside the traditional cancer treatment system that had been created in America, convinced academic and government researchers not to fund Krebiozen research. So the offer of"

take my drug and test it" can only be done if somebody sets aside a few dollars to conduct a study with it.

- 3.74** And I will tell you that, over that 12-year period, there were repeated efforts in which, all of a sudden in a little school, somebody would say, "Oh, you know what? What about that Krebiozen? Maybe we should test that." And then the forces that be would come down and say, "Don't waste your money. Don't do that. Oh, you're better off spending it here on a possibly new and less harmful chemo drug." And so, time and time again—you could do a small book on that—they closed the door to the funding for further research, through 1963.
- 3.75** And then finally, in 1963, the FDA says, "All right, no more interstate sales because there are no clinical studies, double-blind studies, that establish its worth." Now, think of this. In 1962, for those of you who have listened to the course on the secrets of Big Pharma, the government passed its biggest ever overhaul of the pharmaceutical industry, which included clinical trials for the first time that would show efficacy of a drug.
- 3.76** So, before 1962, all you had to do as a drug manufacturer was to show that it was safe. That was only since 1938 and on. Before 1938, you just had to list the ingredients. It didn't matter if people were being killed or they weren't. You had to show the ingredients. You didn't have to show safety. After 1938, you have to show safety. And then in 1962, in this major law that's passed as a result of congressional hearings, you end up getting a law that says, now you have to show for the first time in history that what you're marketing the drug for, it actually works. So, if you're marketing to cure malaria, it cures malaria. If you're marketing it to cure an infection, it stops an infection.
- 3.77** And to prove that, you have to do clinical trials. And we want that information submitted to us, so we can compare the clinical trials and see what the safety profile is and how therapeutic it is. So the FDA gets this new power, and they have the ability to recall drugs. For the first time ever, they can recall drugs that don't work.
- 3.78** They fought with the pharmaceutical companies for the next few years to recall off the market these fixed-dose combination antibiotics that were a waste of time, causing additional symptoms in most cases, and not doing any cures, just adding to the bottom line of pharmaceutical companies. They recall those, and dozens and dozens. But they also don't recall Krebiozen. They stop it by saying, "You know what? No clinical studies. Oh, my God, you're treating cancer. You haven't submitted clinical studies to us. We're stopping the interstate sales."
- 3.79** So, they use their newfound power to be able to, essentially, cripple Durovic from selling it. And then in 1964, he continues to sell it. He ignores the rule. FDA said, "Don't sell it interstate." He's selling mail-order all over the country. And they finally decide to indict him. So in 1964, he is indicted for dispensing a drug that's not allowed for interstate sales. The trial takes two years to get to trial. I'm a non-practicing attorney.

As an attorney in another life, it seems like the legal system is always slow. I'm glad to see it was just as slow in the 1960s as it is today.

3.80 Somebody gets indicted and then they go to trial two years later, and you almost forget about the case before it comes around. And that was the case here. They did—while he was indicted, until the time the trial takes place in 1966—they really clamped down on the sales of Krebiozen. So he, together with four other defendants—another doctor, Dr. Andrew Ivy, who's a medical director at his firm—are charged with promoting a worthless drug, Krebiozen, and misleading the public about his cancer-curing elements.

3.81 Highly publicized. Let me tell you, this wasn't the O. J. Simpson trial in which people were stopping at gas stations to listen to what was being said on television on a gas pump. The country didn't stop, but they paid a lot of attention. It's front page in the Chicago Tribune. It's big news on the network television programs that are coming out at that point and holding media attention. They're following it because it's a cancer cure case, and he is something of a magnetic personality. He's very talkative and very outspoken.

3.82 So, what do you get at the trial? It's fantastic. I read that trial transcript, and I'll tell you, you do get conflicting testimony. So it's very interesting that you do not get what I call one side, where I could say to you, "Oh, my God, they came in, they just destroyed him. They just showed him to be a charlatan." Not at all. There were people in there, including cancer patients, who said, "Krebiozen has cured us." You'd have other doctors saying it has nothing to do with their condition. Then they would argue about the technicalities of a tumor size, and whether the reduction was—and whether Durovic had gone ahead and measured the tumors incorrectly, and maybe they had overreported the size of the reduction.

3.83 So, they get into the technicalities, and this trial plays out. It really is the interplay between what I call regulatory oversight—the FDA—and the scientific advancement, and the need of cancer patients to want to have a cure. There's no question about that. And what happens? The legal debate: constitutional protection. So the court talks about the importance of the constitutional protection of free speech and the public's right to free discussion, even with medical treatments as important as cancer.

3.84 There's a public interest here involved in these medical controversies, and the court talks about this with the judge, this interest in this discussion and debate that takes place about advances that may be able to help people to save their lives. The question is, if you are pushing this forward and hurting his reputation and stopping him, are you also preventing him from possibly saving people?

3.85 The more Krebiozen was attacked in the court, the larger the number of cancer victims who voiced support for the drug. So people who said, "I tried Krebiozen. I bought it, I tried it. A sister of mine tried it, a brother of mine tried it, and they were sick. Krebiozen helped them." They started to have more people come out. That was a nightmare for

the AMA and the medical establishment. Oh, my goodness. They had more people coming out during the trial to say, "I'd like to be able to help you and talk about this." So Durovic, he, in the end of this trial, and his fellow defendants rely on the testimony of the cancer patients. That's who Durovic says, "They are really my evidence, more than any other clinical trial, that these people are here."

- 3.86** And they went through a cross-examination, but they were difficult to cross-examine. The jury returned after four days, acquittal on all charges. Pretty remarkable. A devastating defeat for the medical establishment. And I love this. One of the jurors had a badge that said, "I need Krebiozen to live." They hugged the foreman and kissed him on the cheek. A juror told the Chicago Tribune, "We didn't want to destroy Krebiozen." Now, that's fairly interesting because that's exactly what the point of having the FDA come in and do the original stopping of interstate sales and having the drug marked as a fraud was intended to do, and it doesn't happen.
- 3.87** That's Dr. Andrew Ivy. He's the medical director and physiologist who was head of the department of clinical sciences at the University of Illinois. So, what happened to him and his drug? So, the verdict does cause—and I say—a mini panic; I shouldn't say a panic in terms of people jumping off the—it wasn't like the 1929 stock market crash, and people were jumping out of buildings. But it causes a panic inside the industry because they're worried that, although it's just an acquittal on a criminal charge, it will be seen by the public as a vindication of the drug.
- 3.88** So the public will take the acquittal of this doctor and the medical director and the drug itself and say, "Well, that must be—the drug works, so I'm going to buy Krebiozen." And this flurry of telephone calls and letters, internal documents—the files, when you go through them with these companies, are really interesting. They are in a state of worry about what will happen to their cancer treatments, the ability to mark, as they call it, these fakes.
- 3.89** So the cancer counterculture had tapped into this underlying hostility. There was at this moment a pushback, because the cancer establishment that does treatment was saying that these cured patients, for Krebiozen, were somehow delusional or they were wrong. And those patients are saying, "No, I'm not wrong." And when you tell somebody that it's different than what they experience and what they know themselves, there was even a thought that they had been incorrectly diagnosed and they hadn't had cancers in the first place.
- 3.90** There's an effort to try to explain how they were. And that's why they would have gone into remission. There are some cancers, by the way, if you have them, that go into remission on their own. You'll hear about some miraculous case in which somebody has prayed to a saint, and then they say they had a tumor that was diagnosed, and now the tumor is gone, and the doctor can't find it. They think it's because they prayed to the saint. But some cancers naturally go into a remission on their own, and that's the nature of the tumor.

- 3.91** And it just happened to be, according to the industry, that everybody who took Krebiozen had one of those types of cancers that went into remission on its own. Unlikely. But I think that part of the problem here was not just that Krebiozen didn't go through these clinical trials, but that it's not sanctioned by the medical establishment at the time. So their effort is to do what they've done before, as they did with Rife, which is sort of to destroy it.
- 3.92** How to stop him? Just like Al Capone—the IRS brought down Al Capone—so the IRS brought him down. That's what happened eventually. They charged him with evading \$900,000 in taxes. Krebiozen was a big moneymaker, no doubt about it. And what did he do? He went over to Switzerland. He said, "Thanks, America, for all the sales. I'm going to go over to Switzerland, which will be fine." Meanwhile, his drug, Krebiozen, stayed on sale until 1973. So, pretty interesting that he leaves the US because he owes all this money in taxes, but he continues to sell the drug here. And there would be the occasional media story, questions about his efficacy, but the emerging cancer establishment had won.
- 3.93** Now, there's a small book in how, "Let that be a lesson to any future healer," a Sloan Kettering director wrote in a secret memo at the time. It's the response of the industry that I don't like. I happen to think that Krebiozen, from all the research which I did in the follow-up, was not a miracle cure. It's hard to imagine. And that he had very successfully sought out some patients who did not actually have cancers at the time, but were worried about having cancers. And in other cases, he did fiddle with the figures that came out of the lab in terms of how the success rate was.
- 3.94** So, I don't think that Krebiozen was, in fact, a cure, but I think that if the establishment had taken him up on his early offer of, "Go ahead and test it," you might have found this out years earlier without going through the entire process. Most of the people who end up using Krebiozen in the end are people who are looking for an unorthodox therapy, who don't want to go through the traditional way of dealing with it.
- 3.95** Ralph Moss. He's a former science writer who endorsed this unusual theory that bacterial infection caused cancer. Now, we know that sometimes HPV and other—there are in fact bacteria [and viruses] that do cause cancer. But at the time he said this, everyone thought he was absolutely crazy. He worked at Sloan Kettering, and they fired him in 1977. They said it was because he was promoting a product called laetrile, which became a popular drug in the '70s as an alternative to treat cancer. People thought it might be a cure.
- 3.96** He said, oh, by the way, there's a senior Japanese researcher inside of Sloan Kettering who's done work on this, and laetrile has some anti-tumor activity when used in the lab. But Sloan Kettering said it's absolutely useless, no benefit at all. He was promoting it. It had been in repeated studies to show it being ineffective and was linked to toxic side effects. So he is fired. Now, what happens?

- 3.97** Moss publishes a book, *The Cancer Syndrome*, in which he argues that alternative cancer therapies like laetrile, Krebiozen, and others are constantly suppressed in favor of much more costly treatments. That doesn't mean that his treatment that he's suggesting is right, but he's right about the general idea that they're being suppressed. And although most of the literary critics and science writers just dismissed his book as pure quackery, it developed a cult following among people who were facing a serious cancer diagnosis and were looking for an alternative to surgery, radiation, and chemo.
- 3.98** So laetrile, disproven cancer treatment, no doubt about that, and it is disproven. Plant-derived as a benefit. Naturally occurring compound found in the pits of fruit, like apricots and that, almonds and peaches as well. And popularized here, really, in the '60s, but '70s is when it catches on as this alternative cancer treatment, because the theory behind it is, it would starve the cancer. So, it had an effect metabolically, in which the cancer would not be able to replicate any longer and spread, because laetrile would stop it dead in its tracks. Banned and restricted in the United States.
- 3.99** So despite the popularity, you will not be surprised to find out that the FDA comes in and says not allowed here, which then pushes it into largely Mexican clinics, where laetrile becomes one of the most popular treatments for cancer in the '70s and '80s, with Americans often traveling to Mexico for this end-of-life treatment in the hope that it would change something. Again, this period in which people are sometimes preying on those who are afraid of dying from this disease. Moss was right on many of the points he made about the cancer industry squelching open and honest debate, but he was wrong on backing laetrile. So, gatekeeping logic: protecting patients or protecting turf, maybe a bit of both.
- 3.100** Here's the approval needed if you go ahead on a cancer treatment. You want to get something approved by the FDA, you need to get it approved. Approval rate for conventional chemotherapies. Not every drug gets approved. Sometimes the drug gets up. Eighty percent of the chemotherapies that go before the FDA get approved, 20% don't. They are too toxic. They have some problem. Overall number. Approval for unpatented natural compounds, even to be used as an adjunct during cancer treatment, 20%. Eighty percent are automatically banned out of the gate.
- 3.101** Burden of proof required for unconventional therapies, 90%. You have to show effectiveness, where 9 out of 10 people essentially show some modicum of being improved, less nausea from chemo, items like this on treating the symptoms, 60% for the burden of proof for FDA-approved drugs. So the drugs that are more powerful sometimes, the chemos and that, a lower threshold for proof than sometimes for the alternative.
- 3.102** Patterns and lessons. No question that there are real charlatans in the history of fraud and selling cancer cures. No question about it. And in the bulk of times, those get my blood boiling more than anything else, that people are taking advantage of those who are afraid of dying from cancer. But also, the institutional structures have these high

evidentiary boundaries and guidelines that you have to meet. So, you come up with an alternative, you have to meet a very high institutional standard.

- 3.103 And then sometimes, as you just saw, the institution doesn't even allow you to try to meet the standard because they say, "We're not going to test it." That makes it more difficult. And they do that in the hope you're going to go away, that nobody will buy it, but that doesn't happen, because there is a distrust of an industry that doesn't necessarily look at alternatives and investigate them more aggressively.
- 3.104 Public focus on the dangers of quacks. The structural failures of the mainstream cancer establishment in some part, not failures in terms of how they treat cancer, but the failure that they refuse to move off of these earlier discoveries on surgery, radiation, and chemotherapy makes some people distrustful of the industry.
- 3.105 And science used as a political tool. I'll tell you, you see this happen time and again, where institutions and commercial interests come together, they use scientific language, and they brand something new as not being objective. They try to cut off the funding from the federal government and from academic institutions in the testing, and they do it very successfully.
- 3.106 So, my idea of this is, you approach claims of quackery and institutional gatekeeping with a critical eye. It's all you can do. But these patterns come across case studies, time and time again. And in some ways, what we've talked about in this hour sets the stage for what's coming, which is the war on cancer, which we've heard about many times. The label "quackery"—I think you have to leave with this in mind—is sometimes accurate. But the question is, who gets to apply it? If it's only the gatekeepers, then I'm not necessarily sure it's always used in the best way.
- 3.107 Preview of lecture four: the formal war on cancer under Nixon, the building of NCI's modern empire. We had it created back in '37, but now it's going to become big, blockbuster chemo protocols and survival statistics and how the same gatekeeping structures scale up when hundreds of billions of dollars are at stake. We're at the edge, where we go into the '70s now, where cancer moves from a mainstay medical treatment in American life and a subject of concern and fear among many Americans to become what I call the industrial empire. That happens in the '70s.
- 3.108 Thank you very much.

Lecture 4: The War on Cancer

At a glance — The War on Cancer.

- 4.1 So, right before moving into this lecture, just one half step back for a second. Krebiozen, by the way, when I say that it was proven to be a fraud, people did research on it later, which you think might be late, but they did. And they really were able to show that all it was, really, was creatine monohydrate dissolved in mineral [oil], a variation of that, which in the body doesn't do much, certainly doesn't cure cancer.
- 4.2 And there's a writer, as a matter of fact, wrote a book a year ago, two years ago, about the entire fraud that had taken place, and that sometimes Durovic, there's no question, overstated his case and the cure rate. At the trial, they even talked about the success of one person who was near death. It turned out that that went through unchallenged by the other side of the trial bench, but that person had already died. Nobody mentioned that.
- 4.3 The part that always became difficult to figure out for me was, even when you got through all the fluff, you still came down to half a dozen patients who swore that it had changed their life and it had cured. And the take in the medical establishment is—I'm going to read this to you.
- 4.4 "Many patients in a highly suggestible state and desperate for a cure experience temporary subjective improvements. In one famous case cited by experts"— and this is repeatedly cited later on—"a patient's tumor shrank when he believed he was receiving the drug, but returned when he read in a newspaper that the drug was a sham." Now, maybe that's just coincidental. "When his doctor injected him with plain water while lying and saying it was a new, stronger version of Krebiozen, his symptoms disappeared again."
- 4.5 Now, what I would like to do is bottle that power of suggestion. If you could have the power of suggestion to reduce tumors and then the fear to have them grow back, if that—we talk about the placebo effect. There's no stronger example of it than in those cases. And still, at times, you think to yourself, there's no residue of Krebiozen that could have had a role in it, but it doesn't.
- 4.6 But it does tell you, in those cases, how incredibly important it is. I saw it in treatment when Trisha was going through a treatment, and others. It's not to say that if you go into treatment with your worst fears and afraid of what's going to happen, that the results will be bad. No, but I do think that there is something about the way the mind helps your body heal, and maybe there's no better example of that than Krebiozen.
- 4.7 So, the war on cancer, money, metrics, and myth. We launched that war in 1971. And a reminder, historical narrative, political analysis, not medical advice. Nobody should go out and buy Krebiozen or try laetrile. By the way, there are still occasionally alternative

medical practitioners who work with a ray beam device. You can find them sometimes. They don't pass themselves off anymore as somebody who will cure cancer; they're normally now passing themselves off as somebody who could cure back pain, better than a chiropractor, better than PRP platelets, rich platelets, they will help you. You can occasionally find them.

- 4.8 There had been stories, rumors afterwards, that the industry, the medical industry, in its effort to destroy Rife and his claims, had gone around even with baseball bats, destroying each and every one of the machines that was left. Not quite true, rather dramatic, but some of those machines are still around. They aren't being produced and manufactured anymore, so they're 30 or 40 years old at the very best.
- 4.9 So remember what we did, lecture one, birth of cancer as a recognized disease in early surgeries, and lecture two, mustard gas moves over to chemotherapy with these remissions in Farber's basement. And he was in the basement with his pathology lab. And then Lasker, Mary Lasker, and Shannon over the NIH funding machine. Lecture three, quacks and gatekeepers, Krebiozen, laetrile, and how institutions can control the narrative.
- 4.10 So today's frame, we move from those individuals to what I view as a full-scale war declared by the US government. Remember when Senator Neely offered a \$5-million reward? Big money back then. He wanted, year after year, he can't get a cancer bill passed. Then he finally gets it passed in 1937, and they do the National Cancer Institute. Well, the government's been there and they're funding it along the way, but it's not until this time that you get what I call the explosion in funding takes place. So the war on cancer is not just a scientific program; it's a political commitment in this society to go after, we would think at this point, exposures to carcinogenics and the institutions that grew up around this really become powerful in defending their application of treatments for cancer for decades to come.
- 4.11 Now, why am I showing you Richard Nixon and Teddy Kennedy? Not just because they make a great pair, but the explosion in breast cancer hadn't yet really happened. They're not responding to breast cancer. Think of this. This is the late 1960s, 1970 - '71, we're coming around to. At this moment, there is a warning on the side of cigarettes. They said, by the way, cigarette smoking, and the exact language that they put for the very first one, smoking may [be hazardous to your health]. That was the first warning. It didn't really make a dent in smoking rates at the time. It didn't even cut doctors off from smoking.
- 4.12 You used to be able to go to a doctor in the early '70s, and the doctor would take a smoking break. Doctors weren't worried about it. Patients often thought, "Why should I worry about it?" At this period, as we're getting into '70, '71, the election is going to come up again in '72 for Richard Nixon's second term. What are the issues of the day? Issues of the day are not cancer. Issues of the day are the Vietnam War, the drug culture and hippies, the Black Panthers and the civil rights movement, Martin Luther

King, counterculture movement. Those are the things that, really, political activists are focused on. And that's what Nixon, as president from 1968, is focused on.

- 4.13** He's not even paying—even, by the way, you know when I said a moment ago that there's a fellow called Wilhelm Hueper. We'll talk about him later in a lecture. And he's the head of the environmental cancer center inside the government. He's a great pioneering guy. He's fantastic. He goes after so many different things, and they block him time and time again and everything else. But you know what? He was a smoker, and he actually did not think that smoking led to cancer, even after the surgeon general had put that on. So even the smartest and best of people can sometimes make those errors. So Nixon, at this point, is not really focusing on—there's a voter apathy around the issue of cancer. So the question is, why would Richard Nixon go ahead and launch the war on cancer? Why is he the president who ended up doing that? So something happens behind the scenes.
- 4.14** Lasker, Mary Lasker, has an ally in the Senate, the Texas Senator Ralph Yarborough. He is the person that is her closest ally in moving the cancer initiative forward and making it a priority inside the federal government. He's championing legislative bill after legislative bill, some of which get passed and some of which don't. And he convened an advisory panel of 26 members on cancer. The chairman was a Memorial Sloan Kettering chief. He figures you're going to people who deal with this. Sidney Farber was on. He was vice chair.
- 4.15** Of the 26 members, 24 members were affiliated with Memorial Sloan Kettering or the American Cancer Society. So you've got this as an industry group going forward. That's clearly—I understand, you're going to get advice on cancer inside the Senate, you're not going to pick people who know nothing about cancer, but it was tasked heavily with what I call the official part of the cancer industry. No one complained about any potential conflict of interest because those individuals are also receiving, in the hospital sense, grants from the government. So to the extent that they can say their work is important, it reemphasizes that.
- 4.16** And then the best laid plans get turned upside down by some other event. And in 1970, Ralph Yarborough lost his Senate seat to Lloyd Bentsen, who we might remember later was a vice presidential candidate. And that was a shock because Yarborough was supposed to walk in again, and Mary Lasker was afraid she'd lost her champion in the Senate. She didn't expect it. So Yarborough is gone, not sure about Lloyd Bentsen, whether he really might be able to do anything or not. And so, she has this 1970 cancer advisory panel that Yarborough had put together before he loses his race in [May] of that year.
- 4.17** Now, something happens. Teddy Kennedy drove off a bridge on Chappaquiddick Island in July of '69, where 28-year-old Mary Jo Kopechne ended up dying, and Teddy fled the scene, then turned himself in a day later, reported the accident to the police. A week later, he pled guilty—some of you might recall this—to a charge of "leaving the scene

of an accident that caused bodily injury," and he got a two-month suspended sentence. The fallout from Chappaquiddick would have ended most politicians' careers. But this was Teddy Kennedy, and his brother Bobby had just been killed the year before. Jack Kennedy had been killed six years before.

- 4.18** And although Teddy was criticized, although there was a torrent of bad news about that, believe it or not, there was also this groundswell of sympathy for the last surviving brother of the Kennedys, for having this tragedy take place. And he won reelection to the Senate the year after Chappaquiddick in an overwhelming number, so it didn't put him back in. Now, Mary Lasker. Teddy Kennedy is looking at running for president in '72. He's thinking, "If I can make one run, I might even be able to beat Nixon." And he's looking for an issue that might be able to propel him to the presidency.
- 4.19** So, Mary Lasker meets with Teddy Kennedy, and she sort of talks to him about her idea for this national cancer program, that we need to make it a federal priority at the very highest levels. And she was worried that if somebody doesn't pick it up, nobody's going to pick it up. Teddy comes out of that meeting with Lasker, and he really thinks, "This is the issue that I'm going to latch on to. I'm going to make this the issue to push forward as a national cancer bill."
- 4.20** He puts a bill in. He thinks it's going to set him apart from the rest of the Democratic candidates. And he puts a bill into Congress. You have to admire the breadth and ambition of the name of this bill. Teddy's bill is optimistically dubbed the Conquest of Cancer Act, and it proposes a federal system of unparalleled autonomy. It would remove the National Cancer Institute away from the National Institutes of Health, would set it up on its own. It would give it the most powerful medical research. It would give it a veto power, sometimes, over budget cuts. And that's the breadth of the bill.
- 4.21** Nixon, who had shown no interest as an issue in cancer, he's probably the only senior Republican, he's president at the time, who still is nervous about the Kennedys. He still thinks that Teddy Kennedy could mount a successful campaign, possibly in '72. Nixon's a man who lost the presidency by a hair's breadth to Jack Kennedy, back in 1960. Then he gets defeated in '62 for the governorship of California and says famously in a petulant news conference, "You won't have Dick Nixon to kick around anymore." And then he comes back and wins the presidency in '68. Amazing resurrection and political comeback.
- 4.22** But he's worried about Teddy Kennedy. So he decides, "I'm not going to let Kennedy take that issue, even though he's put the bill in." And Richard Nixon becomes the politician who champions the war on cancer. It happens because of that unusual process where everybody says to Nixon, "Don't worry, Teddy's finished, Chappaquiddick," and Nixon doesn't believe it. So Nixon becomes the champion for the cancer bill. "Time has come in America," says Nixon, "when the same kind of concentrated effort that split the atom and took man to the moon should be turned

toward conquering this dread disease." This becomes the language of president after president.

- 4.23 It's a bipartisan agreement. I know. In the days when Republicans and Democrats could actually agree to pass a bill in Congress. Hard to imagine. You'd think that was 100 years ago, but it was not that long ago. They were able to come together for this. The National Cancer Act gets passed. That's what it's called. It's still called, by the way, in the Congressional Record, the Conquest of Cancer Act. I love that. They didn't change it from Teddy Kennedy's original.
- 4.24 But there's a new cancer panel reporting directly to the president. Memorial Sloan Kettering's CEO became the first unofficial cancer czar. There is a second committee, the National Cancer Advisory Board, [NCAB], and that advisory board is empowered with all types of innovative tools to bypass budget restrictions and cutbacks. The chairman of [the President's Cancer Panel] by the way, was Occidental Petroleum's Armand Hammer. The board members [of the NCAB] include Laurance Rockefeller and Mary Lasker. She gets appointed to that board.
- 4.25 And it has some real power coming away. It's an expansive bill. The belief is, and you hear it from politician after politician in the aftermath of the signing of this bill, that if we could succeed in putting somebody on the moon just the year before, that by the turn of the century, at the very latest, 29 years away, we could conquer this dread disease that nobody had been able to do before. So, it's a level of confidence in the American government, and even in the medical community, because what the medical community realizes is, this opens up the spigots for research money. Big money now.
- 4.26 Now you have money coming in at a level where you can start to expand it. It's pretty amazing to think that a president with bad approval numbers and a divisive war overseas in Vietnam found it attractive to declare a war on cancer, but that's exactly what Richard Nixon did. So there's his quote. "Time has come in America when the same kind of concentrated effort that split the atom and took man to the moon should be turned toward conquering this dread disease."
- 4.27 And the feeling was, among doctors, if you see the people of Sloan Kettering and others, they believe that the infusion of federal money that's about to come in will kickstart the research to find a cure. We're not talking about better treatments; they really are thinking in terms of a cure. So late '60s, public is anxious about rising cancer deaths. Civil rights, environmental movements are in full swing. Getting the EPA, you're getting other movements coming in, competing for the attention of politicians.
- 4.28 Mary Lasker and her allies lobby Congress for a national cancer offensive. They capitalize on that public concern, at least with Ralph Yarborough, the senator from Texas, to think about launching a war on cancer, although he doesn't call it that. In '71, he seizes opportunity here in a war on cancer to balance the news from deteriorating news for the war on Vietnam. And the National Cancer Act is passed in '71, elevates the

National Cancer Institute within the NIH, and creates direct reporting to the president. And then, Benno Schmidt, a New York investment banker, close to Nixon, becomes chair of the cancer panel.

- 4.29 So, you have a merger taking place here that's interesting. The merger is that, in the creation of this large, what I call fatigue from the Vietnam War, civil rights unrest, and that, cancer is a safe area to be behind. Politicians on both sides of the aisle can get behind the war on cancer. Nobody's on the other side of it and saying, "By the way, I want to give up, and let's forget about it." So this is a unified effort. And really, the reason it is important for Schmidt and others like him to come in, they need to bring in the business community.
- 4.30 So it's not just doctors all the time; it's people who also have a business sense of how to make this into what Mary Lasker is asking for, which is a second Manhattan Project. The Manhattan Project was the project of World War II to create the atomic bomb. That's really a top-secret project, all government effort in it. She's asking them to essentially recreate that. Why don't you try to do that again? And I guess, you look back at this, I look at this all the time, in '71, and I think that this was probably one of the few wins for Nixon, coming up close to the '72 election, because everybody had a good feeling about what he had been able to do on this.
- 4.31 The war metaphor. Now, this is a problem that I have with it increasingly, as we get away from this period. So, war metaphor shapes a narrative for you. You go to war to win the war. You're not winning the war in Vietnam, granted. And later we find out that there can be quagmires in Iraq and Afghanistan. So we understand that war is not so clear. But America, for the most part, after World War II, thought that you could have a clean-cut victory in a war. It could be at the end, and the enemy might actually unconditionally surrender, and that would be the end of it. You might be able to cure cancer.
- 4.32 So it takes off what I call— in the industry, we're fighting carcinogenics, and the EPA is trying to control different chemicals being used in industry or food additives or pesticides that could be causing cancers—it takes the emphasis away from that and puts it all into the war. The war is being fought not on the items that might be causing cancer; it's being fought on cancer itself. So that war is being conducted there. And if you talk about something else, you're taking the eye off the goal.
- 4.33 It also implies an eventual victory. At some point, cure becomes the element, not just remission, not just living five years. It is a political pressure at times to say, "We're spending all this money. Are things getting better?" And that's when I said to you before, it's not the manipulation of statistics, but it's an emphasis on statistics that show improvement in treatments, longer living periods, longer remissions, a longer time away, less side effects. Those are the markers you're using.

- 4.34 Metaphors influence who directs the war and who benefits from it. This war framing, to me, channeled money into the high-tech treatment and hospital-centered research that was already there, but it makes sure that that is the status quo going forward. Other environmental causes, you're looking for the cause of cancer, trying to prevent it—that gets underfunded time and time again. And metaphors sort of define what's left undefended, and that really becomes the carcinogens. Prevention was never a high priority for the government. It's really fallen off the radar at this point.
- 4.35 This war on cancer really is a political movement that, for me, accepts that you may get exposure to carcinogens, that that's the exchange for a treatment system that's going to cure you of the disease, in case you get cancer. So we're going to worry about all the rest of that around the time. So these institutions that grew up around this bargain, they have reasons to defend it, and they do going forward. The rest of them remain sort of underfunded.
- 4.36 So, in 1979, eight years after Nixon launches the war on cancer, Arthur Upton, the director of the National Cancer Institute, establishes a program designating 20 comprehensive cancer centers. Now, those comprehensive cancer centers— and we have them in Miami, you may have them in different cities, you've heard about them— those units are the places where people go to do research. Also, they treat patients.
- 4.37 But once something became a comprehensive cancer center, if you were a researcher at an academic institution, or inside a drug company or that, you often wanted to be affiliated with one of those comprehensive cancer centers because they were also the ones receiving the biggest funding from the federal government. So it's a catch-22. They feed each other. The comprehensive cancer centers, by its very name, means we're treating and we're going after and doing the research on the disease like no one else is. That attracts the best talent. And then it's a reinforcing principle. They become the place, more than anything else, where everybody's going.
- 4.38 He's a pathologist, by the way, Upton, a radiation biologist, believe it or not. And he required cancer wards, these dedicated wards to cancer and developing these oncology departments. So you want to be a comprehensive cancer center, you can't just raise your hand. You actually have to have a section set aside just for oncology. You're just doing that treatment and wards just for cancer. So you're now defining what a comprehensive cancer center is.
- 4.39 And also in '79, they become the efficient machines for conducting cancer trials. You have a new drug, a new chemotherapy. If Krebiozen had ever been tested, you'd go to one of these comprehensive cancer centers, and that's where the work is going to come in. Now, you will not be surprised to find out that conflicts of interest abound inside these comprehensive, these prestigious institutions.
- 4.40 On the boards of these premier cancer clinics were drug industry executives, directors of the largest cancer nonprofits. They sometimes were the ones who recommended

treatment protocols for patients. They were the ones who had investments sometimes, or received substantial donations, and then made recommendations for a new product. Did it mean that they were always serving their own self-interest at the expense of patients? No, but by staffing it with a group of what I call insiders, you guarantee that you were getting groupthink, and that groupthink was going to continue to play out with money from the federal government.

- 4.41 Now, there's something interesting about this, but that Nixon declares this war, you have Jimmy Carter come in as president. It's not a big issue on cancer. But from Ronald Reagan and on, every president has emphasized the war on cancer and made it sort of a highlight of the presidency. So Reagan, the personal battles of modern US presidents against cancer.
- 4.42 When you realize how cancer affected them, in the beginning of this, I told you how my wife had been affected with it five years ago, so it makes it very personal. Then some of you have, either yourself or you have friends or family that you know who have had cancer. When presidents have had it as well, it brings it home to them. And as a result, they have the ability to control how much money is spent on cancer and what happens with the war on cancer. So it becomes a very personal issue sometimes.
- 4.43 Betty Ford's breast diagnosis, we'll talk about this later. Only a month after Gerald Ford became president, when Richard Nixon resigned the presidency because of the Watergate scandal, and Ford, this congressman who had been chosen, sort of as the fill-in for vice president, then becomes president, a month after, Betty Ford is diagnosed with that. And she was one of the first women that talked about her diagnosis. It was remarkable. It was a time when people really didn't talk about it publicly, and she did. It framed Ford's thinking for how the government could be a catalyst to raise awareness about the disease for people.
- 4.44 The Reagan family, '85, a lot of people forget this. So Reagan is President. You put this in from '80 to '88. In '85, a lot of people forget that he underwent surgery for colon cancer. Today, if a president went in for colon cancer surgery, there'd be all types of press. Will he or she, will they survive? What will happen? Are they able to carry on the presidency? There wasn't as much of that then. Reagan was not a youngster. But in '85, he has his own cancer scare. In '87, Nancy Reagan, diagnosed with breast cancer while he's still in office.
- 4.45 And the Reagans' personal battle, no doubt about it, if you look at their public statements and what Reagan said, that war on cancer and the funding that goes to the cancer institute during the time that he's president, it continues unabated. Reagan greenlights combination chemo. Now, it's not a result of him being sick, but he's the President who decides to do that.
- 4.46 So, 1984, a year before he has colon cancer, the National Cancer Institute established this policy that was the need for collaboration between the NCI and pharmaceutical

industry to develop anti-cancer drugs. Now, the drug industry loved this because it was a new revenue stream. They hadn't really focused heavily on doing drugs, specifically targeting together with the National Cancer Institute.

- 4.47 They had been running a parallel effort. There would sometimes be chemo products and others developed within drug companies. But for the most part, they were treating chronic diseases. They're worried about blood pressure, diabetes, everything else. They have an occasional addictive pill, which is doing very well. They were worried about selling Valium and Librium and then later Xanax. And now the NCI comes in and says, "By the way, hey, you guys over there in pharmaceutical industry, you want to join us in looking for developing new anti-cancer drugs?" And everybody's raising their hands. So that's '84.
- 4.48 Focus on combination chemotherapy. Following that announcement, comprehensive cancer centers, which had been established back in '79, they begin to focus on developing and testing these combination chemotherapies, mixing two or more FDA-approved chemo agents in the hope that they're going to find a more effective agent. Now, that goes all the way back to the 1950s, when they were testing it without the stamp of approval from the National Cancer Institute. They were mixing chemo agents then in the hope that it might be better, but they were much more rudimentary versions of chemo. So now they're looking to see if they might be able to fight more cancers and somehow, unlikely, if you mix chemos together, will you reduce toxic side effects? Potential benefits?
- 4.49 So, pharma companies, and you can see this in the memos when they're writing about this after the National Cancer Institute gets a hold of them, they are absolutely interested, I should just say, fascinated might be too strong a word, at the possibility of developing a new revenue stream. That revenue stream will be combination chemo drugs. Some of them have patents on it and some don't. But for any combination protocol that remotely could work, it extended the patent that was expiring on an existing chemotherapy drug.
- 4.50 So, how great. You have a patent on a chemotherapy drug. If you look at this in a cold calculus, as drug companies often do about extending the patents, they game the system, they change the molecules, they change the therapeutic delivery, and they get the patent extended. Here, you're about to lose the patent on a chemotherapy. If you combine it with another chemo agent, it becomes a new drug. You get a patent on that, the price can stay high.
- 4.51 Now, the reason it seems so terrible here is because we're talking about cancer and not just extending an antibiotic. And so you think, "Oh, the disease is worse. They shouldn't be doing it," but that's the DNA of the drug industry, always looking for a way how they might be able to push forward on a patent. And despite the focus on that combination, the vast majority, and we're talking an astonishing 98% of the combined chemos from

comprehensive cancer centers are found to offer no trial benefits over existing treatments.

- 4.52** Why is that not a surprise? Oh, my God, I'm sorry to throw my hands up in exasperation, but the drug industry had gone through this before when they did fixed combinations of antibiotics, and the idea, hey, we'll combine antibiotics, two better than one, three better than two, and there's no benefit to it, just more side effects. Here, 98% of these, nothing. And in many cases, they cause terrible and, really, sometimes lethal side effects for cancer patients. So, this greenlighting of the combination chemo, not the greatest move in the cancer industry's background.
- 4.53** George Bush comes in office. In '88, George Bush and his wife, Barbara, they had lost their [three]-year-old daughter Pauline Robinson, Robin Bush, to leukemia in 1953. We talk about Bush the elder, as I call him, and we forget that they had lost a daughter in the family to leukemia, one of those at the time that Sidney Farber is doing his work on leukemia and it is killing children, Bushes had lost that. And that sort of shaped their lives. They had this commitment about cancer that stayed with them. And through his four years in office, the same thing happens in this recommitment to the National Cancer Institute and making sure that money is readily available for research.
- 4.54** The Clintons' personal struggle. So, cancer killed Bill Clinton's mother. They lost his mother at that time, and I think that it shaped his commitment to it. It was always this overhang that he had. And while he was there, Vice President Gore had said this famous line, while they're in office for eight years, the "we", meaning the Clintons, the Gores, new generation in office, "[We want to be the first generation that finally wins that war.]" It's a pretty high goal. Once and for all, this generation will defeat what others have. I think that the Clintons' personal struggle, it helped focus them at times on the disease, but they also drew from those experiences to launch initiatives, together with Gore watching them, on what they really thought might be the end of cancer if they spent enough money on the research.
- 4.55** Barack Obama, his mother died of ovarian cancer. He launched a cancer moonshot in his last State of the Union address, and he tapped Vice President Biden to run mission control. So his initiative was to accelerate cancer research and treatment with the ambitious goal of making a decade's worth of funding, research, and progress happen in just five years.
- 4.56** Well, okay. So what does that mean in terms of the bureaucracy in the government? You know this. They want to get 10 years of progress done in five years. That means they're going to spend 10 years of money in five years. That at least is going to happen. So everybody was lining up for that extra money. But again, we're back to that idea of the war on cancer, and now it's been turned into a cancer moonshot, which gives you the idea that it's sort of like landing somebody on the moon. You might be able to pull it off. It reinforces that idea. It's not just one president saying this. It's a succession of presidents over time.

- 4.57 Biden's personal commitment. Look, he lost his adult son to an aggressive brain cancer in 2015. So I get it. Cancer was personal for him. He pledged in his State of the Union address in, [2022] when he comes in, his very first State of the Union address, that he's going to slash the cancer death rate by half in the next 25 years.
- 4.58 Now, it's one of those promises we've had before, similar promises. You can't make it. It's guesswork. You don't know. Maybe if you'd come up with some tremendous new treatment or you're on the verge of a cure, you'll slash the cancer death rate by half in five years or 10 years, or maybe you'll make no progress, and it'll be just as high or worse 25 years down the road. But it's good for politics. He means it. We're going to cut it in 25 years in half. And so, people listen to that, and that means additional funding as well.
- 4.59 And that's September, on the 60th anniversary of Jack Kennedy's moonshot, when Jack Kennedy says, "We're going to send somebody to the moon," Biden promises, "We're going to do a supercharged cancer moonshot." That idea that all that's holding us back is commitment; all that's holding us back is more federal money; all of these efforts just need a little bit more goosing from the federal government, and we'll be there soon.
- 4.60 So, building the apparatus, cancer centers, grants, and fiefdoms. National cancer centers as flagships. These national cancer centers are academic flagships. They develop and demonstrate the most advanced treatments and diagnostics, no question about it. Big construction and operating grants. Don't forget about that. Building a national cancer center, you can get write-offs and all types of accelerated deductions for the construction, but you can also get grants from the federal government to help to build it, in addition to the philanthropists who want their name on the front of it. And these centers are the magnets for clinicians and scientists, so they form this exclusive partnership.
- 4.61 The NCI is a federation of fiefdoms, as I call it. They have divisions inside the National Cancer Institute for cancer biology, treatment, control, cause, prevention. Those are smaller, each with its own budget items. Each has a senior staff. Each is a bureaucracy fighting within the bureaucracy for a bit more attention, internal competition, and to protect its own turf. That happens without a doubt.
- 4.62 Where the money goes. Large shares go to research, biology, imaging, and lab infrastructure. No question about it. Clinical trials, just a step down, and then cooperative groups. Much smaller shares, and I mean much smaller shares, go to environmental epidemiology. The long-term carcinogenic bioassays and support for regulatory agencies, they're at the bottom of the list. Everyone's looking for results in the treatment end.
- 4.63 Muckraker critiques, as I like to call them, some industry critics calculate that, in the mid '70s, only about 18% of the NCI budget was for cancer causes and prevention. That's about less than 2% of what's funded for the entire bioassay program. The

National Cancer Program at that time had evolved into a vast treatment enterprise, as opposed to looking for those marginal causes for prevention.

- 4.64 Now, when I cite that statistic, or somebody who's an oncologist or involved in research, and they're on that end, they say to me, "It's changed. It's much better. It's no longer just 18%, it's gone up to 20%." And then one year, it had reached 30%, which is pretty fantastic. But still, 70–80% of the budget just goes on that end of treatment and ignores potential causes. Again, my problem is that we're dealing with a disease; it would also be nice to know if, beyond smoking, there are other things that we should avoid that could be causing cancer. We'll talk about that in a lecture tomorrow.
- 4.65 So, here's the NCI growth. That's the building, the National Cancer Institute, in 1971. Got about \$223 million when Nixon declared a war on cancer. That's pretty good. That's it a few years ago. It's expanded a little bit. It looks a little bit bigger. It's got branch offices in other places. By 1978, the year before we do comprehensive cancer centers, the budget had grown from \$223 million to about \$872 million.
- 4.66 And the improvement in the death rate from cancer or the incidence of cancer, the number of cancers that were coming up in the first eight years of the war on cancer, zero. There was no benefit at all. So the war on cancer in the first eight years doesn't make a dent at all, but we're now spending four times the money. And it grows to about 30% eventually of the National Institutes of Health total budget, an extraordinary share for a single disease. In 2025, the budget was \$7.2 billion. And in 2027 fiscal year, which kicks in this coming year, they, the National Cancer Institute, are asking for this large increase to \$11.4 billion.
- 4.67 Now, you say to yourself, and I know this, we have become jaded about money figures. We hear about possible fraud in a state that's \$9 billion, or you hear about \$100 billion in waste and fraud in the government, or you hear about our wars in Iraq and Afghanistan costing over a trillion dollars. So you think \$11 billion is not that much money. But I can assure you, it's a fair amount of money. I often find that people tend to be jaded today about money, and they think, "Gee, I expected that they were going to be receiving \$500 billion." Not the case. They're asking for \$11 billion on the next go and making marginal improvements in fighting cancer.
- 4.68 So the enduring battle against cancer, what I call the unvarnished truths about this. Survival, spin, and statistics. So childhood leukemia, five-year survival rates for selected cancers. So today, 80% of the children diagnosed with leukemia survive for five years. Pretty remarkable, at a time in the 1950s when it was a death sentence for them. So if you look at that, you say, okay, that's progress. Hodgkin's [lymphoma], [89%] survival rate at five years. Breast cancer, [92%] survival rate at five years. Lung cancer, the biggest killer, is [30%], and [14%] for pancreatic cancer.
- 4.69 By the way, on lung cancer, one of the things that's interesting is that there was a study a few years ago in which people who were diagnosed with lung cancer were asked,

when they were undergoing treatment, "Did you give up smoking?" Because those who were smokers, and a third of those who were smokers who got diagnosed with lung cancer said, "No, I'm still smoking." And the survey said, "Why?"

- 4.70 And the answer was pretty much, I'm going to boil it down to paraphrase it, "Well, the worst thing that could happen to me from being a regular smoker is that I could get lung cancer. And now that I've gotten lung cancer, well, that's happened. Why should I give up one thing that I'm enjoying?" So, that's pretty remarkable. So, that's the power of the tobacco industry. They have patients and clients, I should say, sometimes that, even in the face of the worst possible outcome of what might happen, still hold on to their habit.
- 4.71 The war on cancer, unmet promises. A national crusade, Herculean effort, I get it, it's going to split the atom, took men to the moon, we're going to make a difference on this. Big investment of resources, a quarter trillion dollars into cancer research since the mid '80s. Now the figure is getting a little bit better. No longer just \$11 billion, a quarter of a trillion dollars into cancer research. Not all of it federal government, some state governments, some of it coming from private investment, some coming from pharmaceutical companies, quarter trillion dollars.
- 4.72 And what's the reduction in mortality rates? The current [crude] mortality rate for all cancers in America is a third higher than when the government started collecting data in 1930. There have been significant reductions on some cancers, but doctors explain that by saying the diagnosis for cancer is being found earlier—imaging. We have more cancer cases. And so, therefore, there were people who were dying from cancer in the 1950s and 1960s that didn't know they had cancer, or it was never talked about. They weren't listed as cancer mortality.
- 4.73 But again, you follow this figure and you see this decline without any doubt. Look at this, cancer mortality trends, tracking the rise and fall of smoking. It's a bit of what we talked about before. Some tremendous gains in a few cancers, some rare cancers, some leukemias, Wilms tumor, for those of you who have heard about it. But the overall progress, the five-year survival rate is just budging a little bit.
- 4.74 The 1930s, cancer mortality rates begin increasing. They had already become, in the turn of the 20th century, I talked about the fact that cancer is the second biggest cause of death after heart disease. So we knew it was a problem. But they start rising in a way that you can really mark it in the 1930s before World War II comes around.
- 4.75 In the '50s and '60s, smoking reaches its peak. Now, there's no doubt there were still a lot of smokers in the late '60s and early '70s, but the numbers start to come down slowly. That rise in cancer rates largely tied to the increase in smoking. Can you make a one-for-one comparison? No, but you'll see the two. It's very interesting if you track them along. As the country is smoking more, the cancer rates are also going up.

- 4.76 '60s to '80s, smoking prevalence starts to decline. Really, by 1980, you see a fall-off. The health campaigns, the public service announcements, the constant barrage of information, the information that you shouldn't smoke while you're pregnant, that you shouldn't smoke in the house if you have young children because there's secondhand smoke—it starts to have an effect, and people are doing less and less in terms of smoking. 1980 to the present, cancer mortality rates declined, tracking the fall of smoking.
- 4.77 So when I tell you that you can follow the rise of smoking and the rise of the mortality rates, and then you watch the fall-off in the mortality rates, you find this very interesting. It's almost the same type of tracking for smoking itself. Smoking has a major influence on this. But I have to say that I understand why the industry will claim an overall credit in the war on cancer. They'll say, "We have a declining mortality rate because we are spending billions of dollars on fighting this, and we're coming up, and we're closer to a cure." Not necessarily the case. If you were able to take smoking out of the equation, the improvement in the survival rate would be much smaller than that which is often boasted about.
- 4.78 So, prevention, environment, and patients. Prevention efforts marginalized, we know that. Conflicts of interest, they're there. And all of those individuals who are present on those comprehensive cancer centers would absolutely defend vigorously that there's no conflict of interest. But you see the potential conflicts of interest, patients as beneficiaries and sometimes collateral damage, and the environmental carcinogens that are neglected.
- 4.79 Now, global comparison. Cancer death rates per 100,000 for a population. So Japan, 116.9 per 100,000 is the cancer rate. France, 124.4 per 100,000. Canada, just a little bit above that, at 125.7. United Kingdom at 133.1, and the US at 152.7. Now, why is that of interest at all? Because the United States spends three to four times more a year on cancer care, prevention, and research than those other countries. In some cases, 10 times more.
- 4.80 So they're spending less money on the disease. They don't have the war on cancer. They don't have a moonshot for cancer. They don't have the equivalent of the comprehensive cancer centers, yet they have a lower cancer death rate than we do. So, what you would hope for in the US is that we were reducing the death rate. If I looked at that chart—and there are a lot of reasons for the disparity. I get it. It's not underreporting.
- 4.81 By the way, I'm talking about United Kingdom, Canada, France, and Japan. So I'm not talking about countries that don't keep health statistics. I'm not going to some place in which you say, "Oh, the cancer death rate is low because they don't count the numbers correctly." These are places that actually do count these numbers correctly, and they know what the spending is on it. So what should I expect? I should expect to see the US

death rate lower than all those other countries, or at least the equivalent of it. I want to feel as though there's some progress.

4.82 Now, if you go to the American Cancer Society or someone else, and they're doing good work on public information and everything else, I get that. But they don't necessarily tell you that right off the bat, because then the question is, well, how come the other countries have that lower death rate, and they're not spending as much money? Here's how I view this. We have the war on cancer significantly increasing during this time from '71 and on. Through successive presidents, they're building a network of cancer centers in the cancer research, what I call, ecosystem. So that exists.

4.83 What Rhoads thought of when he was at Memorial, and he was an evangelist for chemotherapy, I don't know what his vision was like for the future, but he would have thought this was fantastic, because this was multiple units, all with the stamp of approval from the federal government chasing the same goal.

4.84 The governance and board composition favors, and you won't be surprised with this time and time again, treatment over prevention. The advisory panels overseeing the war on cancer, they're dominated by the very forces that focus on the clinical industries, the drug industry, the industries for scanning, the industries for machines. So they're focused on, how do we detect cancer early? How do we treat it once you're diagnosed with it? And then you become a surviving patient in five years. They're not looking for what might have led to that in almost any instance.

4.85 Survival statistics, used to overstate progress. Now, I think that, in some ways, the impression of great progress against major cancers is, and I'll be careful about saying overstated, because I don't mean to say that there haven't been improvements. My wife is an example that there are improvements. She's alive and well, and cancer is in her rearview mirror. But what I do mean is this, and we'll find this in another lecture.

4.86 The diagnosis of cancer, in many instances, especially breast cancer, has become an earlier event because of imaging. Sometimes it's not that somebody lives longer, their life is extended, they live longer with cancer because it's detected in an earlier stage. So there are these five stages they've developed over time of cancer. Stage one, they have now even a stage zero, which is sort of the very, very beginnings of what might eventually become cancerous.

4.87 And if you're diagnosed even at stage zero, and you eventually are alive five years later, you are a successful cancer survivor and patient. So the five-year number comes in parts, not from better and more revolutionary treatments, as it does also come from the fact that cancers are diagnosed in an earlier time, before they have become lethal. So you aren't dying within that five-year period. And that's why the five-year stat is used repeatedly to establish the point, we're making progress in the war on cancer.

4.88 And then I say this again. I'm going to have a lecture about it afterwards. We're going to talk about environmental and occupational causes and why they are underfunded.

That's not just by accident. By the way, the industries that have those environmental and occupational causes, they're fighting nonstop to make sure that the spotlight is not on them, because they don't want to be sued. You want to know an industry that ended up in the docket and spent billions of dollars? The asbestos industry. That's the example used by everybody, so that's time and time again.

- 4.89 In a war, I sort of think this treatment as too much cancer, the war on [cancer], isn't just a scientific program, but it comes to this metaphor, in which you set the public's mind on the idea that they're going to live to see cancer eliminated, and that's not the goal of most of these researchers, and they'll admit that.
- 4.90 In the next lecture, we're going to explore some examples of how the cancer rate might be affected by exposure to carcinogens in the workplace, like asbestos, vinyl chloride, pesticides, and hormones, and why regulators, industry scientists, and public interest groups fought over each other, fought those cases, and allowed the cancer industry to focus just on treatment. Thank you very much for your time.
- 4.91 How do you think it is that cancer became so politicized with the different people who were—there was a bunch of different people that were working on the research. How did it all become one system, a bureaucracy of people that had the power to say it?
- 4.92 It's a great question. How did it become a single system, this monolithic system? And that's because, although they were in different fields, so for instance, you have the pharmaceutical industry, you have researchers in science, you have people working on chemotherapy, you have surgeons developing new surgical techniques, they also are all allies. So they haven't signed a piece of paper that says, "We're working together in concert to achieve this goal," but they're each supporting each other's work, if that makes sense. They're part of what's the treatment protocol.
- 4.93 And so, they've joined forces, in essence, to lobby Congress and to lobby each president for more funding. And they're fighting over that sometimes, about research funding. There's no doubt about that. But in the most part, they're at least on the same side of the mission, which is to say, "We're going to be treating this disease."
- 4.94 And I will tell you that there are some, and we'll see in another lecture, there are some scientific changes in terms of biologics and what's on the future, what might happen with AI, which might finally turn this antiquated, older system of just surgery, radiation, and poison on its head. It's possible. There are some ideas on the drawing board which are really unique, but I think that, for the most part, they've gained strength by being allies together on this.
- 4.95 Hi. Going back to speaking about Marie Curie and everybody else that was in or had been exposed to radiation, and when she passed away, they buried her in a wood coffin, then eventually put her in lead in, like, two levels, double-leading the coffin—is there any way of eradicating the radiation, for example, like a chemical solution of sorts? Or if she was cremated, would that get rid of that?

- 4.96 That's a great question. I don't know the answer to cremation. Now, you've given me something I'm going to check on overnight. I'm not sure if cremating her body would have eliminated it. I tend to think no, but I'm not sure. But there's no chemical solution that they have discovered that—you can't embalm her or something, and then it'll make it better. At Chernobyl, the site of the accident and the meltdown, the ground there is still contaminated in large parts. They were getting radiation readings at Hiroshima and Nagasaki, the cities that had the atomic bombs dropped on them in Japan, years afterwards.
- 4.97 But the amazing thing about radiation, and you'll get this, there were nearly 100,000 people killed in Nagasaki and in Hiroshima as well when the bombs were dropped. But there are some people, and some people survived and were terribly burned. And then other people who were not that far away from the blast zone, who you would think would be dead in a short amount of time, lived for decades. And so, doctors are still a little confused.
- 4.98 When the soldiers were watching, in the Manhattan Project out in the desert, the first explosions, they were watching through these cutouts, essentially, in what were bunkers without any other protection except for sunglasses, and they were all exposed, and some of them developed cancers, and some did not. So the part that doctors still have trouble understanding completely is why that exposure to radiation will be fatal for some, and the body processes it in a way that starts a cycle of cancer cells that then kill the person, and in other cases, it doesn't. But I will check on the cremation. I don't know if I'll have an answer, but I'll try.
- 4.99 The radiation therapies today, are they offshoots of these original tests with radium?
- 4.100 It's amazing how radiation has advanced in terms of companies producing machines that produce less radiation, more pinpoint radiation, but still the risk of radiation treatment is still there. Trisha, who ended up having surgery, a lumpectomy rather than a radical mastectomy, which, by the way, we had a doctor who was willing to listen to us as well. And so, we raised the possibility of whether a lumpectomy, also an excision of the tumor, might be just as effective as a radical mastectomy.
- 4.101 Doctors prefer doing the radical mastectomy. Why? Because it's more foolproof for them. They're sure they're removing the cancer there in the breast. It's a more mutilating surgery. Then there's also an entire industry that is built up around reconstruction for women after that. And so, that's a simpler procedure for them. The lumpectomy, they have to worry if they've gotten all of the tumor, if they haven't. Trisha knew that radiation treatment, although she wasn't a candidate for chemo, would be part of her treatment.
- 4.102 It turned out that, after they do the original surgery, they send out the tissue to a lab for a pathologist to look at the lab, to see if there are any cells at the margins, if they left anything behind. They said, "You're clear," to her. Then about a week later, a

pathologist inside the University of Miami lab ran a different test on the very same tissue, that normally isn't run by them, with a different solution, and said, "I have some material here that I'm not sure what that is. I'm not sure if that's actually cancer or it's not. It's just some item that we should look at."

- 4.103** So we had that specimen then sent to Johns Hopkins, and said, "What do you think?" Hopkins says, "We really can't tell." And at that moment, you're finished. What I mean by finished is, once they find something that they're not sure is or isn't, you've gone through this entire process of having the surgery, and then the doctor says, "Well, we should go back in and make sure that it's really fine," very few people are going to say, "Forget about it. I'll take my chances that what you found isn't."
- 4.104** And so, she went through a second procedure, fine, and then went for radiation. So, on radiation, it turns out, her cancer is on the left side. Only then did we learn that you had to learn to breathe in a way when the radiation is being given in these little things, so that your heart won't have any long-term effect. It's possible. Otherwise, the radiation of 2021 still runs that risk, even though it's targeted and narrow and everything else.
- 4.105** So you think about that, and you know, I'll tell you, people, I would go over with her. This was during COVID, and then at the end, when she was doing radiation treatment, during the surgery, I had to stay out in the parking lot and call in and find out how she was doing. But during the radiation treatment, I would go over there. And one of the things that she first did in this period of COVID, she went inside the room. They would set you up on this table. She would tell me later, she's the only person in the room, because then the clinicians would leave to go on the other side of this enormous thick glass. No one wants to be there while it's operating.
- 4.106** Then they said to her, "Keep your mask on for COVID." She said, "You've got to be kidding. I'm the only person in the room. I'm about to get radiation. Why should I keep my mask on?" They said, "That's the protocol." She refused. She said, "Then I'm not going to do it." So they let her take her mask off. Sometimes you have to be a patient who's a little bit of a pain. On radiation, there's something very interesting. Last thing, sorry for the long answer.
- 4.107** In doing our research, in doing my research and the deep dive, I found out that they can give 12 to 14 sessions. It depends on how many times they give radiation for somebody. Then they have a thing called partial breast radiation, which is only five doses, higher intensity, but five times, lower overall radiation than the longer procedure of radiation. So the oncologist, the radiologist who first, Patricia, had said, "You're going to go through the longer procedure of radiation." We said, "Why not partial breast? Isn't she a perfect candidate for it?"
- 4.108** Small tumor, totally removed, caught early. You're using radiation to make sure there's no stray cancer cells left. You don't want to give her more radiation than you need. I

know you get paid sometimes by insurance for every radiation emanation given, but maybe in this case, you only want to get paid for five versus getting paid for 12. I didn't quite say it as rudely as that. But they agreed, finally, in a medical consensus and said, "Yeah, partial breast will be fine for her." So they did it.

- 4.109** But if we hadn't asked for it, if we hadn't raised it as an issue, and one of the things that happens is, one thing that has stayed the same, you say, has radiation changed? I'll tell you what hasn't changed is that, for the most part, people get diagnosed with cancer. They're nervous, they're not quite sure what it's like, and they do whatever the doctor tells them.
- 4.110** So if the doctor says, "X, Y, and Z, I've been an oncologist for the last 30 years, and let me tell you, I've treated a lot of patients like this. I know your case very well, and you have to do X, Y, and Z," very few people are willing to say, "I checked on this. I'm just not quite sure that's right. Maybe it was the right thing to do 30 years ago, but what about this new procedure?" Because they just don't want to challenge the authority.
- 4.111** Now, that happens in medicine all the time. But in cancer care, I think it's particularly the case, because you're dealing with potential life and death. It's one of those words. And so, therefore, we were active and engaged patients and patient spouse, much to the aggravation, I'm sure, of the doctors. Remember when I said before, that people thought Raymond Rife, he was an optical engineer, he wasn't a doctor, so you couldn't take him seriously? They couldn't even take Sidney Farber seriously because he was a pathologist.
- 4.112** Well, what do you think they think about a guy with no medical degree at all, who's never been to medical school and is coming in and telling them, "What about this?" So you can see the eyes rolling, but they do listen eventually. They will push back. They don't just roll over, but you can have a discussion with them as an informed patient. And I do think that's important in something like that because the machinery has not changed that much.

Lecture 5: Carcinogen Cover-Ups

At a glance — Carcinogen cover-ups.

- 5.1 Welcome back. Thank you for being here. As we move in, we continue on the politics of cancer. Now, you're still getting—some of you—used to me, but I'd like to go back to something else I talked about yesterday for one moment before starting the latest lecture.
- 5.2 I was talking about US presidents and how cancer had affected them personally and how, during Ronald Reagan's tenure, he goes in for colon cancer in 1985—but the year before, in 1984, he had approved Comprehensive Cancer Centers, this big growth of cancer research. And that's the point at which he had also—you have the cancer centers, but he was the one who said we can do combination chemotherapy.
- 5.3 I had forgotten until I came back last night and Trisha talked to me and she said, "On the combination chemotherapy, you did tell everyone that 98% of those trials turned out to be bad." No question about it. What I forgot to tell you is the evidence of the partnership you were asking yesterday about how all these groups get together and combine. So how does the pharmaceutical industry align so well with what's happening in cancer care?
- 5.4 In the year after the announcement was made—in one year—to do combination chemotherapies, there were 6,000 articles that appeared in medical journals written by doctors and pharma people who were encouraged to write them. It was a record number. So if you're reading medical journals, if you're part of that entire process, if you're in the drug industry, it looked as though there was a revolution. It wasn't just that a rule had been passed, but 6,000 came out.
- 5.5 The applications in the 12 months afterwards to the FDA to get approvals for combination chemotherapies were the highest rate of drug applications ever put into the FDA system in a 12-month period. So it's not just that it was thought of as a possibility. It exceeded everything that had been done beforehand on antibiotics, everything that had been done on anti-anxiety pills. There was a rush into this.
- 5.6 Now, not only was there a 98% failure rate—failure rate means just that there's nothing that comes that's a benefit from them—but the most infamous of the studies was what's called an eight-in-one study. It was talked about a lot afterwards. It is eight different chemotherapy drugs given to children who had cancers that appeared to be terminal. The idea was that they thought this combination of eight would do the difference.
- 5.7 Let me tell you what ends up happening. One in five needed an emergency blood transfusion. Fifteen percent suffered kidney damage. Some lost their hearing, others

developed life-threatening infections, and most of them died soon after the trial. All the eight-drug combination had done was to make their final days miserable.

- 5.8** Now, you would think that the pharmaceutical industry and the people doing the combination of chemotherapies would walk away in shame from this legacy. I'd forgotten—thank you, Trisha, for reminding me—how much afterwards was said in the medical press and in articles that were written, trumpeting the 2% that worked. So it's fantastic. And the 2% that worked, by the way—three months extra time, some of the combinations showed with colon cancer, interestingly enough. And a couple of the combination showed up to six months more with breast cancer. That's what you read about. You opened up the paper, you read about the combination chemotherapies that might do that difference.
- 5.9** And where was most of the testing taking place? At the number one Comprehensive Cancer Center in the United States, Sloan Kettering, where they end up with a board of science that directs them, time and time again, back to pharma. When they first became a Comprehensive Cancer Center, the first director there was a Pfizer director—a board director—who came over and helped Memorial Sloan Kettering. The vice president and the vice chair are Bristol-Myers Squibb directors. So there is this compatible symbiosis that works between the cancer industry and pharma, and they are chasing it here. But I'd forgotten how they spun that 98% failure rate to be a success on the 2%, and that's what you heard about time and time again.
- 5.10** So today, we're going to what I call corporate cover-ups and regulatory collapse. The agencies are supposed to protect us, and sometimes they ignore those environmental and other causes, possibly, of cancer. We've covered the brief history, the creation, the research, treatment, the crushing of the outsiders, and the limited progress on the national cancer centers. Now we're going to flip the perspective and look at what creates or sustains cancers in some ways in this hour, rather than talking about how we treat it.
- 5.11** There are three central themes. Corporations discover their products cause cancer. Let's say they do. And then they suppress or distort the evidence to protect profits. And the agencies that are meant to oversee this are lax. They don't do something or they're blocked at times. And regulatory capture—what I mean by that is, it's when a regulator ends up serving the interests of the companies that they're supposed to be regulating more than they are protecting the public.
- 5.12** And we're going to have three case studies in this: saccharin, vinyl chloride, and asbestos. They each have the same pattern. They show an early warning, some evidence of what's going wrong, and then you expect some more action from the regulatory side. And what the corporations do, instead of, of course, coming up and fessing and saying, "Oh, yeah, we fess up. You're right. Our product is really terrible"—they try to hire people to change the narrative. They go on the attack, and they delay,

as Big Tobacco did, because Big Tobacco could have been one of these examples. They delay as long as possible the day of reckoning.

- 5.13** So the opening question is, what do you think happens when a company realizes a profitable product causes cancer? You'll have your own view of that, and see if at the end of that—hold that thought—what you think happens. Maybe by the end of the hour, it will be better or worse than you expect it to be now. You'll find out and I'll ask you.
- 5.14** A government contrarian, Wilhelm Hueper, is a German immigrant who had a background at DuPont, and he worked at DuPont before joining the National Cancer Institute in. [1938] He's the one who is the founder of the Environmental Cancer Center. Now, Hueper, if you recall, I'd mentioned him earlier because he is also amazing. I'm going to tell you why I think he's ahead of his time in many ways. But he is a smoker who thinks that the concerns about smoking are overblown. Everybody has what I call an Achilles heel. This was his Achilles heel for whatever reason.
- 5.15** But here he is exposing some corporate negligence. When he was working at DuPont, he became convinced that there was a cluster of bladder cancer cases among workers, and he was convinced that it was a chemical that was being used in the process because those workers with higher bladder cancer were working with a particular set of chemicals at DuPont.
- 5.16** So he was warning about that inside the company. He wanted to publish his findings. He had a long list of comparison studies and everything else. And of course, DuPont says to him, "You must be kidding. You're not going to publish an article where you're working for DuPont about that chemical possibly causing cancer," and they stop him from doing it. And when he keeps going, they fire him. That's not surprising. All right, I understand.
- 5.17** So Hueper gets fired by DuPont when he wants to publish this study about what he thinks is bladder cancer linking to. Why does he then go to work for the US government? Because he assumes that if he's working for the US government in the National Cancer [Institute], and they're picking him to be the head of the Environmental Cancer Center, he can go after all of these issues that are bothering him, and there will be nobody blocking him. So this is the idea of what happens.
- 5.18** Now, government service for him is an open door to do more. So his experience at DuPont demonstrates the power of corporate interest to suppress information—I get that—even in the face of what was clear science. He later proves, by the way, nearly a decade later, exactly what he was thinking about: at DuPont, what was causing bladder cancer was in fact the element. They stopped using that chemical, but it takes 10 years for that to happen.
- 5.19** So the uranium mining hazard. In 1948, he becomes the head of the Environmental Cancer Division at the NCI. This is in the heart of the Cold War. It's the same time in

which we're in a race with the Soviet Union to try to build the biggest, greatest, and strongest atomic weapons. Each is building up its own atomic arsenal. So the Atomic Energy Commission also comes into being, and it announces—because we are in a race, as I call it, to build up these nuclear arsenals—it will pay a premium for uranium ore.

- 5.20** So uranium ore is selling at X dollars. The US government is going to pay X plus to get uranium ore. And most of the uranium ore at the time is being mined in the Congo, but the Congo is running a little bit low. So what becomes the area that is the most popular? Sort of this Four Corners area of Colorado, Utah, New Mexico, and Arizona. There's uranium ore there. And so mines are being built.
- 5.21** Now, Hueper has an issue. He had been studying before he came into government service German miners. And he was looking at German miners who had high cancer death rates: silver and uranium miners. They lived on average 20 years after they started working in the mines. That was all. So he believes that in the uranium mines, the cause of those deaths—the culprit—is suspected radon gas.
- 5.22** So what he says, as these labyrinth tunnels are being built and these mines are being built here in America, we've got to check on the radon gas levels inside these mines to see if they're safe or not. He does his own measurements at one point and finds radon gas levels higher than the German mines. And so now he's really alarmed about it.
- 5.23** And what happens? The Atomic Energy Commission and the US government, they're in the Cold War. They have to weigh the value of getting more uranium ore versus listening to Hueper and possibly slowing up the process and doing something inside the mines. So they reject plans to have a different ventilation system. They reject plans to go ahead and measure the radon gas. They go ahead and they reject plans to have what's an extraction system that will make radon gas less. And they block Hueper. He's surprised to find out that not just DuPont will block him, but in this case the Atomic Energy Commission comes in and they block him.
- 5.24** The Cold War politics put worker safety a notch below worker health. It took 50 years before a series of studies—published by the National Cancer Institute, as a matter of fact—[showed] that thousands of miners had prematurely died, and I will quote this from an occupational hazard. Well, I don't know how I would feel if I had a family member who had been working in those mines and later found out that the person who's running the Environmental Cancer Center had wanted to look into the danger of working in those mines and that the government said, "Don't do it."
- 5.25** In the '50s, he's warning about these high radon levels. He continues to do it in the uranium mines, causing lung cancer in miners. And the Atomic Energy Commission blocks him time and time again on these exposure standards.
- 5.26** Now, he's an unusual person because he's not just concerned about what's going on in those mines, but I'm going to give you a little list here of the things that are my

favorites for him. He's looking at a series of things that he thinks are environmental cancers.

- 5.27** So he notes that in South Asian cancers, there's literally a 25% to 30% mouth cancer. It's less than 1% in America and in Europe—among people in the entire European continent and in America. He's convinced that it may be because in Asian countries where that mouth cancer is very high, people are chewing tobacco leaves or eating betel nuts, these nuts that later have a carcinogenic product, and he thinks that's worth investigating. They listen to him, but not with a lot of effort in the beginning.
- 5.28** He also suspects that in Zambia, there's an abnormally high incidence of esophageal cancer. He thought it might be from a popular homemade alcoholic brew, which turns out later to have a carcinogenic effect, and that later becomes illegal in that country. In Guam, there's an epidemic of liver cancer. And it's Hueper who's saying—and he turns out to be correct—that the two local plants that are used extensively in traditional meals had fungal [carcinogens] that ended up contributing to that particular outbreak.
- 5.29** So he's looking for clusters of cancer different from what it is everywhere else, understanding that it can't just be because of where they live. It has to be something they are exposed to. And the environment is then tipping over the genes and the genetic parts of it. It took a decade before a series of epidemiological studies confirmed that the regions with abnormally high cancer clusters—between 70% and 90%—were tied to these obscure [carcinogens] that weren't in other parts of the world but were part of either their diet or what they were exposed to.
- 5.30** Now, what does he put on his list that he wants to look at? He wants to find out why workers who work with nickel and chromium have higher rates of lung cancer. He wants to know why workers who work with benzene have higher rates of leukemia. He wants to find out why people working with plastics have higher rates of liver cancer. Synthetic industrial chemicals and petroleum byproducts and insecticides are on his hit list.
- 5.31** Now, you think of him—he gets the idea of [carcinogens] and what might increase cancer rates—but he's blind to tobacco. And so it really is remarkable when you think of everything else. So I tell you about him on his own and what he's doing. You would think this is a guy who also has Big Tobacco on his list. They're the only ones that aren't on the list anywhere. Nowhere does he think that that's adding to it, and it becomes, of course, one of the biggest factors if you think about it. So that was his blind spot.
- 5.32** Now, he uncovers these. It really is Hueper's work in this period and his government role, even though he had been blocked on the uranium ore. He is the one who really helps to change some of the view inside the government a little bit, but basically among muckrakers, writers and others, that maybe we should be spending more attention looking not just at the efforts to control and treat the disease, but to find out what's causing it.

- 5.33 Obstacles. You know what? He faces corporate backlash and efforts by DuPont. I get that. Government interference with the Atomic Energy Commission coming in and blocking him on uranium miners. The burden of proof. Constantly, this becomes a problem. He has a suspicion that a high cancer rate might be linked to these particular [carcinogens], but maybe he's wrong. Maybe it's something else that's causing it. So there's a burden of proof problem here. You have the suspicion, you see the two things, but you have to find the link. You have to find the causal link.
- 5.34 The burden of proof is difficult. And it would be one thing if I could say to you, it's hard to establish that link. You have to look at people for a number of years in that. But all of these countries and all of these industries that he named, they got going on day one. They set up medical panels. They were looking at it. They had researchers, and it took them 10 years to find it. But they didn't do that. It took them 10 years to find it because they didn't start doing it right away. They delayed. So delay is the process you get into.
- 5.35 Lack of regulatory action. He is—in some ways he's not the regulator here. The regulator is different in each case depending on what the industry is. But there's certainly a lack of regulatory action. Very few are picking up on what he's saying and going ahead and cracking down on these industries.
- 5.36 And delayed recognition. Boy, this is part of the problem in this area, time and time again. The radon exposure and the lung cancer, for instance, in miners comes decades later, done by the National Cancer Institute. By then, those miners were no longer there. The industry was essentially gone, but it becomes a footnote in history, but not for the families who were affected by it.
- 5.37 So the bitter history—sorry for a very bad pun—of artificial sweeteners. And it's very interesting to see this because you think, "Okay, artificial sweeteners, cancer, maybe." So saccharin. Saccharin is one of my favorites because it's so old. It's 300 times sweeter than sugar, discovered back in [1878] by this Johns Hopkins chemistry professor. And he's actually searching for derivatives for coal tar uses. He forgets to wash his hands. I love these accidents. Then he has lunch and tastes something that's sweet. And that's how he ends up finding out that this chemical he's pulled out becomes saccharin.
- 5.38 And it becomes pretty popular at the turn of the century—the turn of the 20th century. By 1900, it's widely used. There is one person who absolutely is against it, Harvey Washington Wiley, who is the person chosen at the Bureau of Chemistry, who's leading the charge to get the 1906 Pure Food and Drug Act passed, which is going to force the labeling of what's inside all of these patent and fake medicines and snake oil salesmen.
- 5.39 When that law gets passed in 1906, Harvey Washington Wiley has the ability, if he considers something deadly or dangerous, to be able to take it off the market because he was also doing studies before the law with what he called the Poison Squad, which was volunteers who would eat meals that he prepared inside the basement of the Bureau of Chemistry with food additives and preservatives, including even

formaldehyde at one point. Two of the students [fell ill], and that stopped the experiments. But he found things that should be pulled off the market because they were so dangerous.

- 5.40 And he was convinced saccharin was one of them. The problem was that Teddy Roosevelt, the president, liked saccharin. And he said, "I don't want you to ban it." So here's an instance in which he says, "Don't stop it." So here's what happens. Harvey Washington Wiley comes in and tries to ban it as a poisonous and deleterious substance. He says it's going to be done. Roosevelt stops it.
- 5.41 Now, World War I starts—1914, '15—you're running through 1918. And what happens with saccharin is that an exception is allowed for use by those with special medical needs, and saccharin is rationed. So President Roosevelt had special medical needs, and others could find saccharin of use. And then after World War I, it returned to normal sales. So saccharin was being sold.
- 5.42 It actually ends up in 1969—you'll find this of interest. There had been an introduction of another artificial sweetener, cyclamates. They had been used in diet drinks and everything else. It was a competitor to saccharin. And then cyclamates get banned because they are considered dangerous and they might be causing cancer. And when cyclamates are banned, saccharin booms. So saccharin is there as a product.
- 5.43 And literally, overnight—20 million pounds a year of cyclamates had been used—saccharin, within weeks, is in Diet Pepsi, Sweet'N Low, Tab. I remember Tab. They've all switched to saccharin. So saccharin now becomes the mainstay artificial sweetener. It goes from a point where you think it's going to be banned after 1906 in the Pure Food and Drug Act, then it survives, then there are exceptions made for it in World War I, and then other competitors come in over time, and it's an also-ran, but then its day comes in 1969 because its biggest competitor is wiped out as not being safe.
- 5.44 So there are these concerns about saccharin. And there are calls from activists like Ralph Nader, who was a consumer activist: "We should investigate it again. We should look at it." They fall pretty flat. And there's Ralph Nader, an activist. Consumer groups are urging the FDA, who has the authority here, to investigate the safety of saccharin. And the FDA, of course—why, you're not surprised—says, "We have a lot of other things on our agenda. Forget about saccharin. We're not going to do anything with it."
- 5.45 And I think the approach, by the way, even of the regulatory agencies is pretty much, if we don't have to do it, we're not going to do it. And that's clearly the case here. And then you get what I call the Canadian study. So the Canadian study is in 1977. This Canadian study is released that makes major news and headlines everywhere, where essentially they say that they had tested on mice saccharin and it increased bladder, breast, uterine, and ovarian cancers.
- 5.46 Now, you would think that pretty much is what I call the height of everything. The FDA responds because there is this part of the law, and I'll explain that to you in a moment,

which says that if there's evidence that it caused cancer in any respect, you have to be able to take it off.

- 5.47** And the food and chemical industries—who was the biggest saccharin manufacturer at the time? Monsanto, by the way, had started their business on saccharin. That's how they got going originally. But the biggest maker now was Sherwin-Williams. These are industrial giants. This is what I call Big Food. And they launched a full-pronged assault against the idea of this Canadian study.
- 5.48** So the Canadian study says: increased incidence of [bladder] cancer, rats fed high lifetime diets of saccharin, whose parents had been fed high lifetime amounts of saccharin. Yes, mice and rats do have parents. So two generations of mice and rats in terms of the experiments—that's what they were saying. Part of the problem was that it turns out, you will not be surprised, that on that study, Sherwin-Williams and others hit gold when they realized that the Canadian study had used an amount of saccharin that would mean that essentially you were having 200 cans of soda a day, maybe 3,500 cans a month of soda, a level that was so high that they started to ridicule the Canadian study.
- 5.49** And what happens is it even became the butt of late-night jokes—Johnny Carson, late-night hosts on shows, would talk about how ridiculous the Canadian study had been and what it had proved. What that does, of course—you have these incredibly high levels of saccharin given to mice, and it's causing cancer tumors. The question is, if you were given normal [levels], or what people might actually be using, will you still get an incidence? Would it be less? You don't know that. So the Canadian study had that, and that allowed the food industry to go after it.
- 5.50** So saccharin and the Delaney Clause, and the retreat from zero cancer risk. So there is this clause in 1958. It's an amendment to the Food, Drug, and Cosmetic Act of 1938. It's called the Delaney Clause after the person who put it up. And it forbids the approval of any food additive—any additive—found that can cause cancer in humans. If you have a carcinogenic risk, according to the Delaney Clause, any risk is unacceptable. That's why the FDA says we're going to take it out.
- 5.51** Now you have the saccharin and cancer studies, which established the bar that says, even if the levels given to those mice were very high, it causes cancer, and so it should trigger the ban. So there were people like Ralph Nader and others saying, "We don't care if you think too much saccharin is given. It's still a cancer-causing agent. It's time to implement the ban. You've got to ban it."
- 5.52** And the FDA moves forward, I have to tell you. So the FDA says, "We don't want to do anything." And then it says, "In '77, we're going to ban saccharin." And the reaction to the FDA—boy, you would have thought the FDA had just gone ahead and said something completely un-American, because people were enraged.

- 5.53 And what does Sherwin-Williams do? It's very, very smart. They encourage a letter-writing campaign—before you could just send an email, in an era before you could send an email to your representative in Congress. And they are swamped with people saying, "How could you do this? Diabetics need it. I need it for weight loss. I need it for this. I'm used to it." And so they come in.
- 5.54 And Congress intervenes. It's Teddy Kennedy who comes up with a compromise, believe it or not. And it's the saccharin study: we're going to go ahead and do a long study. We're going to have a labeling act in which we put a moratorium on banning saccharin, so we won't do it right away. And you get a warning label that saccharin is found to cause cancer in laboratory animals. We're going to study it a bit more. And that overrides the FDA's initial application.
- 5.55 Now, what's interesting about that is you would think that that warning might turn some people off of using it. But the great thing about the American consumer is they ignore warnings time and time again, because they think, if they like a product, "I see that warning, but it doesn't apply to me using it. It probably applies to somebody else who's using it more than they should." So saccharin sales are not dented by that.
- 5.56 Now, what was the position of the American Cancer Society? Oh, my God. This comes to be what I call some real controversy as they're fighting this out, and Ralph Nader is calling for a ban and everything else, and you have a possible carcinogen. So the American Cancer Society says, "Banning saccharin may cause great harm to many citizens while protecting only a theoretical few."
- 5.57 I have to tell you, if your sole mission is cancer, they are very, very good at the ACS—the American Cancer Society—about going to the government and getting more funding for research on improved radiation and chemo and on treatment. But on this, when it comes to a cause that's in the public eye, my belief, from having studied this, is that they didn't want to lose any of their donors who might be fans of saccharin. Therefore, they thought, "We'll take a position that's in between—maybe only helping a theoretical few." I thought, but that was their purpose.
- 5.58 It was at the FDA a record number of letters and telephone calls that were received that said, "Don't do this." Now, there was a compromise for a moment that was suggested. The FDA, as all of this is taking place, says, "You know what? Why don't we do what they do in Canada?" That's a bad idea, first of all, because no one in America really wants to do what they do in Canada. But they said, why don't we do what they do in Canada, which is—and I'm quoting here—"Have it as an over-the-counter drug for diabetics and others who are required for medical reasons"? That's interesting. Bringing it back to those for whom it was medically necessary, as we had before World War I.
- 5.59 And guess who fought that? The industry, Sherwin-Williams, everybody else. Why? Because, and this is my favorite part, they knew it was a nonstarter. You couldn't really have it for medical use, because their own internal studies had shown that a third of a

can of diet soda caused a 15% drop in blood sugar, making people eat more and gain weight faster than those who did not use artificial sweeteners. Since obesity was one of the primary drivers of diabetes, it wasn't actually medically necessary to prevent diabetes.

5.60 They knew those studies existed, so that if they carved out an exception for it, really for diabetes or obesity, it was going to fail once it gets studied. So they said, "No, no, no, no, no—no medical exception like Canada." So they get away with that, and Teddy Kennedy pretty much just puts on the warning label. The saccharin lobby's influence, not surprising. They promoted it as safe and beneficial. It's a PR campaign. They spend real money on it, and funding based on research and public outreach.

5.61 The industry's no surprise to you here, and it will be no surprise to students who have listened to the lecture on the secrets of Big Pharma, that the industry funds many of the studies. So they pay for researchers to go ahead. And lo and behold, what a surprise, the studies that are subsidized by the industry always come out with good conclusions for the industry. It just is a coincidence.

5.62 Mobilizing powerful allies. And so this is where the American Cancer Society and the [American Diabetes Association] come in. I'm going to tell you, the Diabetes Association was major in terms of being at the forefront of saying, "Don't ban saccharin." They come in and they were very strong on it. Swift political reaction. The industry gets a response from not just Congress, but Teddy Kennedy leading the charge on this.

5.63 Undermining regulatory authority. I think that is so fantastic that the FDA comes in for one of the few times on a popular consumer additive to say, "Okay, we're going to ban it." And everyone says, "No, no, no, no. What do you think you are, the regulator?" So the one time they do something that might be right, everybody jumps on top of them, which then makes the FDA, going forward, a little gun-shy sometimes about whether it should come in and actually issue a ban on something. There is a new commissioner at the FDA, by the way, at this time. Not a good time to be a new commissioner. He's the one who proposed the compromise that they had in Canada, but that didn't go very far.

5.64 The rise of aspartame. What is every attack on—if a drug has proven to be unsafe? So when Miltown, the original anti-anxiety mild tranquilizer that was a hit in the 1950s, proves to be more addictive and more dangerous and worse with alcohol, it doesn't mean that we eliminate all those drugs, but it opens the door to a new line of drugs that claim to have fewer side effects and not as dangerous. So Librium comes in. And then Valium comes in on top of Librium because it's an improvement over that. And by the time, a decade later, we're aware of the dangers of benzodiazepines or how they can be abused, Xanax comes in as a replacement for that.

5.65 So the drug industry always uses an opportunity. When one drug—hormone replacement therapy for women on menopause, or high versions of the pill that were causing blood clots—products come out that supposedly don't have that. So aspartame

is coming in to say, "We have a sweetener that is much, much better than saccharin, so don't worry about this." Now, remember, Canadian study, 1977, all of this happens with saccharin just around this time.

- 5.66** In '81, the FDA approves aspartame, which had, by the way, at the time of the '81 approval, a series of articles of concern, of studies that said, "We think aspartame could be linked to different types of cancer and problems inside the blood system." And still the FDA said, "Well, it's okay by us." They did exploit, without any doubt, the food industry, the safety questions over saccharin to be able to say we don't have all of these concerns about aspartame.
- 5.67** Well, of course, it's what we've often talked about before, and it happens in the drug world, which is when a drug is new or a product is new like aspartame, it doesn't yet have the long history of accumulated side effects and horror stories. It takes a while for that to come in. But then we need to test it. The testing is done before the FDA, but what's the real testing for something like this as a food additive that's going to go into millions of products all around the world?
- 5.68** The real testing is us. We become the clinical test. We consume the products, we go out and we take them, and then over a period of years, maybe we see some anomalies on numbers for diseases or cancers, and then we say, "Oh, I wonder if that could be a result of that." Aspartame replaces a billion pounds of sugar in the American diet after its approval. Wow. I'm always impressed by that. A billion pounds of sugar.
- 5.69** Let me tell you, that did not eliminate the sugar market. Sugar is still big here. We eat sugar in a lot of things, even in things that don't appear to be sweets or candies. You look at the back of the label. Labeling will tell you how much sugar is in a lot of stuff. But aspartame comes in, and it's a pretty good hit from the beginning. So despite previous concerns about its safety, the influence of what I call the industry here, they are able to get this approved, and aspartame is on the market.
- 5.70** The ongoing debate. Mid-1990s—after it's approved, it's approved in '81—there are new studies that raise concern about artificial sweeteners and cancer. And in '96, aspartame is linked in this study to brain cancer. So now you have the strongest study yet to say aspartame might be a carcinogen, and it's particularly with brains. In '98, two years later, a trial shows that aspartame can destabilize DNA metabolites. That's one of the items that could cause cell proliferation to go out of whack and start to create cancers. It's the theoretical basis for how you say you could start early parts of cancer.
- 5.71** In 2008, a review concludes aspartame breaks DNA strands in mouse bone marrow. Now, the part that should concern you is that there's a 10-year gap there, which means that all that's happening in between is there are regulators looking at it and there are more studies being done, and the aspartame manufacturers are just doing a successful job of fending it off. They're putting it off, they're putting it off, and then you get another what I call big study that comes up. Same year, 2008, Italian researchers found

increased risk of urinary tract cancer with long-term consumption of both aspartame and saccharin.

- 5.72** But you can still go out, you can still get them. You can get the Sweet'N Low packs, and you can get the blue packs and the Equal packs and everything else, and they're available. I myself like just a little bit of sweetener in my coffee. I use stevia now, which has not yet been proven to be terrible. But if I can't find a stevia pack, or I don't have it with me and I'm traveling, I'll use a little bit, just like a teaspoon of real sugar. I'd rather have the real sugar than the chemical.
- 5.73** When I grew up as a teenager in San Francisco, back in another era when I had no knowledge about this, I used to drink Tab regularly. I'm a living experiment as to Tab, but so far it hasn't caught up to me. But my view is that there are enough questions about this that sometimes, even if the government isn't acting, you need to just have what I call common sense of saying, "You know what? I'm not so sure about that."
- 5.74** Burying the cancer risk: the chemical industry's deception. Boy, vinyl chloride—the effects of vinyl chloride. It's a chemical widely used in plastic manufacturing. What happens is it's a lethal carcinogen. In 1974, B. F. Goodrich announced the death of three of its PVC workers from a rare liver cancer, not the type you normally see very much of. It's linked to vinyl chloride. The problem is that once that happened, then regulators, investigators start to look back and they realize what the trail was.
- 5.75** In the '70s, an Italian toxicologist had presented a study on vinyl chloride that caused lethal cancers in rats. This is in the '70s. In '74, B. F. Goodrich says, "Oh, by the way, we have these three workers working as PVC workers, and they have this rare liver cancer." Chemical companies were funding study after study to discredit the findings on vinyl chloride and the carcinogenic products.
- 5.76** It's not that they go ahead and they investigate it and say, "What can we do for our workers?" It's that this product is important in our manufacturing, and so we really don't want to know if it's hurting workers, but we want to be able to fight the idea that it is something that's bad. In the '70s, the new study confirms vinyl chloride is a powerful carcinogen, but chemical executives bury that report. It's remarkable how they don't publicize it, they don't push it forward.
- 5.77** Now, vinyl chloride: the plastic that ate liver—human liver. So in the mid-20th century, the 1950s, there were the first people that noticed these bone changes—some of the doctors—and there are elements of it listed in some of the hospitals and studies that they did about severe liver damage. They didn't know it was going to lead to cancer yet in these workers working with vinyl chloride. In the late '60s and early '70s—we're talking an early time now—they have identified this rare liver cancer cluster in vinyl chloride workers. B. F. Goodrich hasn't come out with its report about their three workers till '74.

- 5.78** OSHA—occupational health and safety regulators. Here we have the new regulators. It's not the FDA anymore. They set tighter exposure limits, but those limits come literally decades after exposure, like when you were trying to stop the radiation, the radon gas—it comes decades later. So finally, OSHA responds to these decades of studies and pressures, and the industry wants to weaken or delay the standards.
- 5.79** OSHA says, "We think we should have this standard; let's put it at so many parts per million." And the industry says no. They don't do what the drug industry says, which is, you're going to kill innovation and research and development, and we won't be able to cure the next great disease. They say, "You're going to put us out of business. You're going to stop us from manufacturing things so important to national security." These plastics are integral to what happens to an American consumer.
- 5.80** And they argue the limits are not technologically feasible. Well, you hear that all the time from industry: "We can't do it, can't make it. It's beyond our capacity to be able to lower the limits like that." And in '75, those agencies like OSHA often end up compromising. They say, "Okay, the limit will be a little bit higher," and they rely on company-supplied data to reach what the new levels will be, which is never the best way to rely on it.
- 5.81** Then you get this guy, John Stender. He comes in. He's very interesting. He takes over at OSHA, and he's a lifelong labor union official. So he's appointed OSHA's head in the aftermath of the Watergate scandal. And the interesting thing about him—a former Washington State legislator and Boilermakers Union official—is that he brings this perspective to the regulatory agency that's different because he was involved on the workers' end in terms of the union. And he lost part of his own hearing by working inside boilers. So he understood that workplace occupational hazards, as the government calls them, can have physical effects.
- 5.82** And once in charge, he sets about transforming OSHA. He's really the reformer, as I view it—more proactive on his regulatory work. He's going to set about it and say, "We're going to take worker interests over industry interests." He scraps the endless advisory committees. Well, I'll tell you, industry learned that the way you slow something up is you say to the regulatory agency, "Why don't we get an advisory committee and they can study it?" And then the advisory committee is filled up with industry representatives or friendly people to the industry, and they mix it around and they delay it in that. They'd debate these safety standards forever, so he gets rid of the advisory committees.
- 5.83** And he has this focus, as I say; he approved stricter rules for 14 carcinogenic chemicals, including a proposal to cut the acceptable level of vinyl chloride in the workplace from 500 parts per million to 1 part per million. That's pretty remarkable. Not eliminated. So he said, the industry can still use it, but it's going to have to be used in a really safe environment.

- 5.84** And the OSHA lobby. OSHA isn't established as a new regulatory agency until '71. So this is all pretty new. In '74, the chemical industry spends \$90 million—big amount of money today, big amount of money then—to get studies to oppose stricter vinyl chloride standards. And by the way, in '74, when they're spending the \$90 million, remember, that's after the '60s and '70s reports about maybe some liver changes from vinyl chloride. And after B. F. Goodrich has announced that three of our workers, PVC workers, died from a rare form of cancer—liver cancer.
- 5.85** The Nixon administration. In '77, it's the fallout after the Nixon administration: it's Gerald Ford who coerces OSHA to implement what I call these business-friendly policies. And then in comes John Stender. The government picked the wrong guy to run OSHA, meaning that they picked a guy who was actually going to run it as a regulatory agency. My goodness, if the industry knew that, they might have been able to block his appointment.
- 5.86** He approved the rules for these 14 different [carcinogens]. And eventually they end up, by the way, going from 500 parts per million to 50 parts per million. Pretty good reduction still. They don't go down to 1 part per million. But he understood the art of negotiating. So you want to scare the industry? You say we're essentially going to eliminate it or bring it down to 1. And then the industry ends up thinking that they have a victory when they get 50 parts per million. But the workers have the victory because it's a safer level for something that otherwise can be quite dangerous.
- 5.87** Asbestos: a tale of corporate greed and deception. The buried story. In December of '77, The New York Times runs a story, and it's small. This is a small story that says a \$1.8 million lawsuit filed by 28 insulation workers claimed they had developed cancer from asbestos exposure. Now, that's the first time they had sued and they got this settlement. And the Times reports it. It's hard to imagine that in 1977, we as a society were not more aware of the dangers of asbestos exposure. There had been reports on it for decades, but it was still being used with workers.
- 5.88** A corporate dismissal of the lawsuit. Now, when that lawsuit was filed by those 26 insulation workers, they're working with asbestos, consulting on insulation. The companies, [-] Johns-Manville and others, dismissed it in their public statements, saying it's meritless, it's generated by ambulance chasers, there's nothing to this at all.
- 5.89** So the immediate industry response is, "These people are just out after money. There's nothing here in this story." But it really is the start, in many ways, of what I call the beginning of the asbestos industry disinformation campaign. You think Big Tobacco was something when the CEOs went up before Congress and said, "We do not believe that our product is addictive"—although they had studies at the time that showed it.
- 5.90** This was a massive campaign for the industry to shape and distort public opinion. They recruited medical and scientific experts who should have known better, who became industry shills in many ways to go about and talk about how asbestos was not such a

great risk, that it was narrowly limited to a small part of the workplace environment, and they dismissed study after study that said asbestos exposure could be lethal.

- 5.91** The Asbestos Information Association—AIA. Of course, you have to have an association. You have to be able to run your money and your PR and everything else through one association. In pharma, we have PhRMA. That's their lobbying group. Every major organization has a lobbying group. The great thing about the Asbestos Information [Association] is it's not an advocate for the asbestos industry. It's neutral. So you can go and get your information. You have concerns about asbestos, you can go there. But let me tell you something: it's not neutral. It's an industry tool, and it was secretly funded by Johns-Manville and other asbestos manufacturers. We only find that out later, not at the time.
- 5.92** Disinformation campaign: the AIA has this massive effort to shape public opinion by downplaying the dangers and dismissing the studies. They delayed the OSHA standards. OSHA now is talking about standards for permissible levels of asbestos dust in the workplace from the time of 1976 and 1977 under John Stender. And what happens to those proposals for stricter levels of asbestos exposure to workers working with insulation and other products? Ten years, they get pushed down the line. So you have some success on vinyl chloride, you don't on asbestos.
- 5.93** Corporate allies in this deception. Internal documents now available through litigation show that the executive committee of the Asbestos Information Association often congratulated itself. They sent memos to each other saying, "What a great job. We stopped the government from passing this regulation." They were relishing their victories on keeping this. The consequences of the deception is it allowed workers at [-] Johns-Manville and other companies to continue to be exposed for decades to a preventable cancer.
- 5.94** This is a sidelight. This is just one of those stories that is so outrageous that I just had to include it to get your blood pressure raised up a little bit. Dr. Albert Kligman. He's a Pennsylvania dermatologist, and he conducted [experiments]—I call them unethical, to say the least—on prison inmates in the 1970s. And he was injecting asbestos into inmates, most of them Black, without them knowing about it or consenting to it.
- 5.95** He was able to determine—because what was he doing? He had been hired by Johnson & Johnson to see if talcum powder might be safe. And talcum powder had some little residue of asbestos in it, and they wanted to make sure that the levels it was in, would it cause a problem? It didn't kill those inmates, but it caused a whole series of injuries that they had to be treated for. So the use of prisoners for injections to see if talcum powder might be good, not a great idea. He kept his medical license. Never a problem, other than the fact that once that all was exposed, people were infuriated.
- 5.96** The asbestos litigation tsunami. So you get the \$1.8 million in the lawsuit filed by the 28 insulation workers. You think that's the end of it? No, that's just the first. That opens up

the litigation doors. The following decade, [~1 million] lawsuits filed against the asbestos companies. In the '90s, many of those companies—they were asbestos manufacturers, or that was their primary source of revenue—filed for bankruptcy. So it reminds you of Purdue Pharma. My God, one-hit wonder, OxyContin. And where are they? Bankruptcy court.

- 5.97** So this is what happens. Litigation comes in to try to say, "Okay, you've done this wrong." And then at a certain point, the companies go under. And the other thing they do, how about the companies that survived? So they didn't go bankrupt. They're large enough that asbestos isn't enough to take them under, but they moved all their asbestos manufacturing facilities—all 100% most, the vast majority of their asbestos manufacturing facilities—offshore. Let's bring it to countries where we can get workers who aren't yet worried about asbestos and don't have OSHA, aren't worried about that.
- 5.98** The impact of asbestos litigation. This is the estimated cost in the decade following the initial settlement—only in that decade: \$54 billion in total litigation costs, \$40 billion in payments to the plaintiffs, and \$14 billion in litigation expenses. So it becomes a major, major mesothelioma. How many times have you turned on an ad and seen, "Were you exposed to mesothelioma? And if so, contact this law firm"? So once the class-action lawyers get involved, that's also an industry—for another day.
- 5.99** So it's a perfect industrial material with a perfectly predictable disaster. And the effects of that are still being felt today. So asbestos—you think of it back all this time, for decades earlier—really breaks in the '70s, and here we are 50 years later. And the remnants are still being dealt with in terms of litigation.
- 5.100** Now, enter the muckrakers—investigative reporters and activists who come in at this time and start to focus on the industry. So these activists are trying to expose the industries for their bad behavior. Their internal studies are being done by these industries to try to hide the facts of what's happening, and they're prioritizing profits over safety. Now you get a group of reporters and journalists who are coming in who say, "You know what? We know these industries; they really aren't doing the right thing when it comes to workers."
- 5.101** You get Samuel Epstein, Rachel Carson, and Ralph Moss. So Epstein was a physician and university professor of occupational and environmental medicine, and he wrote the book, *The Politics of Cancer*, which I hope he wouldn't mind that I've used for the title of this particular lecture series. He's the one who really talks about that for the first time. It's really a powerful book that ties in the corporate interests and the way that the cancer industry, as he called it, was focused on treatment and ignoring the agricultural chemicals, industrial chemicals that might cause human cancers and how so many of those were unregulated workplace exposures.
- 5.102** Rachel Carson comes in with a seminal book called *Silent Spring*, in which she warns that widely used pesticides might be responsible for a laundry list of illnesses, including

cancer, although she lacked the conclusive evidence, but she was raising the alarm in Silent Spring.

- 5.103** Ralph Moss was a former science writer who endorsed the unproven theory. He is also the person who ends up getting fired later from Sloan Kettering, and he believes in laetrile and other alternatives. But he does think that there's a [viral] infection that can cause cancer, and he thinks the cancer industry is suppressing that [viral] infection. We now know that from HPV and other sexually transmitted diseases that can lead to cancers, there is an element of that that was right. But he's about the cancer industry, as he calls it.
- 5.104** Now, Samuel Epstein, that book is really groundbreaking in many ways because he focuses so much, in my view, on industrial and agricultural chemicals. And he argues that the government's focus on treatment through drugs, chemotherapy, and radiation—not that it was wrong, not that it didn't work—is misdirected, in the sense that too much emphasis is put on that. More attention should be given to what I call finding the environmental and occupational causes of the disease, because we could prevent it. But of course, there was no aggressive effort to do that by anyone other than the muckrakers, really, at this point. They are skeptical, but they're raising the doubt.
- 5.105** So the rise of Epstein: the emergence of cancer concerns in the '60s and '70s. The concerns about these industrial products, like we were talking about before with asbestos and what happened with vinyl chloride. You have Hueper's early warnings. He's talking about the role of industrial carcinogens in cancer. Think of this back in the time when he's taking over the Environmental Cancer Center at the US government, and he's not [recognized].
- 5.106** The publication of *The Politics of Cancer* in 1978 catches the public attention. It becomes a book that not just has a cult following, but reshapes the public discussion as to, are we looking at the wrong thing and not doing enough to prevent cancer? And Epstein's book really challenges what I call the existing status quo of the cancer industry, which was focusing just on the treatment more than anything else. It leads to a shift in understanding by the public of really what I call this very complicated relationship that ends up taking place.
- 5.107** Before this book, in some ways, you think about it, the public would not think, and I'm going to talk here in a second about the public reaction, the public did not think that when you were talking about vinyl chloride over here in the chemical industry, and you were talking about saccharin over here in the food industry, they didn't think it had anything to do with Memorial Sloan Kettering. And it didn't, in the sense that Memorial Sloan Kettering wasn't involved in those.
- 5.108** But it was as if the industrial and chemical industry and Big Food, with some of the additives, were sending business over to the cancer industry: "We're essentially providing you patients in the future." Now, I don't mean to say that any executive sat

down and made that calculation. Nobody sat down and said, "Oh, yeah, let's give him a lot of saccharin, and then we'll help out Memorial Sloan Kettering's bottom line." But that's essentially what it became: those industries that were suppressing news about carcinogenic workplace exposure were, in essence, feeding the cancer industry, which is built around treatment, time and time again.

- 5.109 The public's perception. In this time, studies show overwhelmingly—remember, this book comes out in the '70s, late '70s, '78—the public is still concerned about cancer at this stage as a death sentence. You ask people in these opinion polls that would be done by Gallup if they thought that the war on cancer, which was now 10 years old in some cases—'81—was making a change. And they still thought of it that way because people who had relatives or family members who had cancer often saw them have terminal cancer. They ended up dying. So they still thought of it that way in their own mind. It's not as though all the work on treatment had changed that.
- 5.110 But there was something interesting in these polls, a distrust of the government's progress. The public, even after the war on cancer had been announced, wasn't convinced that things were changing. Books like that one from Epstein start to change the public opinion in some ways to think that maybe there's not an effort to really help us on this.
- 5.111 So: percentage of Americans who believe the government is not to be trusted when it comes to cancer. This was within five years of the publication of Epstein's book. So 65% of those polled distrust government health regulators to protect the public from carcinogens. It doesn't change what's happening in terms of the government, but two-thirds of people had that.
- 5.112 Seventy percent fear increased cancer risk due to government or employer failures. Amazing. It's not that 70% of the public is working inside DuPont or Johns-Manville or handling those materials, but they're worried that they're being exposed to other possible [carcinogens] and that nobody's paying attention to them. And 80% do not believe the NIH cancer-cure claims that there will be a cure for cancer. So people are skeptical. It's not as though the public is buying into, at this point, the war on cancer.
- 5.113 Next lecture, we'll leave saccharin and vinyl chloride and asbestos and corporate cover-ups. And I'll talk about the pink ribbon machine, how breast cancer became a billion-dollar brand. I look forward to talking to you about that. Thank you very much.

Lecture 6: Breast Cancer & Branding

At a glance — Breast cancer and branding.

- 6.1 Thank you for returning. This is just me back looking at some of the items available. If I wanted coffee in the back room, I still have the old standard-bearer of saccharin. The aspartame, still around, surviving all the controversies. We didn't get a chance to talk about sucralose, which is yet another competitor that goes against the three. It's still a multibillion-dollar-a-year industry without any doubt.
- 6.2 Today, I'm going to talk about something that we all see, which is a pink ribbon worn by many people. My wife's had breast cancer, as you know, so we're advocates of awareness on breast cancer. But the pink ribbon has become ubiquitous when it comes to cancer care. There are other now survivors who have tried to turn attention to their particular cancer and to do the same type of what I call marketing and branding.
- 6.3 But there are races for the cure. There are 5K and half marathons and marathons, and there are events. And how it gets here is an interesting story. So from tragedy to global brand. All right, we understand that recap. We've talked about chemo, quacks, gatekeepers. I get all that. They may seem like they're now old stories to us, but we've talked about that in the past.
- 6.4 And so breast cancer goes through this transformation, really. It went from a stigmatized disease that nobody would talk about. The other day, in talking about Gerald Ford, I was talking about Betty Ford. When Betty Ford announced that she had breast cancer and she had a double mastectomy as her surgery, she announced that at a live press conference. Pretty amazing. It wasn't like in a communiqué or a release or a memo from the first lady's office. She talked to people about it and made it very human. I think that you'll see how it becomes recognizably human and then becomes this brand.
- 6.5 The questions that I have are, how did breast cancer become a global marketing platform— which it is? Who controls the pink ribbon narrative? What, if anything, gets lost in this shape? Early detection, feel good campaign. Because one of the things you'll realize as you go along, and it makes sense, companies will get involved. Corporations, they'll say Express Awareness Month. They'll sponsor pink ribbon events. Customers like that. Nobody ever says about a company, "Oh, we think it's terrible that you're doing that." And it's not. They are bringing awareness, but they understand the benefit of associating themselves often on this issue, and this is a very visible issue for them.
- 6.6 So the origins of the pink ribbon, we're going to trace that story of how it starts as really grassroots advocates. It's not like a marketing person from behind the scenes decided to create it. There's nobody pulling, there's no puppet master behind the ropes, decided to come up with this. And it's a sham. It's really grassroots from women who are affected and make this into this symbol.

- 6.7 And then the corporate assumptions about it. They've done studies with consumers, and consumers feel instinctively that if the company is promoting a pink ribbon campaign, it gives a good luster to that company. So you talk about good PR and bad PR. This is very good PR and on an important issue.
- 6.8 So women's empowerment, this isn't specifically about cancer. This is in general. The war had really put a lot of women into the workplace. And after it was over, they were occupying jobs traditionally done by men. One of the things that happened, because I used to talk to Trisha's mother about it and her aunt, and during World War II in England, they had gone and they had worked, and they had done factory work and other work supporting the war effort that had been the province of men beforehand.
- 6.9 Then when the war was over, it was like, "Thanks, ladies. Now you can go back to housework." That type of attitude. They didn't stay in the workforce in those positions because men returned, and then women were viewed in that role. You can go back to that, that's the role that drug makers later were targeting women as housewives and efficient mothers, but not necessarily as part of that workforce.
- 6.10 But I do think that that little part in my own unscientific study of talking to women in Trisha's family who had done that type of work, they really felt as though they weren't happy about going back to the roles they were relegated with before the war. It didn't mean that things were changing overnight. It was not yet a woman's movement, but there was some feeling of that. Dissatisfaction, as I call it, returning to the traditional roles, and that dissatisfaction is there, but nothing is changing overnight.
- 6.11 And then you get this period at which I believe you get in 1966, two French engineers, as a matter of fact, Jean Bensand Emile Gabbay. They invent a revolutionary X-ray device. They end up calling it a Senographe, that we know as a mammogram. That's really the first mammogram that comes out. Raymond Rife, who had come up with this microscope that was such a revolutionary microscope. They've come up with something on radiation that produces these sharp, precise images, low radiation, low radiation compared to taking a normal X-ray, and a significant improvement over the previous generation of X-ray scanners.
- 6.12 When they brought us to market, they really thought—sometimes people invent something and they don't have a feeling for what its effect will be. These two inventors really thought this will help detect breast cancer, possibly at an earlier stage when it's a smaller tumor and be able to treat it at an earlier time. They weren't about the treatment. They were very much about the detection.
- 6.13 One of the things that happens is mammograms are very widely used. Trisha has had a mammogram for every year that I remember being with her, and we've been together for decades. She's one of those people that has chosen to go for mammograms. Remember, I'm not a doctor. I'm not giving you medical advice. So everyone makes

their own choice. In that, nobody ever said to her, by the way, her cancer is what's called estrogen-based.

- 6.14** They do a test later to see if the tumor is estrogen-based. Hers is estrogen-based. So that's what's contributing to her tumor, not caused from radiation exposure. But nobody ever said to her, by the way, for all those decades, when I started to amass the information that the last five years about the cancer industry. I thought that's interesting. There's this low risk, 26 to 59 people. Very specific, right?
- 6.15** They actually do some of these studies to try to figure out what the cumulative effect of radiation exposure could be. There are some people starting at the age of 40, which is the recommended age for mammograms. If you start doing it regularly, it's going to be a very tiny, infinitesimally small number of women who will develop the cancer based upon radiation exposure from the mammograms. Okay, that's fine.
- 6.16** You do a cost analysis, you do a risk analysis in your own head. If somebody had told that to Trisha 30, 40 years ago, 30 years ago, she'd still be doing mammograms because she would say, "It's worthwhile for me to have that detection." That's what made her comfortable. But it would also be good if that was discussed sometimes with patients.
- 6.17** But it isn't because the fear is that even if we tell you that, it's going to scare some people away from doing it. It's not a reason not to do it. It's just what I call a matter that's never discussed. I'm a believer. I want all the information. Everybody's different. Maybe some people don't want it at all.
- 6.18** I want to take it all in, and then I'll decide what I'm going to do going forward. That, I think, is what I would want the cancer industry, as I call it. Mammograms also, the industry estimates that 10% to 12% of screening leads to false positives. I get it. Something appears abnormal, and then you have to do a biopsy, you have to treat. Trisha, over 30 years of mammograms, never had a false positive, which is the good news. So she never went through that experience. Other people do.
- 6.19** Dense breasts are difficult to be able to look at all the time under regular mammograms, and mammograms can be uncomfortable. So this has given the rise of what I call 3D mammograms, and sometimes MRIs, which are now being recommended. But there's a new form of the machinery patented 3D mammograms, which give you a multidimensional view of the breast.
- 6.20** They were originally a higher dosage of radiation than the 2D mammograms. Now they've developed some machines that are the same or lower. The key is the FDA says, "Here's the safety level," and they all meet the safety levels. If you have a choice, you could ask your screening facility if they have a 3D mammogram, if it's a new machine or one that they bought first-generation 10 years ago on eBay. So assuming that it's not an old machine, the newer ones are a little bit better.

- 6.21** The cancer industry, of course, it prioritizes screening. Screening becomes the keyword, detection in earlier time. We're going to come back to screening afterwards, but I want to go off on one thing that you might find surprising. We talk about the growth, eventually, of the pink ribbon brand.
- 6.22** Radical mastectomy. We go all the way back to Stewart Halsted on that. The cocaine and morphine-addicted chief surgeon at Johns Hopkins, who does trials in 1894 and develops the radical mastectomy, which is then used as cancer treatment. What is really remarkable is, to me, that in 1981, the first clinical trial was done to determine the effectiveness of the radical mastectomy, and the results were published.
- 6.23** It's almost 100 years after it went into practice. They had never done a study to determine if that was the best possible surgery to remove tumors on breasts. It was the standard because doctors had become accustomed to doing it that way. In fact, excuse me. In the 1970s, through this period, up to the time of this clinical study, this was the standard practice. At the, quote, "best" hospitals, being Memorial Sloan Kettering, all the cancer hospitals everywhere else.
- 6.24** If an oncologist, a surgeon, thought there was a tumor, they did a biopsy. If the biopsy was malignant, once they got the confirmation of malignant, they would perform in the same surgery a radical mastectomy as a life-saving procedure. So no discussion. Malignant tumor, radical mastectomy. By the way, you leave there. We've done it because otherwise your life was at risk. That was the policy.
- 6.25** Now, that's hard to imagine. You know how I told you before about how we would debate with the doctor? What's the choices? Give me the alternatives. What's the benefit? What's the downside? There's no clear decision here. It's not like you can say that's the right one, but you can take in all the information and decide what's best for you.
- 6.26** If these are the two or three options, there was no option there. It takes this guy, Bernard Fisher, this American surgeon oncologist. He really was somebody who would challenge what was dogma at the time. He did not believe that the radical mastectomy was the only surgical option for women who had breast cancer. When I say he was relentless, I mean he was relentless. The medical establishment was against him.
- 6.27** They were saying here, "You're a troublemaker. You're raising a ruckus for no reason. We've got—it's a great procedure. We've got the radical mastectomy. It's been around for nearly 100 years. It's proven its worth. It's saving lives. Why are you going to go ahead and question it? All you're going to do is put doubt into patients." They're going to think, "Oh, maybe that's not what I should do."
- 6.28** He does a clinical study, and it really is a landmark. In '71, a consortium of academic hospitals conduct this clinical trial based on Bernard Fisher's insistence. Remember I told you before, it's '81. This is a long-term study. It's not something you can easily find out. Ten years, they're gathering data on 1,765 patients across the US and Canada, and

they're trying an experimental surgery on some, a lumpectomy, removal only of the tumor, and what they call breast conservation today, saving the breast.

- 6.29** So could that be as effective on recurrence and on life expectancy as the more radical mastectomy? Eighty-one, the results show statistically identical rates of cancer, recurrence, death, and metastases across all groups. That was stunning to doctors who believed you had to do a radical mastectomy every time it came up. Eighty-one was a realization, as Bernard Fisher would later say, that half a million women—that's not the number of women, millions of women had undergone the surgery, but half a million, was his estimate—had probably gone through a more disfiguring surgery than was needed for them. They could have had a lumpectomy, and it would have been fine, but it wasn't an option for them then.
- 6.30** Now, this ties directly into the rise of breast cancer activism because women's empowerment and advocacy came from the way they were being treated. First of all, you have the power of what I call the survivor stories. So these survivor stories of diagnosis and treatment become very, very important. And they add to the sense that it can be complex at times and the choices can be difficult, but women were not feeling as though they alone were going through a cancer diagnosis if they had it.
- 6.31** They felt as though they were seeing public figures who are coming along and saying something. Then Babette Rosmond. I like her a lot. I don't know her, but I like her. I'm sure I would have liked her if I had met her. Editor of Seventeen magazine. In 1972, she publishes a book called *The Invisible Worm*. It's about cancer, her breast cancer, under a pseudonym.
- 6.32** She talks about her experience dealing—Now, think of this. This is '72. They have started the trial to see if there might be a surgical alternative that is not as radical as the radical mastectomy, and they haven't yet adopted that. They're nine years away from even publishing the results. She talks about her experience with breast cancer doctors, and they were all men at that time. Her message was that she should have the right to decide what was done to her body, at least a role in it.
- 6.33** It's very interesting that she went— She talked about her experience in which she went to a doctor. I'm going to quote from this from her book. The initial surgeon said to her, "Here's our one-step procedure. We find a malignancy, we perform the surgery right away, the radical mastectomy." She said, "I don't want that. I want to wait to see if there might be something else that could be done other than this surgery." He says to her, "You are a very silly and stubborn woman. In three weeks, you may be dead."
- 6.34** Well, she wasn't. She went around and had time to write a book and everything else about it. But that idea, that was pervasive among surgeons. They were not accustomed to patients, to women who are diagnosed with breast cancer saying, "Oh, I don't want this surgery right away. I want to wait and see what my options are." The response was,

"Well, you're a silly woman. You'll be dead very soon." That's not what you really want to hear from your doctor.

- 6.35** The system that was in place was one that was working very, very well for the surgeons because they had an assembly line going. But you had the first people like Babette Rosmond standing up and saying, "By the way, I don't think that's for me." Shirley Temple Black. You might remember Shirley Temple, a childhood star, one of America's most famous stars, and then becomes later an ambassador. She also, in this time, publicly is coming out about her breast cancer.
- 6.36** She has a line that is very interesting. When they tell her, "You have to do the same thing. We have a surgery for this. We do it right away." She said, "The doctor can make the incision, but I will make the decision." So a little bit of time when all of a sudden, women are starting to say to these doctors, "I'm not so sure that I'm just going to follow the rules as you've laid them down." Prominent women talk about their breast cancer.
- 6.37** So really, this becomes the moment. In '72 at the NCI, the National Cancer Institute, they start the Breast Cancer Detection Demonstration Project. We have an outreach, early part of detection, but also talking to women as patients, not just as somebody on the assembly line, and it becomes interesting. You have Betty Ford two years after Babette Rosmond. She comes out. You have Happy Rockefeller. We forget that she was diagnosed when her husband was Vice President of the United States, the wife of Nelson Rockefeller.
- 6.38** NBC News correspondent Betty Rollin. Boy, she talked about her diagnosis and her book becomes a national bestseller. They do a television movie off of it. It opens up a floodgate of women talking about their experience through breast cancer. Really not something they talked about before. The disclosure of these women, the public television, the bestselling book, it's really women learning for the first time—if I can make it, and you'll say this is an overgeneralization, but it's largely true—hearing discussions about self-breast examination that you should check yourself to see if you feel a lump or something different.
- 6.39** It wasn't discussed never on television before. Now you have public figures talking about the importance of it, not waiting necessarily for it to be discovered if you're doing a mammogram, but you might be able to find it yourself. I think that in some ways, Betty Rollin—she was 37 when she found her breast cancer. So young, as we say, as you go along in breast cancer, anything, it's never a good diagnosis, but the younger you are, always the worst.
- 6.40** And First, You Cry. It became her story about this. And I think that these stories of visible women who people thought they knew, you see a public figure, you see them, especially if it's somebody who's in entertainment or in Hollywood. People think they

know the person, and then they experience the trauma of the medical diagnosis with them.

- 6.41 It really started to affect the way that the industry looked at this. I think that what you have here in this material, in some ways, is that it's not a coordinated campaign. It just literally is breaking as a group of independent women who have had enough and are telling their story and saying, "This is what happened." There's no yet rise of what I call breast cancer activism is just starting.
- 6.42 Rose Kushner, she's a medical journalist, and she became an advocate. She didn't have an enormous reading market like Betty Rollin, but she was saying, "You've got to know as much as you can on deciding your breast cancer treatment." She lays the groundwork to empower women with the ability to decide what their treatment might be if they're diagnosed.
- 6.43 Audre Lorde. She is an award-winning poet, and she was one of the first to criticize society's views of breast cancer, particularly the assumption that women should automatically have the radical mastectomy or wear a prosthesis or reconstruction. She said, "Even if you have the surgery, you may not have to have reconstruction." A radical idea at the time.
- 6.44 The key thing from Kushner and Lorde and others who are what I call the activists at the front-end, they are really saying you can take away the decision-making from the industry, basically male surgeons, and you can take back control of that yourself. You are no longer passive when it comes to something as important as breast cancer. These advocates, I also think, they challenge society's assumptions of how breast cancer is as a physical disease, and we're going to treat it as a physical disease.
- 6.45 Now, focus this. Philanthropy and advocacy. In 1981, just at the time when you—and I'll talk more about Nancy Brinker in a moment—you get the clinical study that says, by the way, same results, at least for the women when we studied between recurrence and death for lumpectomy and radical mastectomy, Nancy Brinker, whose sister Susie had died of cancer at the age of 36, says to herself, "I'm going to find a cure."
- 6.46 Now, that is a big thing to take on as your to-do list, but that's her goal. Her sister had died. Brinker founds the Susan G. Komen Breast Cancer Foundation after her sister's name, Susie, Susan. She's the first public effort to raise money specifically for breast cancer. She said, "I want this organization—1982, we're going to be able to raise money just in terms of breast cancer and want to look into this."
- 6.47 Around this time, women with breast cancer were observing the success of AIDS activists. Remember, this is the period in which AIDS activists are demanding that something be found to treat the disease. And the administration ends up passing in record time through the FDA, eventually the first drug, AZT. Those of you who see the lecture about the secrets of Big Pharma will find how the pharmaceutical industry and

Burroughs in particular, gained that to make a lot of money and charge a lot of money for it.

- 6.48** But still, that activism is making the government respond. So people like Nancy Brinker are seeing that, and it becomes a model for them in terms of be noisy, be loud. It doesn't mean you have to take to the streets demonstrating, but make sure the government and the regulators and the people at the FDA and the drug companies know you're there.
- 6.49** The media's role. Time magazine runs a national cover, boy, at this time on breast cancer and says, "One out of 10 will get it if you live long enough." It's a bit of what I said about men, live long enough. Prostate cancer is somewhere there in the background. If you're lucky, you will not know you have it by the time you pass away, even if you get it.
- 6.50** But breast cancer, one in 10 women will get it. The publication of Matuschka, she's a model who had had a mastectomy. And in 1993—this comes later—The New York Times runs a cover story about how tough breast cancer can be, and she poses for the cover of that. Now, I will tell you, Shandrika, if you look at that picture, it's a tough picture.
- 6.51** Today, there'd be controversy about it in a mainline legacy news publication. Back then, that was, I understand, the underlying procedure: radical mastectomy, but that was radical. And what she was saying, what the article was saying is, you can have this surgery if you have to have it, if it saves your life, and that's your battle scar as a warrior who went into it.
- 6.52** That changed the view because everyone was thinking, if you had that surgery, it wasn't something you would then boast about, she was saying, "No, I'm proud of it." Different view, but one that was important. She had that photographed more than anything else. Breast Cancer Kills Women, Wham. She's telling you, vote for yourself, and talks about how as a woman, you can become important in this role.
- 6.53** There's an activism taking place that women are saying, "I'm not afraid of this." Now, building the corporate breast cancer brand. You get the rise after Komen starts. Come back to Nancy Brinker in a second. These national organizations emerge. They say, "That's a great idea." I understand that. They become grant makers, corporate partners. You start to get de facto spokespeople for breast cancer. So now they're going out and they're finding people might be willing to go out and talk to women across the country and talk to medical groups and to pharmaceutical companies. It's what I call these breast cancer nonprofits far beyond just Komen.
- 6.54** The pharmaceutical link. So I say here— and I don't mean this as terrible as it might read if you read it without knowing the full background—when I say pharma companies, breast cancer space is an attractive area for them. I mean, it's not like they're sitting around the table and hoping for more incidents of that, but it means that

they can look at developing drugs for an area of activism and growth that is becoming important.

- 6.55** So is it possible for drug companies to be able to come up with pharmaceuticals that will either take away or supplement the older routines of surgery, radiation, and chemo? When Trisha finished her treatment in terms of surgery and radiation, they said, "Oh, there's also a pill to take." That's very interesting. The pill, aromatase inhibitors, there's a class of them, they compete with each other for the market, given to breast cancer survivors who have had an estrogen-based or hormone-based breast cancer.
- 6.56** The reason is they have clinical studies that say the recurrence of breast cancer in the next 5 years or 10 years is a few percent less than if you don't take this pill. Because what the pill does is it takes the last vestiges of estrogen out of your body. It makes sure that it blocks it as they're coming in. Even if you don't take hormone replacement therapy, the body produces some estrogen naturally.
- 6.57** While you might imagine that the list of side effects for those pills is legendary, 30% to 40% of the patients who start them don't stay with them because of the side effects. It's joint pain. There's a whole litany of problems, and they're difficult pills. But that's how the pharmaceutical industry comes in. You make your own decision. The doctor says to you, "Here's the pill. See if you can tolerate it." You might have a lower percentage chance of the cancer recurring. Then you can look at all the information, decide whether you want to take that pill or not. But that's how pharma comes into that part.
- 6.58** Corporate partnerships, cause marketing. I love this as this develops in the '80s, the idea that you could have marketing for a cause. Pink ribbon products, a portion of your sales, go to charities, a portion of the sales, go to pink ribbon during Breast Cancer Month, gain positive branding. They will do the internal polls afterwards with consumers to see if the consumers felt better about that company having been involved in branding with the pink ribbon movement, and all of those results come in and are very, very good for them.
- 6.59** And politicians in the optics of caring. So supporting breast cancer initiatives becomes an easy win for politicians. But you'll see men and women on both sides of the aisle, bipartisan support. Everyone's against breast cancer. So they're all against a good move that might be able to photo opportunities and bring down the risk of breast cancer.
- 6.60** The dominance of this narrative is really this complex web, as I say. It's relationships between nonprofits, corporations, politicians. They all have a different goal. The activists who create the pink ribbon brand are interested in research and doing something about breast cancer. But the corporations and politicians who may give it a boost are interested in sometimes doing that, but also interested in the halo effect that they get as a result of working with them.

- 6.61** Spotlight on breast cancer. Prior to the '80s, low priority for researchers and nonprofits. Hard to imagine, but true. Leukemia, big. Lung cancer, big. Lung cancer, death rate, still terrible. But have come down with a lower smoking. But very little on breast cancer. '80s, sharp increase in the number of breast cancer cases. More talking about it because there's a wider distribution of the mammogram machines.
- 6.62** So now the question is, you know the magazine cover that says 1 in 10 women may get it in their lifetime. The question is, 1 in 10 women develop breast cancer back in the 1700s or 1600s or 1800s? We don't know. Are we just able to detect it now with the look, the ability to use rays to look into the breast and see tumors at a stage where we might never have seen them because the difficulty here is, and this is always the underlying problem.
- 6.63** Doctors cannot tell. They can do a biopsy and they will tell whether it's malignant or not. But even when they've done the biopsy, even when they find that it's malignant, they can't really tell you and they don't know if it is a slow-growing tumor that would never cause a significant problem for you and you would outlive, or it's a fast-growing tumor, and if it's not out, you will die.
- 6.64** Now, the one change is, to the extent that it's discovered earlier, if it's a less intrusive surgical process after the development of lumpectomy, so that if people have waited and a breast cancer has spread and it's in the lymph nodes or it's metastasized, then you're looking at chemo, and then you're looking at radical mastectomy. Early on, you may be, after 1981, [able] to have a smaller procedure, avoid all of the treatments, and still emerge healthier down the road.
- 6.65** Every decade since the '80s—hard to imagine—the rate of breast cancer has spiked by double digits. Now, you can have a debate all day long, and I've got two boxes of documents sitting in the corner of my home office about the debate over, is that too much diagnosis? Not too early diagnosis, or is that really a development of the risk? And something's wrong.
- 6.66** There's a cluster, for instance, for years. I remember this from growing up in San Francisco. In Northern California, Marin County, a well-to-do affluent area, there's a cluster of breast cancer, the high incidence rate. People have looked into for years why that is. They've never really figured out the reason for it. They've looked at hormone replacement therapy, more of the women were on it, a whole bunch of things, but they haven't figured it out. So the clusters, we don't always know. We still don't have the answers.
- 6.67** Nancy Brinker, back to her for a second. She establishes Komen in '82 because her sister Susie had died from breast cancer. She really becomes— she's indefatigable. She doesn't stop. If she sleeps, I don't know when, but when she first had this, this was a mission for her. She is pushing it and she's telling patients to become advocates.

- 6.68** I really think you see the emergence of the breast cancer advocacy groups like Komen, the National Breast Cancer Coalition. A fantastic group that comes together, and they become advocates together lobbying for more funding because what do we all believe? Mary Lasker believed it when she was pushing the American Cancer Society. If you get more funding, if we get federal funding, we'll cure the disease.
- 6.69** Nixon believed it. The war on cancer, that's triple the budget of the NCI. President after president has believed it. Barack Obama, when he talked about a moonshot and increasing, if we get the money there, we'll get the solutions. It's a tough, it's a difficult disease. But at this point, they're advocating for more money.
- 6.70** Effective fundraising campaigns. Well, I will tell you, the pink ribbon comes into effect. They have generated significant revenues, research, and awareness efforts, and they are having success with that. And they become influential lobbyists. They utilize their financial resources and their public support time and time again to increase more money for breast cancer research, and they are seeing the increase in that across the board.
- 6.71** I view this period of patient advocacy, post-Susan Komen, the breast cancer initiatives that are taking place as a remarkable development in a couple of decades from the time that [nobody] would walk into an office and a male surgeon would say, "That's how we do it. You're a silly woman for questioning this, and you'll be dead in three weeks."
- 6.72** So breast cancer events, 2013. Angelina Jolie, public figures. Public figures can inspire you, and they do. Sharing public stories is fantastic. They make people feel good. Angelina Jolie tested positive for the BRCA gene, which is a genetic mutation. That means you have a 70% or 80% possibility in your lifetime of developing breast cancer. It doesn't mean you will with certainty, but you have a very high possibility.
- 6.73** Trisha did not have the BRCA gene. It just wasn't present. But many women who have the BRCA gene, test positive for it, are then faced with this quandary of, what do I do? Do I just sit back, wait, live my life, enjoy it, and then continue to monitor with annual mammograms, whether, in fact, I eventually have breast cancer, or some started to do a radical mastectomy based upon knowing you had the gene in the thought that that's going to take away the chance of breast cancer by essentially doing the surgery that would be done if I had extensive breast cancer.
- 6.74** Angelina Jolie did that. She didn't have breast cancer. She tested positive for the BRCA gene, and she had the double mastectomy, and she talked about it. As a result, doctors inside the cancer field and in the industry and in hospitals call it the Jolie effect because they suddenly had women coming who said, "I want to get tested for the BRCA gene."
- 6.75** In many instances, they said there was genetic testing caused on this. There's a surge in the system. The health system is spending it because people are thinking, if she did it, that must be the way to go ahead and do it. Really, researchers believe that breast

cancer rates could be reduced if women were more [aware of] preventable cancers such as abortion. We're going to talk about that in a moment.

- 6.76** And the breast cancer industry. So it's embraced a testing frenzy and the uptick in what —now think about this. Preventative radical mastectomy. Boy, that is a radical solution. You're doing it as a prophylactic means to prevent cancer in the future. That's very difficult. Harvard researchers concluded that the press about the Jolie effect caused the health system to have \$14 million in unnecessary tests for people who got tested, were worried, fearful, and they didn't have the BRCA gene. In other cases, some who did, it ended up following the surgical route.
- 6.77** Now, the success of the breast cancer brand gets some pushback because here are some of the criticisms. The movement advocates for other diseases criticize the breast cancer movement because they say, "You're getting funding at our expense." Well, what does that tell you? It's been super successful. That tells you the brand has really worked because when people start to say, "Hey, too much money, too much money being spent on breast cancer. What about all these other cancers over here? We want to have some money on it." That's the best example that you are actually doing it.
- 6.78** An uneven playing field. This is from the National Institutes of Health. This is a slide actually coming in from 2007. I'll tell you why in a second I chose such a back-in-the-day slide. The figures are not different, they're just larger as you go along in time. Breast cancer was getting \$705 million from the National Institutes of Health for research and study. Lung cancer, big killer, often lethal, \$226 million in fraction. Pancreatic cancer, \$73 million, in part because so many researchers thought pancreatic cancer was a death sentence. My mother died of pancreatic cancer, so I know it can be, but very little money going into it. Cardiovascular disease, \$381 million. And heart attacks, the biggest killer. Influenza, \$40 million. In urinary incontinence, \$6 million. I'm not sure how that made the list, but it did.
- 6.79** Now, the reason I mentioned this is this was created by the cardiovascular lobby. They were passing this all around Congress to say, "This is disgraceful. This is terrible. Heart attacks kill more Americans than anybody else. And look where we are on the list, \$381 million." How is it that breast cancer has almost doubled that?
- 6.80** If breast cancer isn't killing that many, it doesn't even affect men. Well, 1%. So that's when you know the brand has been good. They have focused the public on doing that. I will tell you that in terms of politicians and in terms of funding and in terms of corporate branding, it seemed as though there was no risk if you were Apple or General Motors or somebody else co-branding on a breast cancer month with the pink ribbon. But nobody really wanted to be co-branding on pancreatic cancer or urinary incontinence or lung cancer necessarily. It doesn't seem like a saleable thing. The very fact that it's a pink ribbon, very unobtrusive in that sense, nothing terrible. It seems like it's an easier marketing tool for those companies.

- 6.81** Nonprofit funding priorities. It doesn't always correlate or compare with the lethal cancer values. These are values as patients. These are large national organizations who are the grant makers. Here's what ends up happening. Six percent of the money from these large national nonprofits in raising money, they give out themselves, forget the government, 6% goes to leukemia. There are leukemia-specific foundations, but in terms of the nonprofit money that's heading out from that, that's 6%, 8% going to prostate cancer, 9% for colon cancer, 15% for lung cancer, and 33%, one out of every \$3 out of these major grant makers and nonprofits going to breast cancer.
- 6.82** Again, I'm not saying that's a bad thing. I just marvel at the success of the brand. The brand has really come from the days of Nancy Brinker and others. When you're told as an officer, silly woman, boy, there's no silly women here who are creating this breast cancer awareness.
- 6.83** Now, the concept of pinkwashing, as I call it. There's some people now, you know your success when activists are no longer forming the entity with you, but they're criticizing you. There are some activists who criticize Komen and others by saying, "You create a veneer of altruism, and you allow them to do other policies that are objectionable, like using underpaid workers in Asian countries and having workplace carcinogens, and they get away with that because they're doing pink laundry, and so you're essentially washing them."
- 6.84** I don't buy that. Corporations would be doing—they would be still employing low-wage workers in other countries offshore and still delaying fixes to workplace carcinogens, whether or not they were doing pink branding. So the pink branding isn't the reason. But the view is making the public feel better.
- 6.85** Emphasis on profit over purpose. So again, advocates argue that certain corporations prioritize profit over genuine support of breast cancer research and patient care, and they're only using the pink ribbon to boost their public image. I'm sorry to say, but my feeling on that is, "Duh, of course." Do you think that everybody who, as a company, signs up to do a breast cancer month or run or sponsor it—everybody in that director's table makes that decision is really passionate about fighting breast cancer? No. They're worried about their public image, and they've decided that's what works. That's not a surprise for me.
- 6.86** Lack of transparency. How much money is raised through the pink ribbon campaigns goes toward breast cancer research and support. There were some complaints early on about transparency, but I will tell you that the transparency is there. You go to 990s, these tax forms that are filed for groups like Komen, you can see how much money they take in.
- 6.87** You can see how much money they spend on administrative. You can see how much money gets paid to the person that's the executive director who runs it and how much money they're sending out. Now, do they sometimes get their favorites in terms of

where the donations go—hospitals and clinics and that? No question about it. But you can follow the trail of money pretty well.

- 6.88** Environmental concerns. Pinkwashing products that companies aren't looking at carcinogens. That's not going to change if pinkwashing isn't there. The need for accountability. I've seen this constantly. You'd be surprised how many times activists have said, "We want more accountability for the partnerships you work with on moving for breast cancer forward." I think that accountability is there.
- 6.89** Now, you can criticize it if you want to. If you go through all the money, you can say, "You know what? I think you're picking the wrong organizations, the wrong hospitals. You're putting too much into early detection with mammograms. You're putting too much in a treatment." That's a different decision. But you have to get involved in the details to see that. They're raising the money and they are spending it.
- 6.90** Now, the search for a cure and a need for balance. Percentage of breast cancer research funding versus other cancers, clearly very large. The percentage of breast cancer survivors actively involved in advocacy. Remember before we talked about the idea that Mary Lasker had gone ahead and she went to families that she knew or people that she didn't know who were very, very wealthy on the Forbes 400 list. If she knew that they had been affected by cancer diagnosis somewhere in the family, and said to them, "Do you want to fund the American Cancer Society?" So they came in. It was very smart.
- 6.91** Well, people who have had breast cancer become the people that organizations go back to it right away and say, "By the way, you survived. Fantastic. You survived in part because of us because we've changed the narrative. So become a regular contributor." They do become significant contributors over time. Percentage of breast cancer research funding allocated to prevention? Small. Less than half. I understand that. It's the same thing we talked about the other day. A case of cancer prevented is invisible. Nobody knows it.
- 6.92** The people who worked in the plastics industry after OSHA had changed the exposure to vinyl chloride to 50 parts per million. They did not necessarily develop liver cancer in that rare liver cancer, but they're not counted as victories in the war on cancer because we don't know it. You can't say. Yes, you can look at the numbers of workers at the higher levels of vinyl chloride who developed this rare cancer in the liver, and you could say, "Well, it was 3%, 2%. And afterwards, it was 0.1%, a tenth of a percent." So maybe X amount of lives were saved, but it's guesswork.
- 6.93** So the effort on prevention is always less than the ability to save five-year survivors. Trisha—this August becomes a five-year survivor. She becomes part of that statistic that breast cancer and other organizations will say the survival rate is better than ever. And percentage of breast cancer research funding allocated to reducing disparities in treatment because one of the problems is women with access to more treatment, able

to get mammograms, go to doctors, women who are often Medicaid, poor, don't have the access to the information, higher rate of breast cancer. There are criticisms about Komen and others reaching out to all of them as opposed to just reaching out to the women who are making the decision to say, "I'm going ahead and taking care of my health day in and day out."

- 6.94** Now, here's a Komen Planned Parenthood controversy. In the story of the politics of cancer, this is not necessarily one of the biggest by far. Why am I talking to you about it? Because I developed this course just for presentation here at the Peterson Academy, and I get to put it together as I want to deliver it. I find this a fascinating story because I knew nothing about it when I started to put together the story of the industry. Here's what it is after a sip of water.
- 6.95** In 2012, some of you may know this, but I missed it. In 2012, because I wasn't following it, Komen, which is the biggest of all of the pink brands, decides to stop sending money to Planned Parenthood. That's because Planned Parenthood is, at that point, the largest provider in the country for abortions, in terms of what they were doing. It's part of their services for women, having the ability to control their reproductive rights.
- 6.96** That decision really created a firestorm in the industry. Now, why did Komen stop sending money over to Planned Parenthood? Because they're anti-abortion? No, because they were relying on a series of studies that said that early termination of pregnancy, whether induced through a medical procedure like abortion or sometimes—as you know, a pregnancy will just terminate on its own.
- 6.97** Any early termination of pregnancy increases the risk of breast cancer later on. So Komen said, "That's a risk. You're providing the largest number of abortions. We're not sending any money over." Boy, I'll tell you, I'm not sure how I missed that. I'm not sure why that wasn't bigger news. But I do know, I guess, because the minute you say the word "abortion," you automatically are putting people on their sides. They're ready to fight pro-life, pro-choice, and they think it's about that.
- 6.98** It's not. For me, it's about this very interesting question. Is that true? I'm more interested in the fact of I never heard that before as a cancer risk. What's the American Cancer Society say about it? What do others say about it? Even if it's true, it doesn't mean necessarily a woman who's going to have an abortion decides not to, but it's a risk factor.
- 6.99** Remember before when I said a very small number of people might get a cancer induced from radiation from doing detection for decades, over 10 years? The same thing here. Tell me if I'm the patient, and then I can decide if I'm still doing it. If I'm not worried about that risk, fine.
- 6.100** So Cecile Richards, she's the former president of Planned Parenthood. And they are the largest national provider of reproductive health services. So she was in that role from 2006 to 2018. She's a strong leader for Planned Parenthood. She was very, very diligent

about doing her work. She's a key figure. When they stop funding the money from Komen, she doesn't take it very lightly. She went on afterwards.

- 6.101** When she leaves Planned Parenthood in, [2018] she actually formed a political group called Supermajority. She ended up getting the Presidential Medal of Freedom from the Biden administration. She was diagnosed herself with an aggressive form of brain cancer and she ended up dying in. [2025] But she ran Planned Parenthood with a firm hand, politically connected, and she was savvy, so nobody could underestimate her at all.
- 6.102** Here's the issue of breast cancer and possibly an early termination of a pregnancy. Researchers certainly believe that rates could be significantly reduced if women took certain precautions, and on that list is an induced abortion. Now, you say, I've never heard of that before. Here's the abortion list. Not one, not two, not six. I'm talking about numerous studies have statistically linked an induced abortion, especially, or termination of the pregnancy.
- 6.103** It doesn't have to be an abortion. Very important to think. If you're just a woman, you're pregnant, in the first trimester, you lose a child, there's an abortion that takes place by the body doing it. There's an increased risk of developing breast cancer later in life. Now, here's the mechanism of the risk. Pregnancy hormones—who would have known this— actually provide what the studies show as a temporary cancer resistance to 85% on average of a woman's breast tissue.
- 6.104** When pregnant, the hormones that are produced for pregnancy are anti-cancer hormones. They aren't being produced as anti-cancer hormones. They're hormones getting ready for the birth of the child, but that is their effect. When a pregnancy is terminated or ends before 32 weeks, the breast tissue remains partially matured and now has more cancer-susceptible sites. So the sites for a tumor increase.
- 6.105** It doesn't mean you're going to get breast cancer. I don't want everybody who's ever lost a child or had an abortion in the first three months, think, "Oh, my goodness, I'm going to—" Not at all. Not at all. The risk is small, but it's there. It's one of those risks I always say to talk about.
- 6.106** So the long latency period. The average age for a woman having an abortion in the United States is in the early 20s. The median age for a breast cancer diagnosis is 61. That's pretty late along the way. That long latency period is why it's so difficult to make the connection between the two. People say, "I think this is garbage. It's just not true. You're putting together something that happened long ago."
- 6.107** But I think back to Elizabeth [Lane-Claypon], who was the English doctor who had a study early on when nobody believed her that said 500 women who have breast cancer and 500 who don't and decided that not having children is a higher risk for breast cancer than having children. Breastfeeding lowers your risk for breast cancer than if you've never breastfed before. Trisha, we've never had children. She's never breastfed.

So she was in a higher risk category. But did that contribute to her breast cancer? We'll never know. But I understand the fact that there are risk factors like that.

- 6.108** Political sensitivities. No one really in the media talks about the possibility of this thing because it would be considered anti-choice. And I understand that. If you mention it, the fear is you're going to scare away women from making a choice about abortion, and that thrown in the middle of the abortion debate is far too hot.
- 6.109** So sealed documents and material that's now come out in some court litigation shows you that the cancer nonprofits, the ACS and others, they talk about it. You go to the American Cancer Society and you look through their site, you'll find a link about the possibility of early termination of pregnancy and whether it's an additional risk factor for breast cancer. And they talk about it as a link, but not a risk factor. There may be a link, but they don't go to the next stage of actually calling it a risk factor.
- 6.110** That fear is very, very easy. They would lose donations, possibly from a lot of women. They would say, "What are you talking about?" They're very careful. They tread it very carefully. Numerous studies on abortion, breast cancer link from 1957 and on. We're not just talking about it. Statistically significant link between an induced or spontaneous abortion. When I mean spontaneous, it happens naturally.
- 6.111** Those women, a higher risk of breast cancer in the future by the link. The breast cancer susceptibility window, as they call it, is: a full-term pregnancy provides a protection, while the termination before 32 weeks creates more susceptible tissue. The latency period—the long wait between the time that the pregnancy has ended and the development of breast cancer—very hard to make that link solid. Certainly nonprofits do not advertise it and push it forward.
- 6.112** My point about this is, informed consent. I'm one of those buzz kills because I just think you should have all the information. I know that. I know that there are patients. I talk to people. I talk to people all the time about things, and they say, "I don't want to know. Don't tell me. It'll take away my positive attitude. It'll put some fear in my mind."
- 6.113** There were times I was going through this material and preparing this course on the politics of cancer when Trisha, a cancer survivor, would say to me, "I just have to take a break and walk out of the room because it's getting under my skin." So I get it. It's hard to think that I'm not here saying, "Oh, my God, if you had an early—" You didn't follow through on a pregnancy, you're going to get breast cancer down the road.
- 6.114** I'm just saying that that link might be of interest to somebody when they do walk into a Planned Parenthood. It's happened, but they're never going to—that information doesn't get passed along. Only three [days] after Nancy Brinker cuts her funding to Planned Parenthood, she backs down. She restores the money because Komen for the Cure was getting slammed for cutting off money that really worked for thousands of lower-income women who were getting mammographies and everything else.

- 6.115** I tell you that it was a moment at which the pink brand took a black eye. People don't—but if you look at the follow-up press to that, she said in a statement, "We want to apologize to the American public for our recent decisions to cast doubt upon our commitment to our mission of saving women's lives." That was a pretty fast backtrack for her.
- 6.116** Environmental exposures, very, very quickly. The studies think certain chemical solvents and endocrine disruptors to increase breast cancer. So those studies are out there. Occupational risk, what do you think the fields are that are bad?
- 6.117** You don't have to be working in a factory with plastics. You don't have to be working in a mine with radon gas. You can be in beauty, cleaning, agriculture, pesticide exposure, all of that higher risk. Major pink ribbon events rarely highlight those environmental occupational factors. Komen never cut off or talked about having a boycott against beauty salons that were still using chemicals that might be dangerous. There's still some part of that. That narrative fails to discuss the role of environmental occupational risk factors often.
- 6.118** The pink ribbon machine in the malignant empire, for me, this breast cancer branding, it aligns with corporate interests, so it's perfect. Policies, reputations at all costs. Everybody likes it. It's prevention based on environmental occupational risk is minimized. Corporations love that. Hard questions about drug pricing and profit margins are rarely front and center.
- 6.119** So Komen and other pink brand organizations talk often about making care accessible to all, even those without insurance or money. But we're going to find out that some of the latest developments in cancer care under immunology and biology and a whole new set of science that looked like science fiction at times, they're available to the very wealthy or insurance coverage that's willing to pay for it. A lot of people are ostracized from that who don't have that ability.
- 6.120** To me, this is the clearest example of what I call this blending of different allies. The future of cancer care—that's what we're going to be talking about next—emerging immunotherapies, including checkpoint inhibitors and CAR T-cell therapies. I don't usually say it has the ability to revolutionize cancer care because I always think that that is some hack who's a health writer who's been given a PR notice by the company to say, by the way, go ahead and say it's revolutionary. But some of this material is revolutionary, potentially.
- 6.121** Precision medicine, genomics, discussing the rise of personalized genetic-based cancer care and the promise of targeted treatments in the future. Early detection through AI. It does more than just answer a question. If you ask ChatGPT a question, it can be used in serious ways in cancer care and in other medical. And liquid biopsies, we'll talk about that. Shifting the focus to prevention, which I would like to see. That's the need that I talk about. And patient-centered care and shared decision-making.

- 6.122** Nancy Brinker, Shirley Temple Black, the women who had cancers affect their family or themselves, and went through this process and gave women a role as patients at the table to be able to confront the doctors and to change the medical community to say that if you need surgery, not everybody needs surgery, but if you need surgery, you have an option of breast conservation as opposed to what had been used for 100 years beforehand.
- 6.123** There are no choices here that are easy. Having gone through this as an observer, I can assure you that if you go through it yourself—and it's not just breast cancer, it's not any cancer—all you can ask for as an informed consumer is to say to the doctor, the oncologist, the treating physicians, "I don't like the fact that I have cancer. I'm afraid of cancer—you can say that—but I'm not afraid of information. Tell me what you know and what you don't know. Tell me what the latest is. Tell me that there's a treatment that I should consider, and then what the risk to that is."
- 6.124** Then you can go home and with loved ones and friends and everybody else, you can try to make some sense of it and figure out how you proceed and how you are comfortable. None of these, but it is remarkable to me that the industry, the surgical industry, kept a monopoly on the most radical of the surgeries, which can save lives in many instances, but didn't allow anybody else to consider a clinical trial to see if it might be just as good.
- 6.125** Thank goodness, that was eventually done. It took until '81 to get the results, but that was a great thing. Thank you very much.

Lecture 7: Biotechnology Breakthroughs

At a glance — Biotechnology breakthroughs.

- 7.1 Something that Trisha reminded me of when I finished the last lecture, and I'm glad she did. She said, "You know, by the way, when you're talking about Angelina Jolie and that she in 2013, based upon her BRCA test—her grandmother died of ovarian cancer, and her mother had breast cancer and died of ovarian cancer. And 3 weeks after she had her surgery, her maternal aunt died of breast cancer." So it puts that risk front and center. Having the surgery dropped it. They said your BRCA1 gene makes you 87% likely to have breast cancer in your life, and after the surgery, to 5%.
- 7.2 So I wanted to make clear that the problem with the Jolie effect was the aftermath. I'm not criticizing her decision, fair enough, but that people then rushed into the system based upon what a celebrity or public figure did, as opposed to actually thinking about the consequences of it. She probably waited very, very carefully.
- 7.3 Now, one other thing, if I can, to go back before starting number seven. I can't believe I forgot this from the very beginning because it was the first thing I was going to tell you today. So what am I doing last night at 10:30 at night? I'm starting to research whether, in fact, they cremated Marie Curie's body. Would it still have been radioactive? The answer is yes. But of course, I'm incapable of just getting that little bit of information. So I wish it was. I mean, Trisha is saying it a half an hour later, "We have to get up early. We have to get up early."
- 7.4 Of course, I forget, radium is an element. So heat, burning it, doesn't destroy it and doesn't neutralize it. If you cremate it, you actually lose the mass, the bones in the body, and in some ways, it's more concentrated in the ash. It's possible to be dangerous if it's inhaled in the ash.
- 7.5 Crematoriums across the country know this, that if somebody has been treated with the radionuclide treatments for cancer, in which a radioactive isotope or that is put into their body, and therefore, they emit low levels of radiation. Depending on the isotope that's used, it's from 8 hours to 10 hours to a few months. There's a period where if somebody passes away after being given one of those radioactive treatments, they, in fact, are radioactive. And so Marie Curie would still be radioactive after all of this time.
- 7.6 And there are rules in the crematorium to follow. What, of course, I was wondering was what happens if somebody's been treated with a radioactive isotope and they're emitting radiation and they've passed away and they end up not at a crematorium, but at a funeral home where nobody knows that they were just treated with that? That's the rabbit hole you sent me down last night to answer that question about Marie Curie.
- 7.7 It turns out there's no requirement or regulation that makes hospitals inform funeral homes if people have passed away, even inside the hospital, that that person may, in

fact, be radioactive. It is considered an occupational low risk for funeral home employees that they might deal with corpses that they aren't aware are actually emitting low levels of radiation. There you have it. More than you could have said. Thank you for the question. It was fascinating to find that out.

- 7.8** Here we're talking today about biotechnology—promise, perils, and the pace of ethics, because sometimes technology moves so fast that you won't be surprised that the ethical considerations lag behind by a little bit. Setting the stage. We know the early treatments—surgery, radiation, chemo—and the political war on cancer. Then we go through the whole part of the regulatory cover-up, the [carcinogens] that the industries won't deal with, the pink ribbon brand. Then we're into a whole new world of immunotherapy, CAR T, which I'll explain to you, and CRISPR, as it's called.
- 7.9** Blunt tool versus targeted magic. Boy, this is really it. Here's the question. Think about this. Chemo poisons fast-growing cancer cells. It's really what it's doing. It's killing them off by poisoning them. Problem is [also] affecting other tissue or cells nearby. Collateral damage is what it is. The new biotechnologies often want to harness your own immune system to fight the cells that have gone errant and are replicating without being controlled. The idea of this, in theory, is fantastic.
- 7.10** So maybe we could reprogram T cells. Maybe we could make the DNA itself change its course and stop this wildly replicating group of cells that are going to eventually become cancerous. The idea of this really is something that's come of age in some ways. The idea that we could have specific medications that could target DNA mutations, if you think about it, possibly targeting even mutations inside the tumor. So you have a tumor, you don't need to necessarily excise it anymore through surgery. That's some of the theories on this. But you might be able to attack the DNA changes inside the tumor, offering the chance to essentially kick start the patient's own immune system to destroy and have the tumor essentially disappear.
- 7.11** Now, if I said that to you 40 years ago, you would have thought I was absolutely crazy, and that I was a science fiction writer. Actually, 1980 is when you start to have the biotechnology age, but still it would have seemed a far way away from that. First of all, before we get into some of these, the remarkable history of cancer immunotherapy. I do like this because you know by now that I like history because history sets the predicate for where we end up.
- 7.12** That's the father of cancer immunotherapy, William Coley. You know when he's doing this, when he's beginning experiments? 1891. Immunotherapy is something you think, God, that has to be happening really like now. It's not something that seems so long. He's an American surgeon at the New York Cancer Hospital. He has a patient, an Italian immigrant, whose name was Zola, who has a tumor on his right tonsil that Coley describes in one of his reports as the size of a small hen's egg.

- 7.13** Now, Coley comes up with this idea that other doctors thought was absolutely insane, and they often criticized him for it, of taking a live virus, or in this case, Streptococcus, which, if you get infected with it, can in fact cause this severe blistering in skin infections. So that's the result of it. Because in his mind, if he's taking something that really causes—it's called erysipelas, the disease or the outbreak that comes from Streptococcus. He thinks if he's able to create that on this tumor, infect it with something more virulent than the tumor itself, it will attack the tumor, which is the idea of immunotherapy, and be able to reduce it.
- 7.14** Now, Zola knew if he wasn't treated, he would be near death soon. But I'm sure that this was not a comfortable procedure as it was done over a period of time. It's 5 months of trial, so of injections over a period of time. And finally, the patient develops a full-blown infection in his throat and is very, very sick from it. But the remarkable thing is the tumor once thought to be inoperable and unable to be saved starts to dissolve and soon disappears. That is what you would call a medical miracle. He went on to live for another 8 years. That is also fairly good. And then the cancer returned, and this time it was fatal. But it really was after that point.
- 7.15** So Coley takes that as the starting point, as what I call this father of cancer immunotherapy, and he starts then to work on other patients with large tumors. He has 10 other patients. He puts them through the same Streptococcus bacteria treatments. But you know what? This won't surprise you. Infections caused by agents like Streptococcus are notoriously impossible to predict accurately, so you can't control them. So sometimes that worked to bring away the tumor that was in Zola's neck. But in other cases, that infection spread. Two of his patients on two different occasions end up dying.
- 7.16** He changes from the live Streptococcus microorganism to a dead one, what they've done on the flu shot, what they did on the polio vaccine afterwards, to avoid those possibilities. But still, it's very controversial. I don't want you to think that anyone at this point was running to him and saying, "You know what? This is a great new area you've opened up." They all thought that he was on the edge of practicing a form of medicine that wasn't very good. But that's really the idea of immunotherapy, and it goes back to him.
- 7.17** So, by the way, 30 years before he was doing anything, these German doctors really made the first breakthrough. Wilhelm Busch and Friedrich Fehleisen, they observed tumor regression in patients who were afflicted with the very same breakout that had been caused by Streptococcus. That's where Coley gets his idea from.
- 7.18** Then he, cancer researcher and bone surgeon—but remember, when I'm talking about the 1890s, don't give too much credence to cancer researcher because there's not much that they know about the disease other than what they can see. And he starts injecting that live bacteria into those tumors. And he gets regression. The big case is his first case.

- 7.19** Then you go forward 50 years. They're actually called Coley's toxins. So something is named after him. And those toxins are cancer treatments that are used as infectious agents. Doctors really don't understand how they work, but they understand that it fights the cancer, and it comes around the same time as irradiation therapy and chemotherapy are coming out.
- 7.20** So nobody views it as a major pathway forward on treating cancer, but they know it's there as this fringe treatment. And today, after all these years, not just in this presentation, but Coley is widely hailed in the medical community as having been this pioneer in developing immunotherapies.
- 7.21** So one of the first modern ones that really comes into use are interferons. Some of you may have heard about interferons over time. They're a class of—and it's interesting because they really end up being a class of cytokines that are these proteins inside of cells that produce what I call a response in the cell. If you have a viral infection, they start to produce the immune response. And they're discovered back in the 1950s. Their antiviral and immune-regulatory properties were recognized pretty early, but it's not until the late '60s that interferons are used by researchers in cancer therapies to say, can they also reduce or suppress the growth of tumors? Maybe they can if they can activate some part of the immune response to those tumors.
- 7.22** There's a series of studies that take place at a bunch of different hospitals for a period of about 15 years, and it's in 1986 that the FDA approves the first cancer treatment, this IFN-A. It's a form of interferon. It's an immunotherapy for treating hairy cell leukemia.
- 7.23** I will tell you, there are diseases named after all types of things. Whoever decided to call it hairy cell leukemia was clearly a doctor looking through a microscope at what looked like hairy cells. It's a rare form of leukemia. It's a subset of leukemia, and it's very virulent, and it's rather rare, and it has the worst name of all. So not only, if you are afflicted with hairy cell leukemia, do you have all the worries about worrying about it, but you have to tell friends and family that you have hairy cell leukemia seem somehow even degrading to that point, but they've never renamed it. So that's what it was originally approved for. Sorry for that side rant.
- 7.24** And it has its side effects, but it remains this part of cancer immunotherapy that is being explored with other agents. So interferons are there on what I call the drawing board, and they get an FDA approval.
- 7.25** In 1986, biotechnology was in bloom for the following thing. So here's the '80s in biotechnology. Genentech creates the first—they are a biotechnology firm without a product, and they are able to create—there's a massive story beyond this of how they finally do it, but they create synthetic insulin. Pretty amazing in the early '80s. That is the beginning of the biotechnology and pharmaceutical business.

- 7.26** Now, as a sidelight, we don't have time for it here in this part of the lecture about the politics of cancer, but there is an hour lecture somewhere along the line about how is it possible that a company created a synthetic form of insulin and the price didn't go down? We used to have to take it from—right? You have to get cows and everything else, and now we can do it synthetically. We can do it in the lab. We can just manufacture it off the assembly line. Boy, the price of insulin will go down to nothing.
- 7.27** But you get a patent on it, you can control it. All the companies come in, they say, "Now let's play nice." Play nice means don't start to undercut us. No one wants a price war here. You can make a lot of money with your synthetic insulin. You're priced near the price of the insulin that we're manufacturing over here. And so there's a process in which insulin prices still stay high, even though this potential supply is unlimited.
- 7.28** Recombinant DNA technology. So biotech. By the way, Genentech, before it even finished the synthetic insulin, its stock price doubled on its first day of the IPO. It was up 500% within a short amount of time. There was a rush. There was a rush of startups and venture capital money to create biotech companies. They didn't know what the products would be, but the idea that one company had done synthetic insulin meant, let's go, this is the future. Biotech was the buzzword in the '80s. And that's when you get the approval of the first interferon. So this idea that they are starting to engineer, work with bioengineered living cells and recombinant DNA technologies to develop new therapies, they don't know where that leads.
- 7.29** Launch of the Human Genome Project. So the National Academy of Sciences in [1988 recommends] this ambitious program to say, let's do the full human blueprint, figure out the entire blueprint of the human body, and that will be the key for some of these technologies to treat people later for diseases and everything else. It takes 13 years to do that. Well, I'll tell you, it's a long time. It's a big project.
- 7.30** It's not until 2003 that you're finally getting some of that work being done. It really was the basis, as I say, for some of the biotech products that come in the future. You get the first synthetic insulin. That's the beginning of it. You have the recombinant DNA, launched the Human Genome Project, and that idea of the new possibilities for personalized medicine has taken a grip inside the cancer field as well.
- 7.31** Interleukin-2, IL-2. In the 1970s, before the biotech companies were around, researchers had already identified interleukin-2, as it's called, as a T-cell growth factor, and they found out that it plays a crucial role in stimulating and activating the body's natural immune system. That's a discovery they have.
- 7.32** Now you come into the 1980s, the age of biotech, and Dr. Steven Rosenberg, that doctor there at the National Cancer Institute, is doing clinical studies to see if they administer high doses of IL-2 to patients with advanced late-stage melanoma and kidney cancer, there's any effect. And those trials really show potential for IL-2 to lead to the partial, and in some cases, complete remission of tumors. Now, we know enough

by this point to know that remission doesn't mean cure. Fair enough. But remission is the one method used constantly in terms of biotech tools to determine whether they are successful or at least being effective.

- 7.33** The treatment that he ends up getting, he develops a thing called the adoptive cell transfer, in which he takes these interleukin-2 cells, extracts them from a tumor, grows them in large numbers in a lab, and then administers them back to the patient. So the patient's getting like the cavalry is coming in, an extra dose of what they had in their original body taken directly from the tumor in order to attack the tumor this time.
- 7.34** Hundreds of academic centers are doing the same research as he's doing at this time. So this is across the board. It's not like he's doing it on his own. He's not like Coley out there in the wilds with this idea. His work is being followed. And he's the first to use genetically engineered immune cells known as CAR T cells—we'll talk about that in a bit—to treat patients with aggressive lymphomas. It really is, to me, in some ways, the first breakthrough on treating immunotherapy in a direct way by reinfusing patients that did not respond to standard treatments.
- 7.35** Here's another little caveat. When they were looking for patients for these trials, they're looking for patients in the worst condition. They're looking for patients who have these aggressive cancers and have been treated with other and didn't respond, and the tumors didn't shrink. That's why they get the ability to see they aren't treating them for the first time with this. They're coming back to it.
- 7.36** The promise of CAR T therapy. I call it an extraordinary biological breakthrough, and I really think it is. You are engineering in a lab, synthetically, a patient's own T cells—here's the theory of it—to recognize and destroy cancer cells. It's pretty impressive on paper. The initial hype surrounding CAR T's potential is fueled by clinical trials which really did push the envelope about what was happening.
- 7.37** I'll tell you that the way that it works, we found out about it in 2012. We, meaning the lay public. I remember reading the stories at the time that a 6-year-old girl who had been seriously ill with an acute form of leukemia had been cured through a trial application of CAR T therapy. Now, that 6-year-old is today turning 20. She'll be 20 this year. Pretty remarkable. So it's not one of those stories, like I've told you so many times in the past, in which somebody is temporarily cured, has remission, and then 6 months later, they're dead or the cancer returns. She's still alive. I have followed her story on and off for a number of years to see that she's still alive. The longer that she's alive, the more I start to believe, maybe CAR T can actually make a difference.
- 7.38** Now, this is the father of CAR T cell therapy. He's the one who does the 2012: Carl June. He removes the patient's immune cells, then he comes back and engineers them in a lab to recognize and attack the cancer in that patient and puts the cells back into the body to do the job.

- 7.39 It took him until 2017 to get FDA approval for two different cancers: childhood and young adult acute leukemia, and a few rare lymphomas. So he's treating adult patients with CAR T at UPenn back in 2010. He gets the breakthrough with his young patient in 2012 with great results. First time he treated a pediatric patient—more at risk, the drugs can have a greater toxic effect on a younger and smaller patient. And then 5 years later, he gets the FDA approval. Now, there are seven approved CAR T therapies with around 35,000 patients who have been treated around the world.
- 7.40 Here's the idea on this. When I said before the pluses and minuses of new types of therapies, this has a tremendous plus. You can see it right away. The idea of it in theory works fantastic. And what the CAR T pioneers, like Carl June, loved is they thought it would work against many types of cancers. It may—it hasn't been tried against all types of cancers. They think it could even be used against other illnesses that you have. And the thing is, everybody, all of us, can potentially benefit from it because we all have T cells. So that's what they're working with. They're working with our own T cells. You have T cells if you're human. Those T cells can be activated.
- 7.41 I love this. In some of the early promotional material, they would talk about the companies developing CAR T therapies: "Your T cells become a magic bullet." That's the idea, right? You don't have to worry about a drug, a pharmaceutical. The T cell will turn around and do the work for you.
- 7.42 Now, why are you not surprised that, boy, startups were running for CAR T therapies, tens of millions of dollars in venture capital? Those are all the new biotech. The big 900-pound gorillas in the business, the pharma companies, they weren't just sitting back and saying, "Oh, wow, what a revolution. I guess we'll have to put aside all of our products." Novartis, Celgene, Amgen, Merck, they open up biotechnology laboratories, companies fully versed and heavily funded. They either buy up small biotechs or they're creating themselves within the companies. This is the point at which I say, you know that biotechnology is here to stay because now it's been enveloped into the top 10 pharmaceutical companies as well.
- 7.43 Now, more than 100 companies today, 2026 as we're talking, are developing CAR T therapies. So we've only seen the beginning of this. We're going to see more. You're going to read in the future about something being a CAR T therapy, and you're going to say, "I remember when I was at the Peterson Academy, and this lecture was talking about CAR T being fairly early, and now there are 600 CAR T therapies for a whole bunch of approvals from the FDA." And I think that that will happen.
- 7.44 But something to know, this is a very tough procedure. So you think it sounds like you're just taking somebody's T cells out and then you're reengineering them and you're putting them back in. You can be reinfused with engineered cells, no doubt about it. The oncologists have to create room for the new cells because otherwise they're overwhelmed by having some room in the bone marrow. So they have to take

some of the bone marrow out first, which means a bone marrow procedure, so it's very painful in the preparation for it.

- 7.45** Then they also reduce the bone marrow sometimes with high doses of chemotherapy before you get the CAR T treatment. You think you're avoiding necessarily chemo. You are in terms of curing the cancer, but they're preparing the bones to lower the white blood cell count, they want initially to have less bone marrow in there. The chemo preparation is so debilitating. The FDA has [eligibility criteria for] that part of the preparation for patients over 70. It's just too tough on patients over 70, so you [have to] consider that.
- 7.46** Now, that doesn't mean it doesn't work, but it does mean that there is no simple free lunch here. The real challenge for CAR T will be in the future having a procedure that is not so debilitating, whether they can do that, and they're working on it. That remains to be seen.
- 7.47** Here are the limitations of CAR T. After telling you what a wonder it is, and a revolutionary drug, now I'm a buzz killer. I'm going to tell you a little bit about the side [effects]. It depletes normal B cells. So one of the things it does, it depletes B cells that the tumor is relying on, that it's living on and growing. But like other [therapies], it also depletes normal B cells. So it kills leukemic B cells. Fantastic. We're standing up and applauding, but it also destroys the normal B cells in a lot of patients. So they are left severely immune-compromised and susceptible to a host of infections which could be fatal if they are not watching carefully.
- 7.48** Risk of organ failure. A massive number of cells killed by CAR T can overload the kidneys. So what happens is it leads to a release of toxic cellular debris, as I call it. The kidneys, you're getting billions of cells, literally billions of cells, cells are measured in the billions, infused back into the body. It's such a large amount at once that for some patients who are already immunocompromised, their kidneys can't process the infusion of this cell routine. And as a result, if the doctors aren't watching carefully and it's happened in some of the trials, it can lead to organ failure. So we're talking about serious side effects, the possibility. There's a downside here as well that you have to watch out for.
- 7.49** Now, CAR T cells, bespoke. Bespoke has become one of those words. Everything is bespoke nowadays. It used to just apply to a tailor—Trisha's father was a tailor in the East End of London. He could make a bespoke suit, but now bespoke is everything that's considered good. Bespoke cures, but at a high price.
- 7.50** Toxicity challenges. The key word is this cytokine release syndrome. It's actually what they call it. Systemic inflammatory response, neurotoxicity, confusion, seizures, brain swelling. Intensive care has to be applied to this because you're really keeping an eye on what takes place in terms of the patients.

- 7.51** One of the ways, by the way, that they try to deal with this, doctors, is they treat lymphatic patients who have leukemia, who are getting CAR T, with a weekly intravenous infusion of gamma globulin. We've heard about gamma globulin. It's used to boost somebody's immune system. Now patients who are getting this are getting a weekly dose of gamma globulin.
- 7.52** But since CAR T has only been in use for a decade plus, we don't have the clinical evidence yet to really know with certainty whether somebody's normal production of their immune B cells will return when they stop CAR T therapy and they're away from it long enough. How long will they have to be on immunoglobulin injections? Is it just 3 or 4 years after treatment ends, or is it going to be a lifetime? Not sure yet on that. Has CAR T depleted those cells and hurt it in a way that's permanent? Not sure.
- 7.53** And that, by the way—what I'm talking about here, and I'm going to show you a few other things—the toxicity problems, those difficulties that have come up, and they are, is one of the major factors, it is the major factor, as to why all of the talk about CAR T being used on other cancers has stalled. It isn't over, but all the enthusiasm in the beginning—we can treat leukemia—is going to apply to cancers across the board, and maybe, in fact, one day it will, you'll read about it, but it stalled all of that. It's put the big yellow flashing warning signs on. You've got to make the procedure better and less invasive in that sense.
- 7.54** Staggering cost. Well, I'll tell you, hundreds of thousands of dollars for a single CAR T infusion. So you didn't think it was going to be cheap, right? There's no bargain sale on this. No blue light special. They understand that this is a special routine. The complications of management, I guess a big medical team that's around you, they're watching it all the time, is a challenge in the health system, but it is an expensive proposition. \$500,000, \$600,000, \$300,000. Insurance often covers it. But this is a cost factor.
- 7.55** This is a first-world problem. In America, we're dealing with it. In some European countries. You're not having this in large swaths of Africa or in the poorest sections of South America or in Asia; that's not even a possibility. So it's being tested and seen here. The cost is keeping it from any type of adoption, what I call in a wider sense in the world.
- 7.56** Narrow access and inequality. So it improves. Approvals so far are just limited to these specific blood cancers, and just literally a couple. And to me, it does create this two-tier system of health care, but that's part of something this expensive.
- 7.57** The hype, the venture capital and the trial failures. So CAR T and IPO has boomed, and companies have targeted with very limited success on solid tumors. They've had some luck, but it hasn't been nearly as effective as it is on blood cell tumors. There, the T cells, they're supercharged, come back in and they're different. They haven't seen that happen on solid tumors.

- 7.58** Now, let me tell you it's interesting here because doctors have developed what they call problems with CAR T, and they have names for them. If you're inside the business of cancer treatment using CAR T therapies, they have a thing where the CAR T cells stimulate T cells to attack. They're supposed to do, but in the wrong position. They don't go after the cancer. They go after something else in adjoining or somewhere else in the system. That is called, inside the practice, off-target killing.
- 7.59** Sounds like something a military euphemism would have come up with—collateral damage. But off-target killing, if you ever have CAR T therapy and they tell you that your patient has off-target killing, that's not a good thing. It means that CAR T is using [T cells] to attack somewhere else in the wrong location. It causes heart attacks and permanent respiratory illnesses. The most unintended consequence, a little discussed, to CAR T therapy is this moment at which it overwhelms the kidneys. We talked about that before, these dead cells are kidney accumulating.
- 7.60** Finally, there's this cascading of variation is what doctors call, and here's the name for it, a cascading cytokine storm. I'll tell you, they took that from the best pages of meteorologists. A cascading cytokine storm. Not quite sure what that is, but you are—because what it actually is an over-stimulation of the immune system that causes the immune system to attack itself. It's a bit of what happens with [rheumatoid] arthritis or [autoimmune] arthritis, where the immune system is attacking itself and causes this arthritic pain and development on bone. Same thing happening here, but on a much larger scale. I like the potential of CAR T. I'm just saying that it's not quite there yet, without a doubt.
- 7.61** Now, in 2020, the Nobel Prize in Chemistry went to a French microbiologist, Emmanuelle Charpentier, of the Max Planck Institute for Infection Biology, and an American biochemist at the University of California, Berkeley, Jennifer Doudna. They, in 2012, the year that CAR T had cured this 6-year-old patient, had a discovery of a bacterial immune system that they called CRISPR to edit DNA, sort of the molecule of heredity. They were awarded. They made scientific history really as a result of this. And they're the only science Nobel ever won by two women at the same time. I like that footnote. Marie Curie, the first woman to win it. Here, the first time, two women won it.
- 7.62** The Washington Post is typical of the coverage of CRISPR: "the most important innovation in biology in 30 years." Now, they have a better name for it. They call it molecular scissors. That's interesting. That gives you the idea of what it really does. Molecular scissors means that they can start to engineer and do things on gene editing. That is truly the edge of the possible. Now we're talking almost in the realm of science fiction. It allows doctors not just to treat symptoms, but directly repair or disable the genetic instructions that caused the disease. That's the promise of this.
- 7.63** One of the things that happens, which is very interesting, is that the two creators of it, they believe that it was possible that their discovery could make changes not just in

cancer care, but could make changes in a whole realm—if I could find it here—of different things, from providing food in crops that you might be able to engineer and push out more food production.

- 7.64** It could create a difference on genetically inherited diseases by making sure that—now you're in the realm of ethics—that the embryo doesn't have a whole host of genetic diseases that they might otherwise have and might not be carrying any of those for the future. And so they're talking about it, not just—they are actually saying that it could alleviate hunger, food supply shortages. CRISPR could bring species back from extinction. You might be able to genetically create a species that had been extinct and have it return to the Earth. It's fairly impossible.
- 7.65** So you talk about you could repair here, in theory, the idea of everything from sickle cell anemia to muscular dystrophy. So we're not just talking cancer. So molecular scissors and CRISPR has the potential for really upending the entire business. So, CRISPR basics: gene-editing tool, works like a highly precise pair of scissors for DNA. And the reason I say that with half a smile is I assure you, I've lost brain cells reading the technical reports of this. CRISPR is a complicated technology. They didn't win the Nobel because they just said, "Let's come up with a molecular pair of scissors."
- 7.66** But that is the best way to think of it and describe it. Scientists identified the exact gene that they want to change, and a small piece of genetic material that they call a guide RNA is programmed to find that specific dot in your DNA.
- 7.67** Then that little guide that they put in is like a GPS token ringing off to the medication. They load a coding protein at the right location, and the protein cuts the DNA, and when the cell repairs, they can remove it, disable it, or replace the faulty gene. Pretty remarkable. It is the point at which you think science has advanced. Boy, go back to the days when we were just developing the radical mastectomy as the choice to remove cancer.
- 7.68** There will come a time in which what we're doing today, and I'll talk about this more in the next lecture—100 years ago, what they were doing to treat cancer seems pretty prehistoric in some ways an oath. But what we're doing today will seem like that to people 100 years from now. What we're seeing here are the early days of what I call this change.
- 7.69** Where CRISPR is real. So these therapies have been improved for FDA approval for genetic blood disorders. Sickle cell, one of the first. In oncology, trials are testing whether it can knock out genes that help tumors, what I call, evade immunity or sensitive cancers. So there's a whole series of clinical trials underway right now on those.
- 7.70** Safety, scares, and setbacks. Why are you not surprised if some of this would happen? Nothing's a free lunch. Gene editing can cause off-target changes, so that happens as well. Unintended consequences. Regulators have paused, and they're scrutinizing some

of the gene editing results because a Columbia professor, as a matter of fact, who was trying to cure hereditary blindness, went in on what should have been a fairly simple, not simple, nothing simple with CRISPR, but thought it was simple to be able to repair this blindness by changing this genetic mutation that caused it, and in fact caused a series of adjacent cellular changes that caused additional problems. And so that put the yellow flashing lights up.

- 7.71 Personalized gene editing in extreme cases. So there are CRISPR therapies for these rare conditions, and they show me the promise of what might come and also the ethical considerations. So if life can be transformed, it's those who can afford to transform it can do it more than anybody else. And in cancer, this raises the question about the equity of treatment, and it will in the future do this because these are very expensive.
- 7.72 Conflicts in the pipeline, all the leading cancer hospitals, I mean, millions in payments from health care companies went to a key official, and we'll talk about this. It's that inside part of the system that continues to feed itself. They see a new therapy, and they're jumping on top of it to feed it as quickly as possible.
- 7.73 Now, one of the reasons that CRISPR is so particularly controversial is it raised the possibility, and it was done in China by a scientist, that you might be able to create the perfect human straight out of an embryo. You might be able to create that by doing all the work molecularly on that embryo and do it. That's essentially what happened in a Chinese case.
- 7.74 Now, in the West, the response to that was to say, "Okay, we're going to ban embryonic experimentation for the time being. You're not going to be able to work on embryos." We don't want to start to think about engineering the perfect race. We were back there 80 years ago when another country thought they could engineer the perfect race, and that didn't end very well.
- 7.75 So imagine if the Third Reich, who thought that there was racial superiority in the Aryan race, had CRISPR molecular scissors to be able to take embryos and make sure that nobody was Slavic or Jewish or black or whatever else, and they could just have Aryan all the time. They would have loved that idea.
- 7.76 That's not to say I'm not saying CRISPR is a Nazi ideology or it's used to create a master race or anything else, but there are ethical considerations on this. In China, a scientist was able to use them right at the edge of the boundary. And in the West, for the time being, at least, if there's somebody working on it, they're doing it against this regulation and agreement that we're not going to do it. It doesn't mean there won't be some scandal on it at some point coming up down the road.
- 7.77 So risk and concerns. It can lead to destruction of thousands of DNA base pairs. That's what I was talking to you about at Columbia, where the university geneticist used

CRISPR to repair the cellular mutation for hereditary blindness, and it led to genetic havoc in about half of the chromosomes.

- 7.78** And that's the real problem, because once that happens, those are the changes and mutations in the chromosome that it was causing as a side effect that can lead to the growth of cancer. So you think that this could be a tool to stop cancer, but in that case, it could actually be leading to cancer. And so I think that was one of the things that brought immediately up the idea of we might be moving a little bit fast on CRISPR.
- 7.79** The malignant empire, when cancer care, pharma, money, and ethics collide. So rapid pace of biotechnology achievements, no question about it. CAR T, CRISPR, breakneck speed. A lot of implications, good down the road. Challenges, though, keeping up with the ethics of it. Patient safety, unintended consequences, and the boundaries of genetic engineering. Where are those boundaries? How do we set them? Guess what we're doing in real time? Regulators have never been good about setting the right boundaries in real time. So why should we think now they're going to be fast on top of it? The science always is ahead of them.
- 7.80** A need for, I say, need for cautious and responsible development. I would be thrown out of a planning meeting at a biotech firm. Genentech would fire me if I was working for that firm. Cautious and responsible development means slow. It means the competitors beat you to the market time and time again. They're in a race to develop these products. So I'd like to see them uphold the highest ethical principles, but I understand it's a marketplace in the end. So we have to strike the balance between what I call these transformative possible treatments and the considerations of oversight and thorough testing and a commitment to patient safety. I'm hoping that that is what the FDA does all the time.
- 7.81** However, the FDA has what I call "special passes" for cancer drugs. Now, I put that in quotes. That's not what the FDA calls them. You can't imagine the FDA would say, "By the way, we have a special pass for cancer drugs. Come over here and form a queue right here, and we'll send you through that door and give it to you." But that's really what it is.
- 7.82** It allows pharmaceutical companies to sometimes shorten the clinical data on a fast track. Here's the part that works for the pharmaceutical company's benefit. This is the same, not just on cancer drugs, but across the board. Really astonishing to me. You do the studies as a company. If it's not being done by an outside, you sponsor the studies, you have it done in the lab, and you get the reports. When you submit to the FDA, you have to submit your clinical studies, but there's no legal requirement that you have to submit all your studies. You get to submit the studies you want to.
- 7.83** Now, if you have taken a study that shows there's a lethal outcome to the drug that you're proposing and you hide that, you're eventually going to be sued and be bankrupt. So it's not that you can hide everything, but you can make sure you can just

rejigger the numbers enough that what you're submitting to the FDA is the drug that you want. And to the extent that there are a few more speed bumps along the way, you don't have to give those in. And that's a great loophole in the FDA.

- 7.84** Volunteer subjects must be in good health. Now, think about this. You volunteer for a clinical trial, you can't have comorbidities, and you have to have normal organ function. That means that almost all the people used in clinical trials for cancer drugs are healthier than real cancer patients who have a comorbidity cancer, who are often undergoing some other form of treatment. As a result, you almost want to test on people who are already ill, but you aren't. So therefore, the drugs are showing their effect, not always showing you how it will be for somebody who is actually undergoing cancer.
- 7.85** Pharma can report the results on evaluable patients only. Now, this is—I tell you. If I made it up, you'd think I was just joking. But it's true. You become part of a clinical trial. You're being given a new drug. Cancer drugs often very, very difficult on the body. Toxic side effects. And the effects are so terrible, you have to drop out of the trial. Guess what? Pharma doesn't have to tell the FDA about that. It doesn't have to say 15% of our clinical trial patients dropped out because they were getting terrible migraines every day, or they were getting terrible bone pain or whatever else. They're not considered. They're not through the entire trial. Therefore, you don't have to report those results, only on those that are evaluable.
- 7.86** Researchers can fish for subgroups to show drug effectiveness. Even if it doesn't work on the originally targeted patient group, they will sometimes look for another patient group in another trial to be able to get that drug through what I call the special pass of the FDA. It's a lenient process. The FDA will tell you it's diligent, it's rigorous, it's costly, that's for sure, but it's also lenient to allow these companies on drugs that really can be life-saving, should be life-saving to get there.
- 7.87** Here's the troubling stats on drug trials. The percentage of clinical trials reporting favorable results for drugs being tested: 61% of all clinical trials, as of a year and a half ago, report favorable results for the drugs coming in. It's pretty amazing when you think that 39% do not. All right. So even the drug companies are trying to put the square peg into the round hole. It doesn't always work. Still, sometimes they can't get a good clinical trial out of it because that drug really has problems.
- 7.88** But lo and behold, what a surprise. You've got to be kidding. Eighty-nine percent of industry-funded trials, they have good results. What a coincidence. It just must be a coincidence that the industry pays—the pharma companies are paying for those particular ones. They aren't being done in an academic center by somebody who has no relationship with the pharma company. Sometimes that's done. But when the company is paying for it, 89%.

- 7.89** Non-industry-funded trials, 50% approval. It can't just be a coincidence. I'm sorry. There's something about it when you are able to be doing those trials without necessarily getting the paycheck from what I call the pharma companies that might make you look at the clinical trial a bit harder. That statistic tells me when the industry or the pharma companies are funding the trials as opposed to an academic or NIH or government, you're more likely to get a drug through.
- 7.90** The incestuous relationship between cancer research—boy, I say it's this web of conflicts of interest. I told you before about Memorial Sloan Kettering, the first comprehensive cancer center, and how the original head of it was a Pfizer board director, and the president and chairman were Bristol Myers, and they've had pharma executives on their board for a long time. These academic researchers are intertwined often in what I call this alliance of moving forward on bigger, better drugs, gains for researchers, lucrative payments, lucrative funding. You build your careers sometimes based upon this.
- 7.91** So a case study. Memorial Sloan Kettering, leading research institution in the United States. Certainly the one with the most funding, has the biggest reputation name. And this little example will show you how what I call ground-breaking cancer research, big financial flows of money, lack of good oversight, and the challenge of maintaining integrity in this high-stakes world of cancer.
- 7.92** And this case defines the ethics gap because for me, it's the intersection of science, capital, and prestige. It's all of it. It's not just a little cancer clinic out in the middle of the wilds in some state. You're talking Memorial Sloan Kettering, which you might think might be a mistake to make that assumption, but people think because it is the most prestigious, it will be the strictest about not allowing anything to take place that could be untoward or have the appearance of impropriety.
- 7.93** Failure of oversight. When oversight fails at the top, then the entire cancer care system absorbs the distortion. I'll show you this.
- 7.94** And what I call blurring of ethical boundaries. So the stakes have risen with these new cancer therapies because they are so expensive and there's so much money at stake. The reputation of scientists and researchers are on the line that I think, if anything, the stakes are higher than ever.
- 7.95** It has systemic implications because the exposure of these conflicts of interest and the transparency problems it reveals are industry-wide in many ways. Memorial Sloan Kettering, most powerful: 125 labs it has running. You think about the size of it. It's not just one single little laboratory in the basement: 1,200 oncologists, \$5 billion annual revenue, and a \$4 billion fundraising system. So it's a brand.
- 7.96** Selling hope. We lived in Manhattan for 25 years, 27 years, almost. We knew neighbors of ours and others who got sick with cancer. They all wanted to go to Memorial Sloan

Kettering. We knew doctors who worked at Memorial Sloan Kettering. It has that reputation.

- 7.97** And then a series of stories come out. The New York Times, ProPublica, it hits the news big. And what it is, it's about this man, José Baselga. He's the chief medical officer at Memorial Sloan Kettering. He's the president of the American Association for Cancer Research as well, and the editor-in-chief of the journal Cancer Discovery, which runs articles about new cancer drugs. He is the person who, more than anybody else at Memorial Sloan Kettering, is responsible—they have committees—for helping to lead and create what I call these oncology trials. They're central to how the industry moves forward.
- 7.98** Now, this is what happens. What does it turn out The Times and others are able to report 2017 and later? Millions of dollars come from Bristol Myers Squibb, Roche, and others behind more than 100 articles published in the journal that he is editing. They are more of promotion pieces from the press department than they are clinical pieces.
- 7.99** But they have the trappings of an academic article because they have names attached to the serious researchers. That's the old tool for those of you who saw the lecture on Big Pharma that Arthur Sackler developed for the drug industry. Put articles out with people's names who look as though they have a great reputation and the public will believe it. Board memberships, undisclosed in over 100 articles. So there would be members of the boards of different groups of pharmaceutical companies, of biotechs, never disclosed in the byline of any of the articles that were run.
- 7.100** Startup equity. Baselga and others have startup equity in a lot of companies, over 100 that might hit gold in terms of biotech product, and therefore the stock might double or triple or quadruple overnight. You would think that that's something you need to disclose because if you're doing an article on that technology, you happen to promote that company, somebody would like to know you have a financial interest in it.
- 7.101** Now, the failure. The honor system, it didn't work. Disclosure relied on trust. At Memorial Sloan Kettering and some other leading comprehensive cancer centers, they believe that if you held this position of trust as he did, that you would disclose voluntarily any potential conflict. What we as lawyers were taught in law school to avoid even the appearance of impropriety. I know plenty of attorneys who violate that all the time, but that was the idea in a technical sense. Don't even give something off that could be the appearance of impropriety. That's the honor system. It wasn't working here.
- 7.102** Minimal enforcement. The disclosure rules were lax. He technically, and I do mean technically, didn't violate any of their rules because they didn't have any rules in place. They didn't have to worry about these conflicts of interest as it went along.
- 7.103** Rare penalties. Nobody had been punished before for going ahead and having their hands in four different pots at the same time. The medical journals were failing to

uphold the ethical standards. It's ProPublica and The New York Times breaking the news. The medical journal should be doing it. There were rumors for a while about Baselga and what he was doing.

- 7.104** The internal reaction, comparing the public's concern for patient trust versus the institution's concern for drama, 85% of the public who knew about this instance when later questioned said this is about patient trust. It makes me worry that the care I'm getting has already been gamed somehow for financial reasons as opposed to being about my best interest. Ninety-five percent of those questioned inside the industry were worried about brand damage. So it's different. It's where your eyes focus and what you think is important.
- 7.105** Here are the consequences. September 2018, Baselga resigns from Memorial Sloan Kettering. So head chief medical officer, big thing, he has to step away. After his resignation, he retains his medical journal and all of his roles in the Cancer Society, so he doesn't have to give those up. He just leaves Memorial Sloan Kettering. And then my favorite, he's hired by AstraZeneca to lead their oncology research and development. So we talk about a revolving door. It's fantastic. Somebody who has to resign from Memorial Sloan Kettering, the most prestigious cancer institute in the country, because of potential conflicts of interest, AstraZeneca says, "Come over here. You must know a lot about cancer. You can run our R&D department." So he lands on a golden parachute.
- 7.106** Industry-wide admission. Let me tell you something. The Dana-Farber Cancer Institute in Boston, the Fred Hutchinson Cancer Research Center in Seattle, all these major cancer industries, said no one thought it was a conflict what he did, but now that we see it in daylight, we realize how it could appear that way to the public. So they all modify their rules. The conflict-of-interest rules that didn't exist before, now they're rolled out in real numbers.
- 7.107** And policy changes, the other cancer centers come in and they even present—Memorial Sloan Kettering, they make some changes. One of the changes, there's a whole list that they end up making. Employees who represent Memorial Sloan Kettering on the boards of other cancer institutions, they can't accept personal compensation like stock options and equity stakes. Baselga was getting equity stakes in some of these companies for helping them out. Can't get that anymore. But that would have seemed common sense to me at that time.
- 7.108** So the chief risk officers are enacting these policies, finding other institutions. Stronger disclosure rules, belated reforms, better tracking of the potential conflicts of interest, still largely reported. But now there is a risk that you will be humiliated and run out of your business.
- 7.109** And why this matters? The advent of immunotherapy, CAR T, molecular scissors, all this to me, it just ups the stakes. More money involved than ever before. So you really have

higher stakes today than you had before. Higher profits, astronomical price tags. The financial incentives for somebody to do the wrong thing are higher than ever.

- 7.110** And weaker guardrails, the system that failed to prevent these ethical breaches, still relies largely on self-regulation. And to me, that's an open hole for leaving patients vulnerable. So as cancer treatment becomes more advanced, my wish is, and it's only my wish because I can't enforce it as a policy, for more vigilant oversight and stronger disclosure, if I was running these centers, I would have mandatory disclosure like you have—I was going to say—like you have with politicians, they have to file financial disclosure forms, and yet still that doesn't stop corruption at times and back dealing and everything else. There's no sure fix to somebody who wants to game the system and is going to do something corrupt. But it would be nice to at least know that they have to provide this information as they go forward.
- 7.111** This is where we end on this, because I do think that we have this promise of immunotherapy. We have the promise of CAR T. We see what molecular scissors could possibly do. All I know is that 5 years from now, we're going to have more approvals from the FDA on some of these new technologies, and we'll find more cancers that may be responding to it. I'll probably have a longer list of potentially terrible side effects. But that's the nature of the business and treatment. Thank you very much.

Lecture 8: Cancer's New Frontier

At a glance — Cancer's new frontier, the finale.

- 8.1** Thank you very much, again. Getting to the final hour of the Politics of Cancer. There's this great Columbia oncologist. She's written a book, Azra Raza. [She] says something that I liked. In [her] book, [she] said, "We boast of magnificent, godlike technological advances, editing the genome efficiently, turning genes on and off at will; yet, cancer treatment, for the most part, remains Paleolithic in comparison. The issue is not as much that there has been little progress in cancer research. The question is why there's so little improvement in treatment."
- 8.2** That comes back to that issue that I had before when I was telling you that historians say that cancer treatments were primitive in the past. Historians in the future will say cancer treatments are primitive now. We just don't know where they're going. So, reimagining. I have to tell you that I have a personal bias against that word because it's used too often by politicians, that we're going to reimagine HEW [U.S. Department of Health, Education, and Welfare]. We're going to reimagine this and that. Normally, it's just a way to try to throw you off.
- 8.3** But in this case, I used, against my own good intent, reimagining the malignant empire, because maybe it says a goal that we won't be able to do anything, reinventing cancer care toward prevention, transparency, and better cures. I'm going to look at you with some options in this.
- 8.4** 2019, total patient economic burden. The year before COVID, because COVID throws the numbers off a little bit, the national patient economic burden of cancer care was a little over \$21 billion. What I mean by that is, \$16 billion is out-of-pocket patient expenses. It's a lot of money, one of the highest of any of the industries because the care is so expensive. Then they make an estimate, hard to figure out, of \$4.87 billion, almost \$5 billion, on what they call patient time cost. It's the value of time that patients spent traveling to and from health care, waiting for the care, receiving the care.
- 8.5** I know, from going with Trisha to the five radiation sessions that she had, there's a process of getting there and then preparing to go in and then waiting there and then getting it done. Then they have you wait a little bit afterwards because they want to make sure you're not lightheaded or anything's wrong with you. Then you're able to go home. There's that process. It was nearly an hour away for us to get there as well. That happens all the time. People go through that constantly.
- 8.6** Looking ahead from inside the empire. We know about the fear of money, politics, branding, all the chemicals that aren't looked at, the regulatory capture. Today, I'm going to try to focus. At the end, I might touch on a little techno-optimism, but for the most part, I'm going to leave the techno-optimism behind. Actually, I think that the people who create molecular scissors, or CAR-T, might not even call me an optimist

because I am somewhat enthusiastic about the technological promise they hold, but I'm also very realistic about the downsides so far of what they have, so they may not consider that that optimistic.

- 8.7** But what I think we need to change, sort of what I call these three lenses for the future. We have new detection screening tools, liquid biopsies, multi-cancer early detection. We'll talk about liquid biopsies in a second, but liquid biopsy sounds very, very science fiction. It's really just a blood test. They give you a blood test. It's liquid because it's a blood test. Sometimes in that blood test, they see markers of DNA or white blood cell counts or others that indicate there might be a problem.
- 8.8** The great thing about liquid biopsies is, you can do blood tests repeatedly. It's not like doing a biopsy or doing an actual exam like a mammogram, where you don't want to subject somebody to radiation too much. Liquid biopsies are easier, but they've been perfected.
- 8.9** AI, big promise on AI and data-driven screening and risk prediction. Political and structural reform, that's the hard part. Prevention, transparency, and governance. Boy, I'll tell you, the first two will not seem as difficult as the last part.
- 8.10** So, key question, at least the question I ask you—if you had to choose, would you rather see a perfect cure, a real cure for late-stage cancers, or a world where half of those cancers never develop? So, tough, it's a Hobson's choice. So we say half the cancers will never develop. But in the other half of the cancers that develop, there's not going to be a cure. They're not giving you that cure option; it's just going to be this same gauntlet.
- 8.11** Or we won't prevent the cancers at all. So everybody, the maximum now, will come down with cancer. They'll eventually develop it, but we'll be able to cure it at the late stages. It depends on how you fall into that bracket. If you're one of those who has a cancer prevented, you'd like the option of preventing cancers before, and because you'll never be into the system in the first place, but it's not easy to know that.
- 8.12** The World Health Organization. Boy, I haven't cited a statistic from the World Health Organization since before COVID because I've had a lot of problems with the WHO at different times. I don't necessarily always view them as they're politically driven. We could have an hour lecture just on the political problems in the WHO. But they estimate—because they're one of the few doing it, because the US estimates about the US, France estimates about France, Germany about Germany—the WHO estimates that 30% to 50% of cancer cases worldwide are preventable through tobacco control, environmental protection, diet, and vaccination, HPV and vaccinations like that, not COVID vaccinations.
- 8.13** And transition to new detection. The next waves of innovation really focus less on cutting and poisoning tumors than on finding them earlier and maybe being a little bit less invasive. That promise is one that I would like to see fulfilled as you went along.

- 8.14** So, a couple of screenshots, as I call it. Now we are in the final hour of talking about the politics of cancer, and I want to talk to you about who ends up getting cancer. Shouldn't we have done that in the first hour? Maybe so. But now I want you to think about this in light of the industry that's been created around it.
- 8.15** So, myth. Cancer is mostly genetic and largely unavoidable. A lot of people believe that. All cancers involve a genetic mutation. That's where some of the confusion comes because cells are mutating, but most are not inherited. You don't inherit the genes that necessarily give you cancer. Believe it or not, and I know you'll believe it because you trust me by this point, only 5% to 10% of cancers are driven by inherited genetic mutations.
- 8.16** The type of mutation that Angelina Jolie had in the BRCA1 and was concerned about, which gives you a much higher incidence, only 5% to 10%. The majority—and this isn't just, by the way, me putting this on a thing, this is based upon a series of studies and [clinical trials] over the last 20 years—arise from genetic mutations that are caused by tobacco, diet and obesity, alcohol, infections, and pollution, and occupational exposure.
- 8.17** Now, here's one thing. If your genes, they may not give you an absolute guarantee that you're going to develop a cancer, but you could have genes predisposed if you are exposed to one of these carcinogens to developing cancer. Why is it that, out of all the tens of thousands of workers in the plastic industry who ended up working with vinyl chloride, why didn't all of them come down with liver cancer? The majority did not.
- 8.18** Not just because those who came down with liver cancer were the ones who worked the most with it, but they had a genetic disposition that wasn't going to cause cancer without the exposure to the vinyl chloride, but the vinyl chloride made, they were more susceptible to that exposure bringing them over. It's the same reason why somebody can be a cigarette smoker for 40 years and still be alive and kicking and doing well, and somebody else will have lung cancer who hasn't had or has minimal exposure to smoke. So, your genetics load the program, and then the environment triggers it, if that makes sense. Random mutations exist, but they don't dominate cancer risk in most major cancers. Now, here are approximate causes of cancer in high-income countries. That's Western Europe, Japan, China now falls into that in terms of studies and that, the United States. Start with the big blue. Environmental and lifestyle factors, so roughly 75%. Inherited genetic mutations, in this particular study, 8%. Then the part that will drive everybody crazy, including me, random somatic mutations. That literally is how the scientists define bad luck. They can't figure it out.
- 8.19** You haven't been a smoker, you haven't been around secondhand smoke, you haven't dealt with vinyl chloride, you don't have anything to do with asbestos, you haven't used talcum powder, you're not a heavy drinker, you're not obese, and you develop a cancer, and you're going to the gym 5 days a week and everything else.

- 8.20** They're not quite sure why a cell inside your body starts to do that mutation, starts to replicate, and loses control and keeps replicating. And that is what they call random somatic mutations. That particular area, in my view, will be one of the great scientific advances. There is a Nobel waiting for some researcher who is able to figure out why that takes place, other than just bad luck. I don't like the bad luck theory of a disease like cancer.
- 8.21** Now, approximate contribution by category, high-income countries again. This is what is broken down in terms of the studies. Environmental lifestyle factors, 70% to 90%, modifiable at the population level. Tobacco exposure, certainly 25% to 30% active smoking plus secondhand smoke. That's a remarkable figure, to think that still up to a quarter of cancers are a result of smoking and come from smoking. Remember, it can lead to lung cancer, but it also is a contributing factor to bladder cancers and colorectal cancer, and so a series of things.
- 8.22** Diet, obesity, and physical inactivity. Strong links to colon cancer after menopause and endometrial, pancreatic, and liver cancers. Alcohol, 4% to 6%, no safe threshold, but they believe it's a contributing factor to some cancers. Infections. Remember those researchers early on who would say, "We think there's an infection, some bacterial infection is causing cancer," and people would say, "You're just absolutely crazy"? Now we know that's the truth. So HPV, HBV, HCV, H. pylori, lower than the global average. But those infections now controlled often with a vaccine that's given early on.
- 8.23** Environmental and occupational carcinogens, only 5% to 10%. Now, I don't mean to say that 5% to 10% of people getting it, that's low risk, but you would tend to think it's higher in low-income countries. In the developing world, the risk of developing cancer from environmental pollution, asbestos, benzene, those products are still often used. They haven't been banned. But now in the West, where we've, finally after decades, realized how bad they are, try to control some of them, it's down to 5% to 10%.
- 8.24** This issue where you finally get to the point of medical iatrogenic exposures, diagnostic radiation, treatment-related secondary causes, a very, very small number. Now, inherited mutations, look at this, 5% to 10%. BRCA1 and 2, Lynch syndrome, APC—these are all genes marked to much higher risk of cancer, TP53. These increase susceptibility, but they don't create cancer on their own. So there's a misunderstanding of that.
- 8.25** People think if they get positive for a BRCA test, for instance, or they're positive for APC, that means that that's going to cause the cancer. No, it just means you have a gene that is more predisposed in the body to develop the cancer at a higher rate, and therefore, exposures to environmental [carcinogens] may, in fact, trigger that at some point.
- 8.26** Random somatic mutations, or, 5% to 20%, what doctors call replication errors. Replication? What's the error? Oh, your cell is doing a replication error. It is continuing

to repeat and grow and not be controlled and not die and continue to expand. So it arises during normal cell division, not inherited, and it's often misused rhetorically to minimize preventability, because those who don't like my idea of prevention say, "Oh, why do you waste all this time? Why are you worried about [carcinogens] that you're exposed to and worried about cigarette smoke, and you're worried about everything that takes place in the workplace and at home? Because you know what? If you have bad luck, you're going to get it anyway, no matter what you've done." That's not a reason not to have prevention. It's understanding the disease. I get that.

- 8.27** But it's most relevant to some leukemias, pediatric cancers, and certain rare tumors, least relevant in lung, skin, bladder. But that's the somatic possibility. Now, you get a cancer at some point, and you're in this point where, what's the future going to have? This German oncologist, Klaus Pantel, and a French professor of oncology, Catherine Alix-Panabières, coined the term "liquid biopsy" back in 2010. So, it analyzes biomarkers in blood to detect one of those cancers that has developed before.
- 8.28** It has no idea, by the way, if it finds a cancer, what caused it. None of us do. But you know that you have the cancer. It looks for the characteristics of it. Blood tests. You will not be surprised to know that companies have raced to have patents on their own liquid biopsies, and they all market them as better than the other. We're more attuned to it. We're a little bit more sophisticated. There's one, Guardant Health Shield, detects DNA signals associated with tumors, one of the most sensitive.
- 8.29** Multi-cancer early detection. There's a test, Galleri, that analyzes cell-free DNA patterns in the blood to detect cancer signals before symptoms appear. Pretty remarkable, before you would even otherwise know that you have the cancer. The hopes and hazards. It's a potential to expand early detection, but there's a risk— you won't be surprised; it's the nature of all of these tests—of false positives. It sees something that turns out not to be cancer, and therefore you've gone through a battery of tests and overdiagnosis.
- 8.30** It's been integrated into existing screening and coverage, but the findings aren't always as clear-cut as you would imagine. The dilemma is, would you want a blood test that detects a possible cancer, but can't pinpoint it? It just knows there's a cancer somewhere in the body. It doesn't know where. Done a blood test for it. Now, they come back with the results and they tell you, "We think you might have cancer." Now, you have to start the process of finding out where. It is a help, but it's not necessarily as precise as it sounds with a liquid biopsy.
- 8.31** The promise of AI and cancer cure. AI is going to revolutionize a lot of the world for better and worse. I saw Elon Musk the other day at Davos, the international gathering of the bankers and everybody who's got a nonprofit of some cause and that, and he was being interviewed, and he said that he thinks in the future, we will live to see more AI robots on Earth than people. And he sees nothing but benefits from that. They will be able to take care of elderly parents and free children up from taking care of their

parents. They will assist in everything from helping us at home to helping us at work and everything.

- 8.32** He might be right. Look, he's a visionary on a lot of fronts, so I dare not say he's wrong, but it's a leap for me to suddenly think that it will be a planet with 7 billion robots and that that's how we'll be handling all of our crises. But it's very interesting because people like Musk and others who are at the forefront of creating AI large language models, they see uses for it down the road that's difficult for us to see. We use ChatGPT, or you use Claude, or you use one of the products that are out there, or Gemini, and you ask it a question, and it gives you an answer back. Now it seems like a smarter version of Google, but that's very rudimentary to the people working in AI, especially in medicine.
- 8.33** So, working with what we have. There are decades of cancer research, and I can tell you, you're talking millions and millions and millions of pages of clinical trial studies and papers, this immense volume of material. No matter how good your memory, you can't remember it all. Unleashing the large language models on that will be fascinating because it may find conflicts or things that were missed, that I would consider a footnote, that changes our view on how something was approached. And that's important, what I call getting through the massive information we already have.
- 8.34** So, no question, no single person can do it. It's possible these large language models can. They excel at data processing. This is one of the things that—this I can get my head around. I understand how they can recall and contextualize and aggregate. I think back sometimes to books I've done in the past in which I'd spent 9 years on and off, and looking into the finances of the Vatican at one point, and the amount of information in terms of financial information, offshore information, banking information, it was vast.
- 8.35** I said to Trisha any number of times, "I'm unable to do this. I can't get through the middle of this and figure out how they did the connections." I finally did. Took a long time. Torturous. I often wonder what it would have been like if I had been able to take those 50,000 pages of documents and scan them in and have a large language model take it on and ask it, "How do they really transfer that from the Bahamian bank over to Liechtenstein?" and see if it could give me that answer.
- 8.36** Would it have saved me 6 months, 7 months, of that type of work? Possibly. I look at it in that very small world. In the world of cancer care, you're talking about just reams of information over a period of time. So leveraging that AI, we might be able to accelerate progress in cancer research by making better use of the existing information. So that's part of the challenge, is making better use of what we already have.
- 8.37** So, iterative science and the future of cancer research. Iterative science just means that's the nature of medical care. You're not doing things just once and leaving it. You're doing it again and again. You're coming back to do it again. It's that type of process. It's iterations on top of each other. Can we make better use of the things that

we have? Cancer research is no exception. The future potential of this is to streamline studies, maybe trial designs, analysis. That's going to be one of the areas where I think it could be useful, the promise of AI.

- 8.38** Hard to imagine, but true. In 2026, there are AI-powered cancer detection tools that are already revolutionizing the field of identifying cancer cells that can be overlooked and invisible to the human eye; and in some cases, catch the detection, as we'll see in a moment, of potential cancers, even when genetic testing has missed it. They have advanced algorithms, and they reconstruct vast amounts of what I call medical imaging data from CT scans and MRIs to detect the subtlest patterns and anomalies that could indicate the possibility of growth of a cancer cell that the human eye will miss.
- 8.39** Now, what does that mean? You understand this. You can give the answer now. You know it also is going to mean, at some point, false positives. There's no doubt, it's not perfect. Nobody's thinking that AI can do this and see things that nobody can see with the human eye by putting together all these MRIs and scans and looking for the little anomalies on the subtlest wave pattern. Sometimes it's going to pick that up and it's going to turn out to be nothing. It's going to turn out not to be a malignant tumor or that. There will be people that will get a temporary scare, but it may also find cancers that would otherwise go undetected. To the extent that better treatments might be available, that would be fantastic.
- 8.40** Now, AI's efficiency. Processes an enormous amount of data. I get that; it overwhelms human researchers. It can take care of that. Identifies patterns, patterns and relationships within the data that is almost impossible for regular oncologist researchers, academics, to otherwise piece together. Makes the predictions. Here's the fantastic part. Not fantastic. Here's the machine part. Doctors are sometimes afraid to make that prediction because liability, malpractice. They tell you something, and the lawyers are two doors behind, ready to come in after them.
- 8.41** AI hasn't learned yet, may be programmed at some point, to be risk-averse in terms of litigation. Oh, they'll be sued at some point, I'm sure. It's only a matter of time until Grok and OpenAI and everybody else is sued for a wrong diagnosis, or they miss something, and somebody ends up dying on a cancer that was missed, and they're going to sue AI, I guarantee you, because the AI companies have a lot of money. And so, therefore, lawyers go after where the deep pockets are. So it's only a matter of time until that happens.
- 8.42** And then maybe at that point, they, meaning the programmers, who put the guardrails up for AI, so if you go on and try to discover how you can make a dirty bomb, it's not going to tell you necessarily right away, step by step, how to do it. The same thing here. They may be able to give it a little bit more reticence in predicting what the treatment outcomes will be, but we are in the Wild West right now. We're in the period now where we can get the unbridled information, and cancer researchers are using that.

- 8.43** AI can carry out the statistical and computational analysis that would be not time-consuming— I'm not worried about that—but would be error-prone for human researchers. So as of yet, the machine turns out to be better sometimes than the researchers. It has an unparalleled amount of data processing capability, and it's hard to overestimate that.
- 8.44** So, could AI help overcome some barriers to cancer screening, like proximity to specialists, socioeconomic factors, and medical literacy? Absolutely. AI could level the playing field. If you don't have easy access to a screening facility and you're not able to do it, you might, through a liquid biopsy and other tests, be able to have AI fit your history, your age, your genetic makeup into the system, and analyze and tell you what the risk factors are, and whether, in fact, it's worth a 2-hour trip to see an oncologist.
- 8.45** What I call it, an early warning system detection can identify patterns and relationships between health, environmental, and behavioral factors to say you're at a higher risk of cancer. And personalized prevention, it provides a customized screening recommendation based on an individual's profile. One of the things I thought when going through the process of watching Trisha go through the machinery of the cancer system was that it can save your life, they do save your life, but it's a machine, and you're on an assembly line.
- 8.46** Sometimes you have to remind them, the best oncologists, "I'm here, I'm an individual. It's different," because their pattern is pretty much the same. You had a lumpectomy, and then you get radiation, and then you get an aromatase inhibitor, and everybody can't be the same. We know that. People vary a little bit. The possibility and the promise, when I talk to oncologists about AI being able to individualize it to a greater extent, that's one of the items that they're looking forward to. It holds that promise, at least.
- 8.47** Now, it can, by reviewing—here's the part. You're going to want this. What does it need access to? Talk to the promoters of AI within the medical community in oncology, and they'll tell you, "You know what? We can really make it powerful. We just have to feed up all the patient records, the diagnostic tests, all our studies that we have. We'll put it all in there, and it'll be able to really come out with fine recommendations."
- 8.48** I don't know necessarily that I want all of my patient records and diagnostic tests sitting up on a server with OpenAI, that somebody's going to hack it, or it's going to become available, or it's going to be used by other medical companies for marketing to me in a different way. I'm going to start to receive all that information. Now we're at the point where you have to think about not ethics as we talked about last time in terms of no conflict of interest, but what I talk about, the privacy concerns about giving everything over to the machine. Large language models, fantastic, but you're giving it all up, and your privacy, for it to hopefully make an earlier detection.

- 8.49** If you do that, the oncologists who are promoting this say it will see the unseen relationships between health, environmental exposure, lifestyle risk. It wants to know where you worked. It wants to know what age you were when you started to work at that dry cleaners. You ever work in a nail salon? You're going to have to give up everything because the more information it has, the better it can be on predicting your cancer risk; the less information it has, it's operating in the blind, so you have to give it all up.
- 8.50** Personalized risk assessment. The great thing is, if you give up your whole life information to a large language model, it will give you a personalized recommendation about how you should go forward. I'm not necessarily sure that I'm ready to do that. Automated screening solutions. So it can streamline the cancer screening process and it can reduce the barriers to early detection; that is true. These technologies are going to continue to evolve. There's no doubt they're going to unlock new pathways to identify high-risk individuals.
- 8.51** The problem is, if it identifies a high-risk individual, put all this information in an AI model and it says—I do it for myself—and it says, "You're high risk. You're at high risk of developing whatever, prostate cancer," so what does that do to my own mind? Do I live in that fear that Angelina Jolie had about the BRCA1? Do I want to know necessarily that I'm high risk? Am I going to change any part of what I'm doing in my lifestyle because I'm high risk? I'm leading what I think is a healthy lifestyle, anyway.
- 8.52** So there are points at which you say information is king. Information is wonderful. We're going to provide the information through AI about whether your lifestyle and every other genetic part of everything you've gone through in the medical system makes you a high-risk patient for possibly developing cancer. And yet, the question is, then what do you do about it? Have more testing every year? Be worried about it all the time? I'm not sure necessarily people assume, the oncologists assume, the pharma people I talk to, the people who are developing AI, all assume that everybody wants that information. I don't know if everybody needs that information, but that's a different issue.
- 8.53** AI's role in diagnosis. It is true that it can compare data and make assumptions like veteran oncologists do in the diagnostic process. You know, oncologists are concerned sometimes. Remember, the surgeons were concerned about the introduction of radiation because they thought it cut back on surgery. Then in radiation, they're worried about chemo coming in because it's cutting back on the radiation. If I was an oncologist, I'd be worried about a reduction in the number of oncologists in the future, if the machines can replace them and do the screening almost as well.
- 8.54** Limiting invasive procedures. It might be able, like liquid biopsies and the interpretation of those biopsies, for less invasive diagnostic tests like actual needle biopsies. If it is used by experienced oncologists as a tool to augment their knowledge, then that's where it's going to become the most powerful, as opposed to letting it take over the

system. Now, we talked before about this deep-level information in genomics, and the idea that we have now the entire DNA of the human genome. That's where AI comes in. That's where it can be specific about what could happen with things like molecular scissors.

- 8.55** Radiation dose optimization. One of the things I was always worried about is, what's the dose of radiation that Trisha is getting when she's going for a treatment? They set doses. I asked them. They would tell you what it was. Then, of course, my next question is, when's the last time you had the machine calibrated? Because I want to make sure when they set the machine to give 300 units, it's actually giving that as opposed to more or less, but that's a different issue; that's my concern.
- 8.56** But in some instances, some of the early studies show that AI may be able to say this is the precise amount of radiation needed in these five sessions or 10 sessions for this patient, based upon this tumor and the entire history of everything found about this type of tumor and its genetic makeup. It may be less or more than what doctors do, because doctors slot you in to this one category, and then they fit you in based upon it. This will be a little bit more precise.
- 8.57** Surgical assistance. There are some instances in trials in which it's actually providing assistance to surgeons in terms of information in real-time adjustments to surgery. Now, that worries me because I'm thinking to myself, what if the machine is not making those adjustments to surgery? You mean the surgeons are just going to do it the way they were doing it, and it wasn't quite as good? I'm sure that happens all the time.
- 8.58** Again, if you think, and maybe it's my bias as a lawyer, when I read that, I think that's potential for the future in terms of cancer surgeries. But it also is a sure bet that AI will be the company's defendant in litigation. Because once one of those surgeries goes wrong, is it the real-time hands-on tool assistance being done during the surgery that caused the problem, or was it the surgeon's negligence? You know what lawyers do? They assume all.
- 8.59** The AI in cancer research and drug development has the potential for being used by biotech companies really to take this process of machine learning and revolutionize drug discovery. I'll give you an example. Predicting pancreatic cancer. In 2023, they did research to see if AI might be more efficient in finding early detection of pancreatic cancer. Traditionally, a tough cancer to find.
- 8.60** One of the problems with pancreatic cancer is, as opposed to some other cancers, because it's a silent cancer—you don't know you have it until late in the process when you have terrible back aches or you look jaundiced, you're turning yellow—it's too late often to treat it. That's why it doesn't have a high survival rate, not many people making the 5-year point. The idea is, if we can find it earlier, there are some surgical procedures that are big procedures.

- 8.61** We have a neighbor in Miami Beach who survived, is 13 years a survivor of pancreatic cancer, found fairly early. They do a procedure called the Whipple, which is a major surgery, but essentially it gives you a chance at survival. If you were able to pinpoint pancreatic cancer at an earlier time, which patients were at the highest risk for pancreatic cancer, it would be a major breakthrough.
- 8.62** In 2023, it learned the diagnostic risk, went through millions of cases of cancer, and then of pancreatic cancer fed into the machine, and came out with this curve, identifying high-risk groups. The study at Harvard and the University of Copenhagen found that, in almost every instance in which they were able to take somebody in a very small group, we're talking a tenth of 1%, so not many patients, in which it thinks extremely high risk for pancreatic cancer, they would go ahead and do further testing, and they would find out that AI had sort of been what I call, it's the crystal ball of looking in and you think it's making a prediction, but the accuracy was uncanny.
- 8.63** So there's a case in which they think it could be useful in the future in identifying early on pancreatic cancer. It matters because what happens is, those early signs are missed. You're not giving CT scans. You're giving a CT scan eventually to find out if there might be some tumor or something like that, but you're not doing any of that until you get the predictor from the AI that says it's there.
- 8.64** Now, AI-assisted thyroid diagnosis. In one case, a woman with a thyroid lump that wouldn't go away underwent an ultrasound exam, and her doctor orders this invasive biopsy procedure, which turned out to be benign. But she sought a second opinion from a doctor using an AI-powered thyroid ultrasound analysis, without the need for the invasive test. Save the patient time and biopsy discomfort.
- 8.65** What's interesting is, the first doctor does the traditional thing. "Let's do a biopsy of it. Let's find out if, in fact, something is malignant." Second doctor says, "No, we can do it with AI. We don't actually have to do the biopsy." They say, "You don't have—this situation of the thyroid lump is not malignant." Now, it turns out, she's alive, she's well. This is a case in which, if there's a mistake made by AI one time and that turns out to be a malignant tumor, I can guarantee you again, it's going to change the nature of it.
- 8.66** One study that looked at this type of diagnosis found that an AI-based system could defer 14 out of 28 biopsies, with 13 of those being surgically confirmed as benign. In this instance in which they did a study afterwards, they do the biopsy anyway; you don't get away without the biopsy, because they want to see if AI is right. They confirm it in 13 of the 14. The 14th, one case, not correct. But that deep learning model, it has the potential, oncologists believe, of reducing biopsy rates if it's widely applied by up to 40%. It would be pretty amazing. Biopsy, no easy process, so these studies are in the early days and they're going to continue to play out.
- 8.67** Now, you know me well enough. I can't just praise something as the next greatest thing since sliced bread and not tell you a couple of caveats even early on. Let's talk about

the AI systems. AI-assisted mammography. Studies show that AI mammography can match or exceed traditional double reading while reducing radiologists' workload. Great, so radiologists don't have to work as [hard]; they can do more of the readings at the same time. Lung cancer screening interpretation, same type of thing. The risk prediction models based on imaging plus clinical data can predict individual risk of developing tumors, potentially guiding better treatment.

- 8.68** What's the downside? AI-assisted mammography. Performance can drop on data from different populations rather than those used for training. It's trained. Remember, it's a machine. It's fed information. It's given all the data. If it's fed that information just from one group, it's only fed European or fed Americans or Canadians or whatever else, it might miss people who are Guatemalan, El Salvadoran, or Thai. So you have to train it on all different populations. It's only as good as the information you put into it. And so, that's one potential downside.
- 8.69** Liability questions on the lung cancer screening. What if it misses a tumor? So who's responsible? You need to have continuous calibration. The doctors have to take responsibility for what the machine is telling them and make sure that it's not giving them the wrong information. And the risk prediction models. Tech and pharma companies see AI as a way to lock institutions into these proprietary platforms and cross-sell other services. Some AI vendors may oversell performance from very limited studies.
- 8.70** Let me tell you, those of you who saw the Secrets of Big Pharma understand that one of the reasons OxyContin was such a great success, and I mean success not in a good way, but a lethal drug, was the salespeople. The men and women who went out for the sales force for Purdue were excellent salesmen. They could have sold cars, they could have sold homes, or whatever else. They happen to be selling OxyContin.
- 8.71** And the same thing here. You're getting AI proprietary systems with representatives, always drug representatives or AI representatives, but they're salespeople, going out and trying to convince a hospital, Memorial Sloan Kettering or a cancer center, to use this particular AI model, which they have branded and they're making money from. And they have very small studies, and they're telling them that it can be revolutionary, that possibility of overselling is a very real possibility.
- 8.72** So, what the future holds. Some remnants here in the future are appearing on this. I love this, the smart bra. You never heard of that? Oh, yeah. Worn for an hour a week. It's developed by a company that, of course, has a patent on it. And what it will do is, it will measure the temperature of the body through the breast tissue. It will pick up this little point cellular changes caused by the appearance of the first cancer cells. This smart bra is going to let a technician know somewhere that that particular person might be developing cancer before it can even be detected on a mammogram. Now, I'm not sure necessarily the smart bra is really going to be something that takes off, but there's money being spent on that.

- 8.73** Experimental pills—this is fantastic—that you swallow, and it literally collects cancer cells in its earliest stages, if they haven't progressed to the point where they've metastasized. You'll collect them in the body and you'll excrete them in your urine. That's pretty amazing to think. Now, if you think that the cure for cancer is going to come through that pill, you haven't been paying attention. But I guarantee you that it will be sold to somebody, will make some sense along the way, and then we'll find out afterwards that it's not just taking the cancer cells possibly out of the body, but has other problems. But that is one of the items the future holds.
- 8.74** A cocktail of genes, literally, that pinpoints cancer cells so that a handheld scanner is passed over the body. So, they pass over the scanner, over the body, full body scanner, but it's handheld. You've already had this cocktail of genes that you take. It highlights, like fluorescent in essence, the equivalent of it, if you think of it, of the cancer cells; and they find out if you have cancer cells, and where are they in your body.
- 8.75** And finally, a small magneto-nanosensor that could identify surrogate markers for malignant tumors from a prick of blood takes liquid biopsies and makes it look like it's old school. Let me tell you, they have in these biotech labs, and this is just four of them, a list of items that sometimes read like science fiction, and some of them will never make it to market. Some will never get FDA approval. Medical devices, others, they'll fall by the wayside. But they are thinking about how to move in the future with AI.
- 8.76** So, ethical considerations. Serious concerns around the ability of insurance providers to use AI predictors to determine the cost of coverage. Boy, you give all your information and medical records and diagnostic testing and your work history and everything else to AI, then what's going to stop the insurance companies from buying that information from the AI company, so they can determine your risk and if they really want to cover you for all these things?
- 8.77** What if you're one of those individuals who has a very high risk of having a cancer that's treated by CAR-T or by molecular scissors that's going to cost \$600,000 or \$700,000? Novartis, by the way—one of the things I have to tell you, I forgot to tell you before when we were talking about the cost of these products—was fantastic. They provided a product under CAR-T that's very expensive. It was one of the highest-priced ones. They actually said to the insurance companies and the hospitals, "You get your money back if it doesn't reduce the cancerous B-cells." That's almost a money-back guarantee. That's amazing.
- 8.78** They don't say cure the cancer. They don't say reduce the tumors. It's going to kill cancerous B-cells, but they know that's what CAR-T does, anyway. It doesn't say it's also going to possibly kill good cells or any of the rest of the things. But I love the idea of some marketing people that said, "You know what? It's a lot of money. Let's give them a money-back guarantee." So here, insurance coverage, regulatory bodies, and compliance codes, boy, they're going to have to match the growth of this type of information.

- 8.79** AI models are right now niche models. They are complicated, they're expensive, and making them really useful for consumer-level adoption so you'll be able to get information, that's a challenge. It's not going to be easy to do because the AI companies and the medical providers, they want to control it. Social impact. Bias and fairness issues arise, the biases of those who create the models, and the interpersonal responsibility when a person is removed from the equation needs to be addressed.
- 8.80** You have a machine that's helping you decide how to deal with a life-and-death issue. And that machine has zero ability to empathize. It can't reason, it has no feelings. You are just part of the mathematical equation and operation that it's doing to determine what it says is a risk for you, without any feeling as to whether you're a 3-year-old child who has an inoperable tumor or whether you're a 95-year-old person who is suffering from arteriosclerosis. It's going to give you what the odds are.
- 8.81** And the coldness of that machine—doctors aren't always known, I don't mean to tar the medical profession across the board, for their bedside manner. The oncologists often are tough. They go through a process of seeing a lot of people who are very sick. And so, we want to go to a good doctor, not necessarily one who's warm and personable, but it would be great if it was both. It'd be nice to have a doctor who was good, top of the game, and at the same time made you feel better about your condition and what was going to happen to you in the future. They don't always go hand in hand, but you're certainly not going to get that from AI.
- 8.82** To the extent you have medical students in the future and doctors and pharma executives who come in, who rely more on the machine to assist them, the possibility of less interpersonal relationships with patients just seems to me that it becomes a little bit bigger. So, the ethical considerations— addressing the readability, scalability, and social impact are as crucial for me as anything, as AI becomes further embedded into cancer care.
- 8.83** The politics of reform. Reform is a bad word when it comes to cancer care because no one wants to do it. This framework that I talk about, about prioritizing public health over profits, this is the idea, system of choices. The problem is always a choice about incentives, governance, and whose interest takes priority. Is it the industry's priority in treatment, or is it the patient's priority in terms of what they get?
- 8.84** So, the existing incentive structure, and you've seen it since 1971 in terms of the war on cancer, has been to prioritize profits over public health and prevention—certainly, the institution of cancer, which becomes this industry there. The governance and decision-making processes inside of the cancer care system are influenced by corporate institutional interests, often, as I say, at the expense of patients and what the community might need.
- 8.85** So, whose interests matter? Current system is designed to protect the stakeholders. That shouldn't surprise you. That's the nature of big business and bureaucracy. I

understand that. We don't like the idea to use the word business and cancer care because it seems as though they should be different. But we should've learned a long time ago, it is in fact a for-profit business in most instances, especially with pharma companies, cancer centers. And dismantling the malignant empire, unlikely to happen.

- 8.86 But I do think there's a system to realign the incentives and the governance and change the focus of it in some ways. So, here's what I think of as some takeaways. You as patients, me as a patient, read the cancer claims skeptically. The next time you read about the next great CAR-T or molecular scissors, marvel at it, but realize that you have to get past the initial wonder-drug stage to find out if it really is.
- 8.87 Follow the money. I always say that. You put it together like you do with Memorial Sloan Kettering, nonprofits, political institutions, pharmaceutical companies. You see the aligned interest sometimes, when you were asking the other day about why do they all work together and they're in different industries, are working together for the same goal? And recognize the corporate interests. They serve the interests, largely the corporations and the institution, not always of the patients.
- 8.88 So, I asked a group of people that I talked to about this at a local group once, about a year ago, how they felt about the politics of cancer. About 70% were skeptical of cancer narratives. 80% understood the influence of corporations. 85% said they'd be willing to have a commitment to prevention over treatment. And 90% said it made them have a desire to drive systemic change. Now, that's 22 people, so we're not going to change the world. I understand that, but it was a way of getting some measurement.
- 8.89 Now, systemic changes for prevention, transparency, and cures. Rebalance the funding priority, easy for me to say, but it would really require political will to be able to do that, from late-stage treatment to a bit more primary prevention. Expand independent public funding for trials. Right now, the drug companies and the new AI companies are funding much of the research. I'd like to see most of that out and see more public funding of those trials.
- 8.90 Increase transparency in research. Require full publication of all clinical trial results and raw data sharing after a reasonable period. So the companies say, "We can't give you our raw data because that's proprietary. That tells you what we're doing." But after a certain amount of time when the drug is on the market, that should be available to everybody. The intellectual property controls over that should be gone.
- 8.91 Reform the conflict-of-interest policies. Make them strict. Make it tough, so that if you're serving on the American Cancer Society, serving on a board of a comprehensive cancer center, you can't get stock shares, or can't get options inside a new company developing a revolutionary breakthrough for cancer treatment.
- 8.92 Strengthen the regulatory independence. Boy, I want firewalls between the regulators and the industry advisory boards. It doesn't exist. They cross over all the time, back and forth. And reduce the revolving door between senior regulators and the companies and

the industries they're supposed to regulate. This would require more than just the cancer industry requires, pharmaceutical requires, defense industry; goes across the board.

- 8.93** There's no appetite in Washington to do that because the people come in for each administration. This is a [bipartisan] agreement. Democrats and Republicans, thousands of people come in to serve in the new administrations. And then when they leave the administration, they go on to private industry often, at very, very big spots. Everybody wants the former Secretary of Defense, whoever that is, on their board, if you're General Dynamics. Everybody wants the former head of the FDA, for the most part, involved in their efforts on it. So they go straight into there and they don't want to close the door, none of them.
- 8.94** Empower patients and communities with real veto power on trial design. Boy, that would be something, if a trial was taking place, and you asked for public input before the trial took place. Pricing agreements and access decisions open to the public. Possible. So rebalancing priorities, preventable cancer cases worldwide, they exist. Current funding for prevention, very low. Potential reduction in cancer death through prevention is possible. And public awareness of prevention strategies would be very, very good.
- 8.95** So, shift the funding and priorities toward evidence-based prevention strategies, my suggestion. Increase transparency and independence in cancer research and regulation. Empower patients, communities, and public health advocates to shape the cancer care agenda. That's never going to happen. I understand that this is my wish list; I'm entitled to a wish list at some point, even though I realize the industry is going to throw it in the rubbish bin. Enact strict conflict-of-interest policy to reduce corporate influence.
- 8.96** Incentivize the development of affordable, accessible cancer interventions. Boy, I tell you, that runs against the entire industry DNA. Affordable, accessible cancer initiatives? The better the innovation, the costlier it is. And build a healthcare system that prioritizes public health over private profits, that also is on the wish list that I understand is not going to happen anytime soon.
- 8.97** So, patient councils. They've been used in some local communities, sometimes on cancer care. They get feedback from patients. It's passed on to the powers that be, can have an effect. Not everybody has the power of the pink ribbon brand to have an influence. So local communities sometimes on unusual diseases and cancers can have an effect.
- 8.98** Whistleblower protections. Well, I'll tell you, protect whistleblowers. They come out with information about double dealing inside the cancer industry or a comprehensive cancer hospital, I want to know about it. Community engagement and transparency reforms. Disclosing conflicts of interest and data. Cancer system, this is it. So reforming

the malignant empire. Embrace new tech. I'm willing to do it. I like liquid biopsies and multi-cancer [early] detection. I get it. We have to use it. The future is here.

- 8.99** Harness that AI responsibly. Screening while mitigating risk. Prioritize prevention, if you can, by shifting funding to a little bit of that. Promoting transparency, empowering the patients, strengthening the regulation. Now, path forward is not an easy one. I get it. It's not going to be simple, and all of a sudden, we have a reformed politics of cancer. I do think some final thoughts for this.
- 8.100** Cancer is a medical disease. I said to you in the beginning that this is about, cancer is at the heart of this, but this isn't about cancer. It's about the structure that's built up around it. So, it is a medical disease, but the outcomes are shaped by political and economic decisions often. What gets funded, approved, reimbursed, and delayed determines who benefits and who waits. It's the way the healthcare system works, even on disease as critical as cancer.
- 8.101** Progress does not fail by accident. It fails when incentives override evidence, and sometimes they do, still paying to go down the wrong path. Patients are not just the recipients of care. In the world that I view it, they are also the stakeholders in a system that governs it. And understanding this system doesn't cure cancer, but it changes what's tolerated and what we, as the public, should demand. And this, for me, is where accountability begins, in patients who are treated with some dignity and lives extended.
- 8.102** The structure of the cancer system itself—I made some notes. I was thinking about this last night, separate from this. This profit-driven treatment, the marginal prevention, the suppression of low-profit alternatives. It's not why America is losing the war on cancer. It's not because we have a lack of brilliant scientists or technology or money being spent on it. It's that we're approaching it just strictly as something to make markers of extended survival times. And the metrics for improvement and progress are too low-bar.
- 8.103** When I talk about having regulations to prevent prevention, one of the things that's great for industry in the cancer industry, is that you have—here's some of the groups, EPA, FDA, OSHA, CPSC, USDA. They're all looking at additives and food and everything else. They're spread across the group, and that's why there's so little regulation on this. There's an unreasonable risk that takes place because it's inconsistent. There's no unified policy on it.
- 8.104** A future-oriented cancer policy, in my view, is less about moonshots and more about what I call the hard bureaucratic work that's required for coordinated testing, for preventative infrastructure that focuses on patients, so that any of us, if we become part of the cancer care industry at any time in the future, can be confident that, while we still have a serious disease to deal with, that we're dealing with a system that has

provided us an alternative in a list of treatments that are not there because they are politically the ones that were pushed forward.

- 8.105** They're not there because somebody, the FDA, decided that this was the one they wanted to have more than anything else. And they aren't there because the largest company with the largest lobbying dollars put them there. They're there because they really are quality, and maybe they give us a chance at a new future. So, cancer care, having been through it with Trisha, is a process of hope. It's a process at times of anger and frustration. And you talk to any cancer patient that's been through it, and they will go through their ups and downs, and they have their moments. That's the way the system is. I get it.
- 8.106** But they are dealing with us, and they, meaning the system, at a time when we are the patients at our most vulnerable. Because still, that very same disease, it used to be called the Big C, the very same disease that Mary Lasker was told, "You can't mention on the radio because you can't say the word cancer," even though we can talk about it openly now, and the risk factors and everything else, it still has, in the back of our mind, it has this ring that it has of the seriousness of that illness.
- 8.107** And that's still the strength of the industry, is that it knows for the most part we are willing to accept whatever they give us because the hope is looking for treatment and survival. But at the same time, I have found it's not incompatible to be, I would not say a difficult patient or family, but to be an informed consumer. So if you get there, you go in wise, you do your own research, you question them, and then you will have your best possible outcome.
- 8.108** Thank you, by the way, very, very much. Politics of cancer, not an easy subject, not easy as students to come out to the class and listen to eight lectures about the cancer industry. There are other industries that we could talk about that might be a little less heavy. This is a heavy one because many of us have somebody that we know, we have a family member. Almost, it's impossible to be in America or in any other country and not be affected or know somebody, even if it's outside of your own family.
- 8.109** So you immediately see that and you think to yourself, "Politics of cancer, I can't listen to 8 hours about cancer." I will tell you, in preparing it, and I know this from Trisha, who would occasionally say, when I would say, "What about this?", she would say, "I just need a break from it for a few minutes." So if she needs a break, and she works with me on every project, on every interview and everything else, I know it's a tough subject.
- 8.110** So, I want to thank you very much, because I'm hoping it also opened the eyes of students here and others who watch it to the idea that, yes, it's an industry that can cure, but it's also a machine. Thank you very much.

- 8.111** Let's say, just for the sake of imagining, that some AI system or some quantum computer or something creates something that's maybe not a cure, but equivalent enough to at least make cancer irrelevant. What happens to this big industry?
- 8.112** It's a great question. In my work as a journalist, I often run into people who have a theory, a conspiracy theory, about a given industry. One of the ones I run into the most frequently about cancer is, people say to me, "Don't you realize you're wasting your time? You're talking about the politics of cancer. They have a cure." They, it's always they, the mysterious they. "They have a cure for it. They just haven't put it out because there's too much money to be made from it."
- 8.113** I think about that. No doubt that this chronic care system, as I call it, this care for cancer, if there was a cure, or it was largely eliminated, you would eliminate a \$220-billion business overnight. So there's a lot of money there. It's the same reason pharmaceutical companies prefer to put out drugs for chronic conditions like diabetes, heart disease, and everything else. You take a pill every day for the rest of your life instead of putting out antibiotics, which you take for 5 or 7 days.
- 8.114** So all the drug pharma testing people are looking for an ongoing chronic situation, even though with cancer, patients may sometimes be cured or not, other patients are coming in all the time. So the question is, is there a secret cure that somebody has in the drawer? One of the things I think back to is Steve Jobs, who was the person who made Apple. At the time that he died, just having turned [56], he had a cancer that he had waited a while on to discover. He's got billions of dollars. And he could buy, if that cure existed somewhere, he could essentially have billions of dollars go toward that cure. He's unable to extend his life.
- 8.115** I think that, in the future, if there was an AI model together with a group of scientists who came up with a form of stopping cancer cells when they are detected very early, and making them regress and essentially disappear, and there's no major side effect, so it was the end of cancer as we knew it, they would patent that, and the rest of the industry would be crying about having lost their business. But that group would make tens of billions of dollars a year in profits.
- 8.116** So the idea that there's a consortium and that somehow they've come up with that cure, but they buried it in a drawer so that everybody can share in a piece of the pie, real capitalism, you find that type of discovery, you win the Nobel Prize, and you make more money than anybody could imagine. But the problem is with cancer, with all the work I've done on the latest material, it's not an easy disease. There's so many different variations of it. So it's not as though there's a single cure, and that's part of the problem.
- 8.117** So I do think, and there's been tremendous progress against some small genetic cancers and leukemias, I think we'll see more of that in the future, and some cancers

will be less and less visible as they disappear, but other cancers are going to prove more difficult. I think that, if that cure existed, it would be marketed.

- 8.118** Hi. Earlier on, we were talking about people who have been given placebos and their power of positive thinking. In some cases, people's tumors had just been left and they weren't there anymore. Then we were talking about the chart that had causes for cancer, most of it being environmental and the small amount being hereditary, and then you had a 17% of random. Has it ever been looked into with links or correlations between maybe having stress and high cortisol levels that maybe produce something that could contribute to cancer?
- 8.119** So I know that there have been a lot of studies done on the deleterious effects of cortisol and how stress—you know, I love the general advice. They say, "Reduce stress." We should all reduce stress. It's a good health item. It doesn't always happen. I say, "I'm bringing my stress level way down. I'm not going to sweat the small things," and then something goes wrong on my notebook computer, and I can't get access to the internet. My blood pressure is going like this. We break our own advice sometimes on stress.
- 8.120** There have been a lot of studies on stress, but nothing that I have seen that looks for whether those spontaneous, the bad luck incidence of cancer, are results of people with higher levels of cortisol over a period of time. It would be fascinating to find that out because I often think in the back of my mind, you talk about powerful medicine. I think of the Krebiozen case, of that one person whose tumor reduced. And then when he read about the fact that it might be a scam, it came back, and then a doctor injected him with water, and it went away again. That's pretty remarkable.
- 8.121** Now, that's unusual. They believe, oh, well, it's a spontaneous remission. Just like there are spontaneous cases in somatic developments, that must have been a spontaneous one. But I have no doubt about the power of the mind. It would be very interesting to see if those who had that situation, you were able to link to higher levels of stress or cortisol, but I've never seen that work done.
- 8.122** Under that effect, if I could just add one last thing, you would tend to think that Buddhist monks who have gone off at ashrams and they're trying to achieve nirvana and they're praying for the rest of the world would have very low rates of cancer. But I'm not sure, because I've never seen a study based upon cancer rates with Buddhist monks, but I'm going to look for it afterwards, as I looked for whether Marie Curie, and I'll have to report that later, unfortunately, to you.
- 8.123** As far as a patient being an advocate for his own personal health care, how would you suggest going about getting credible information by which to make an informed decision, when Dr. Google and AI are not always reliable?
- 8.124** Yeah, no. That's a great—boy, we're accustomed to Google and AI giving us bad information on a question that we ask. That's fine, but you don't want it on cancer care.

So, I wish there was a shortcut. There isn't. It involves going to PubMed. It's an online database of clinical studies. You can look up your particular situation that you've been diagnosed with. You can find out what the latest talk is in the medical community and clinical trials that have taken place.

- 8.125** And so, it's a slower process because one of the things that, for instance, I was looking for when Trisha got her diagnosis of her tumor, and they suggested to her that she should be using this medication, an aromatase inhibitor, I was doing searches on all the clinical trials on PubMed to see what the latest was. And one of them said, remember we talked about before the side effects, you get side effects with that drug and people just stop using it?
- 8.126** So there's one study done by Europeans who said you could take—the normal dosage of one of these aromatase inhibitors is 2.5 milligrams. Doses as low as 1/33 of that daily dose kept the serum levels of the drug at the same level in the body. So I thought that's fantastic. So that means you could take a fraction of what they are prescribing and go ahead and have the same results. I mentioned that to the oncologist, who then said, "But that hasn't been studied. We don't know if having the same serum level is the same as having that dosage."
- 8.127** Then the next part of it is, there's another study that says, even when you reduce the dosage to that level, people still complain about the same side effects. So, even when you're doing the research, you're in what I call the deep dive, you don't always know if you've covered every part of it. There is a human element to that. And on the one hand, I don't want to rely on the machines.
- 8.128** So what I would do, what I did before, and what I suggest you do or anybody else does, is, you do some, you go online. So PubMed is there. It's easy enough to do. You get some studies that you think might be of interest, and then they're written in medical language, and they're impossible. And then you put those studies, the names of them, into an AI model, and you say, "Is there anything that contradicts the conclusion of these studies? And provide me in three paragraphs what the bottom line is for a layperson."
- 8.129** Now, it's less likely to pull the false information out of the ether. It doesn't mean it's foolproof, but you use AI as the assistant to take some material you already have, and instead of spending 6 hours going through it and having to look up every fifth word in life sciences or that, you get the AI model to help you through.
- 8.130** There's one thing I know without a doubt. When I would start to question the oncologist or the surgeon, they would immediately pull out what I call the MD card, which is, instead of speaking in lay terms, they'd use the medical terminology. They would try to have a thing where I would then have to look at them and say, "What are you talking about?" They would say, "What are you, a fool? You don't understand that

particular thing?" So, I like to, of course, become versed in that, so that I can ask an intelligent question and then try to understand the answer.

- 8.131** It's very difficult because the same way that Betty Rollin was told by her doctor, "You're a silly woman. You'll be dead in 3 weeks," they don't do that anymore, or they didn't at least in our treatment, but they use what I call the MD wall to keep you at a distance, to make it very clear that you're just a layperson and you're sort of questioning them.
- 8.132** But if you show them that you're not just going to stop, that you have a question and you're going to persist and you really want to know, and you're still going to have treatment with them, but you're just asking this out of curiosity, almost every doctor we dealt with eventually came around to actually having a serious discussion with us. Then that discussion, they would talk about the pluses and the minuses, and that's exactly what we wanted.
- 8.133** Because there's no situation in which we say, when you do research, in which you're going to come up with an answer that's so different than what doctors are offering you, that you can go in and revolutionize it, unless you're dropping out of the system, unless you're going for something like laetrile, or you're going for an alternative, Krebiozen. But if you're going within the traditional medical system, there are only so many option treatments, whatever the disease or cancer. And what you want to know is the pluses and minuses.
- 8.134** It's that part we talked about disclosure. You want the full information, at least I do, then I can make a decision fully informed. I don't want to find out afterwards, "if I had known that." I never want to be in a situation if I say, "If they had told me that, I would have made a different decision." I can blame myself later for making the wrong decision, but then I own it. I wish there was a shortcut. I don't have it for you.